

TRIANGULATING POINT SETS

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Discrete, Combinatorial and Computational Geometry
Universitat Politècnica de Catalunya

TRIANGULATING POINT SETS

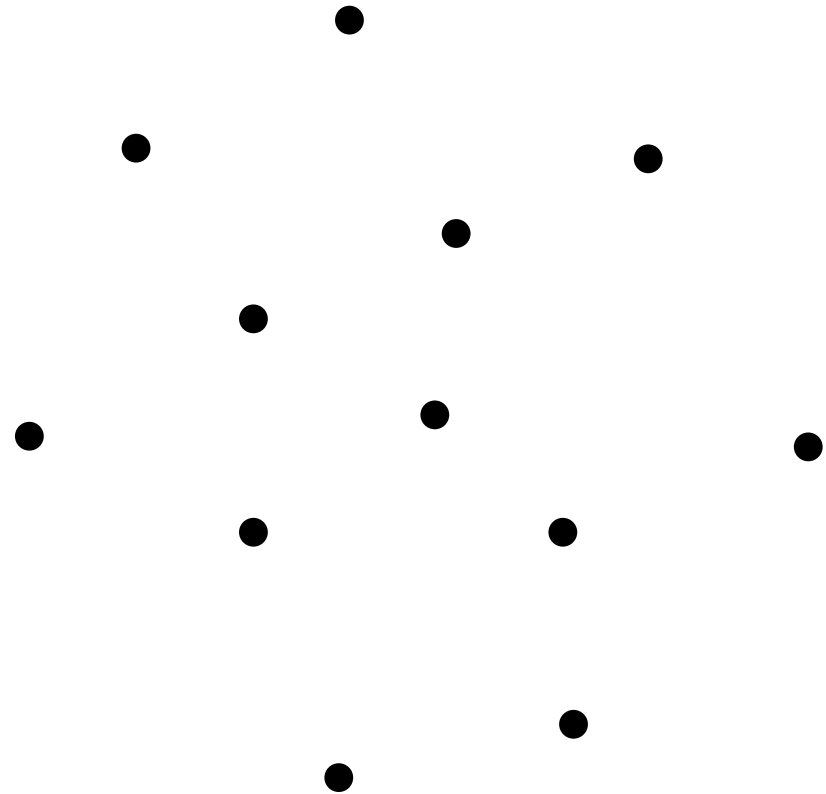
DEFINITION

A triangulation of A set P of n points in the plane is a graph having P as set of vertices which is rectilinear, planar, and maximal in the number of edges.

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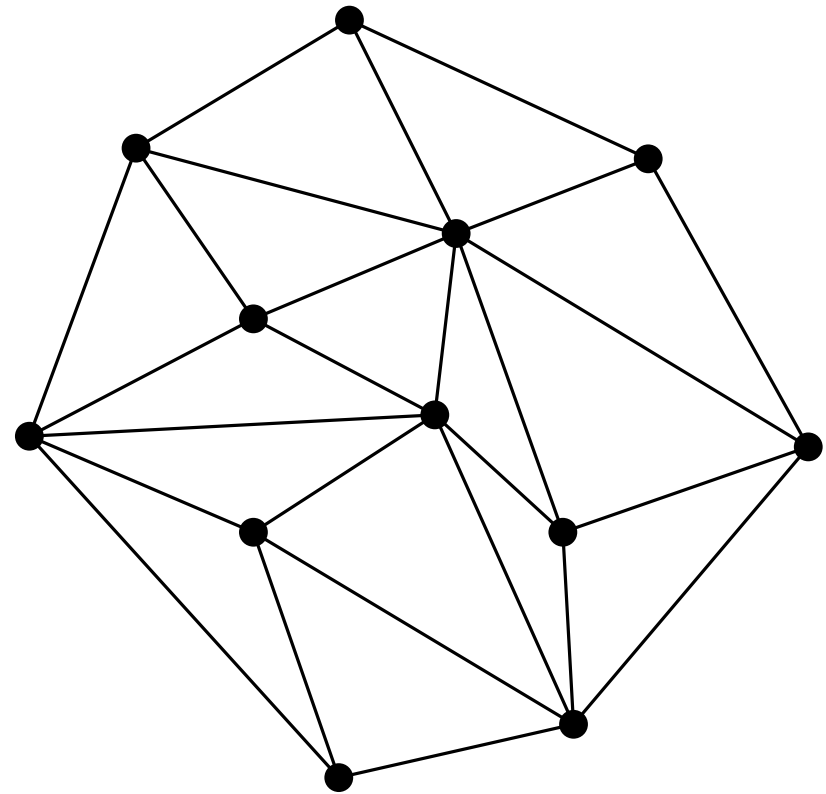
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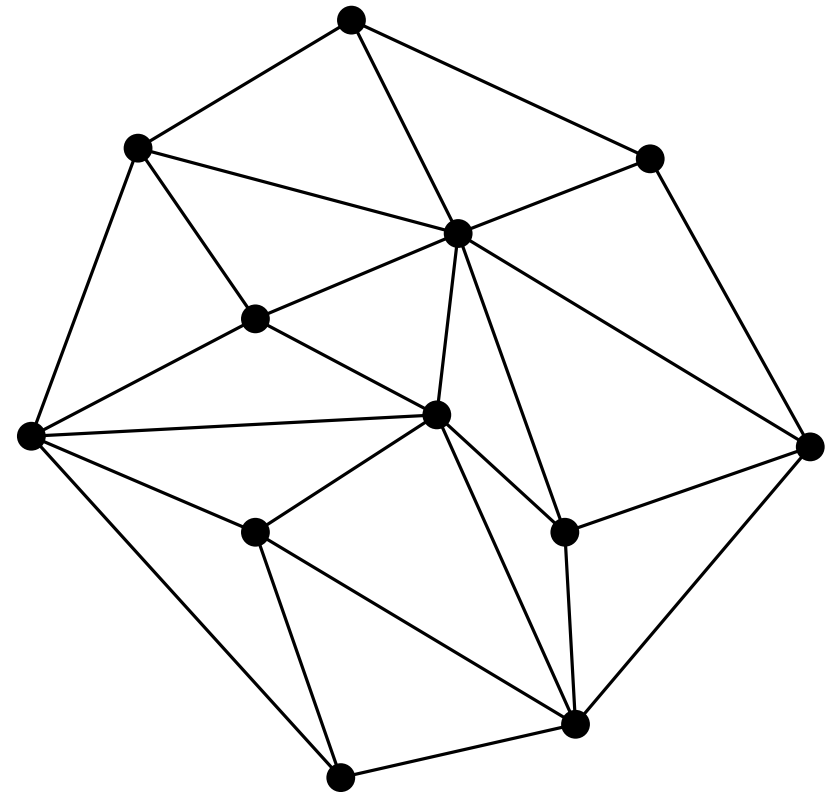


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A triangulation of A set P of n points in the plane is a graph having P as set of vertices which is rectilinear, planar, and maximal in the number of edges.

Corollary. All the faces of such a graph are triangles, except for the unbounded one, which is the exterior of the convex hull of P .



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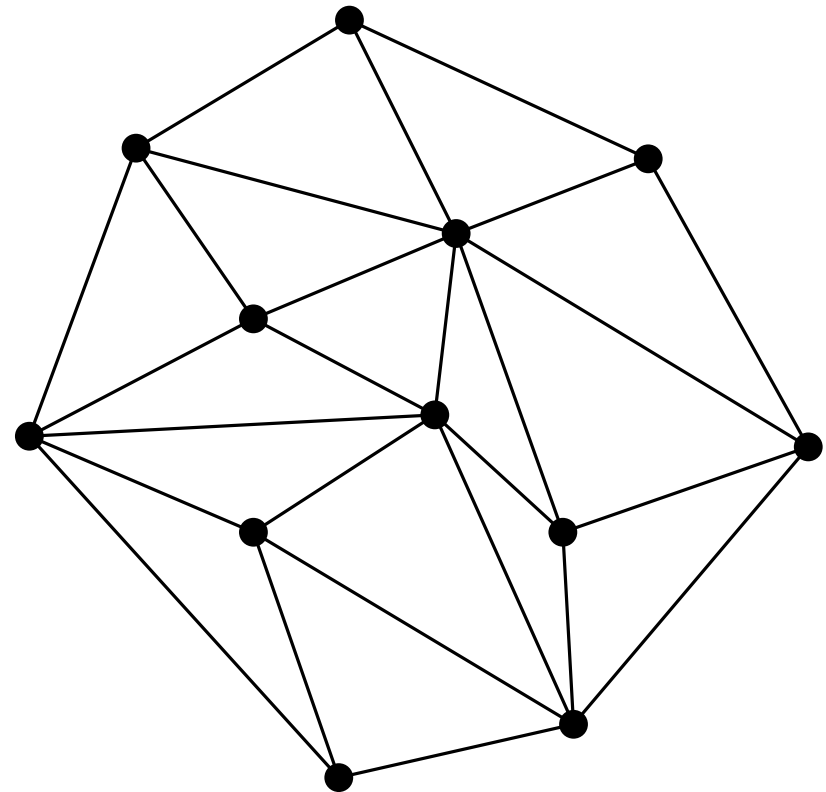
COMPLEXITY

Every triangulation of any set P of n points has:

$2n - h - 2$ triangles

$3n - h - 3$ edges

where h is the number of vertices of $ch(P)$.



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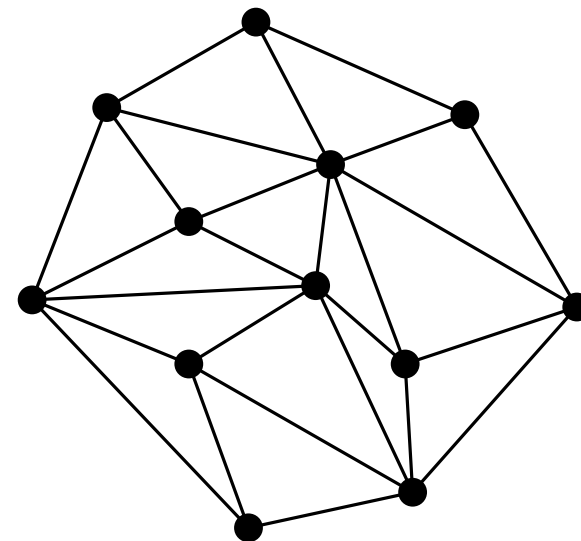
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Proof. Each triangle has exactly 3 edges. Each internal edge belongs to exactly 2 triangles. Each external edge belongs to exactly 1 triangle. Therefore, $3t = 2(e - h) + h = 2e - h$. According to Euler's formula: $n + (t + 1) = v + f = e + 2$.

Combining both equations:

$$e = n + t - 1 \Rightarrow 3e = 3n + 3t - 3 = 3n + 2e - h - 3 \Rightarrow e = 3n - h - 3$$

$$3t = 2e - h = 6n - 2h - 6 - h = 6n - 3h - 6 \Rightarrow t = 2n - h - 2$$

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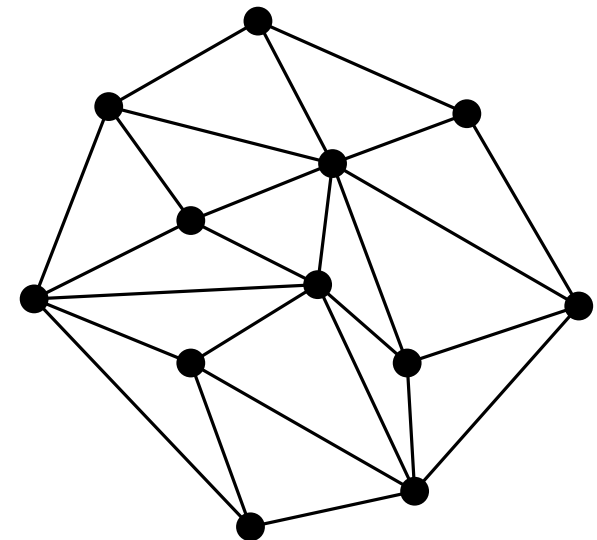
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DEGENERACIES

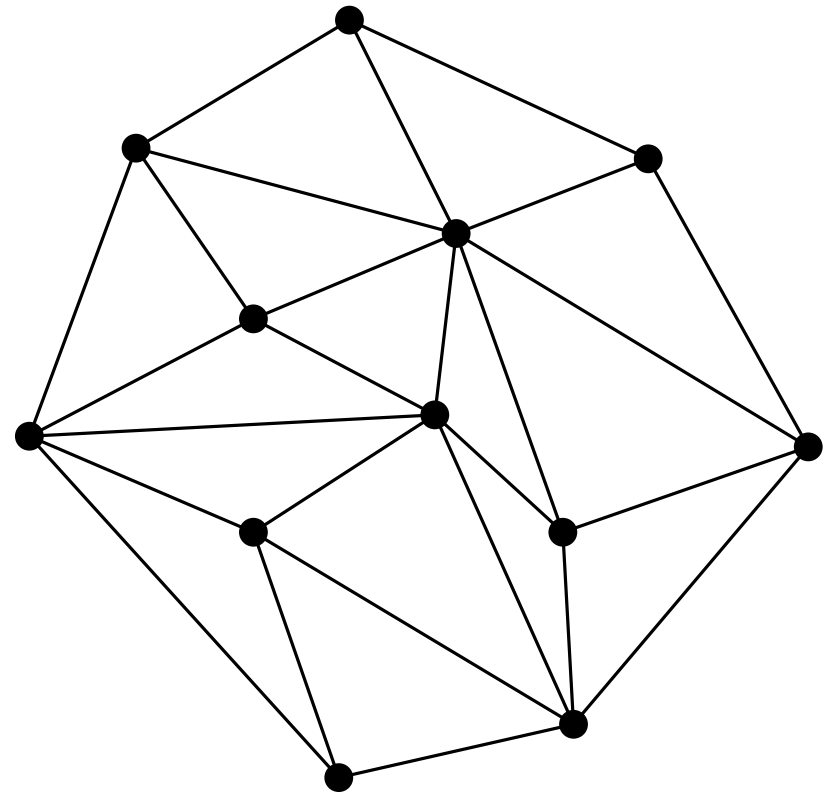
As you may have noticed, we are assuming that the set P does not contain three or more points on a line. The assumption is hold along the entire chapter.

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DATA STRUCTURE

We want to answer the most usual questions for any *decomposition of the plane*:

- For any given triangle, report its edges/vertices.
- For any given vertex, report the sorted list of edges/triangles incident to it.
- For any given edge, report its endpoints and its adjacent triangles.



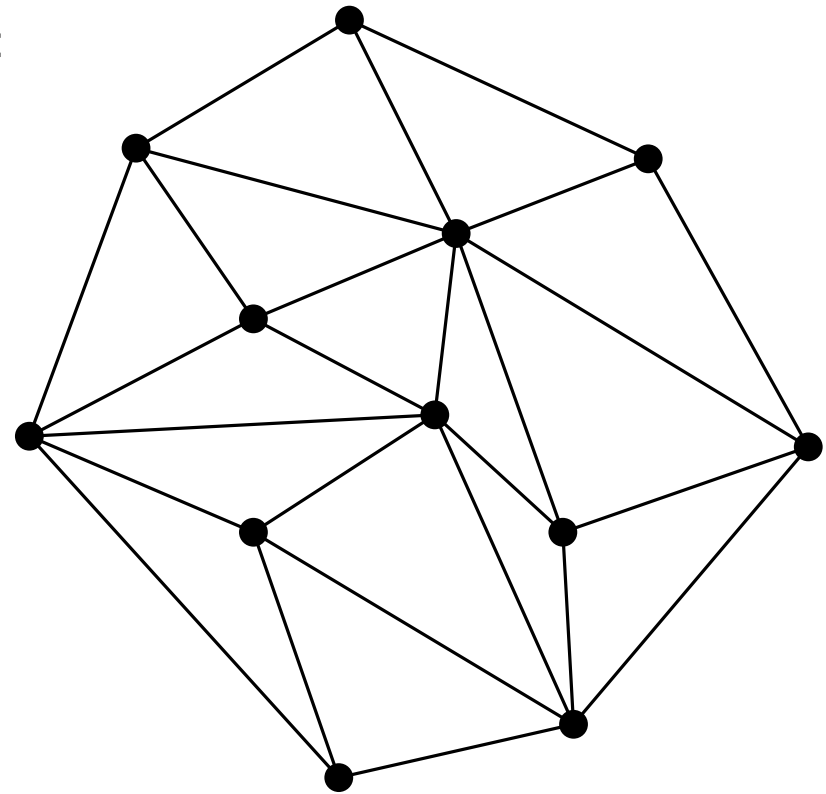
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The appropriate structure is, once again, a **DCEL**:



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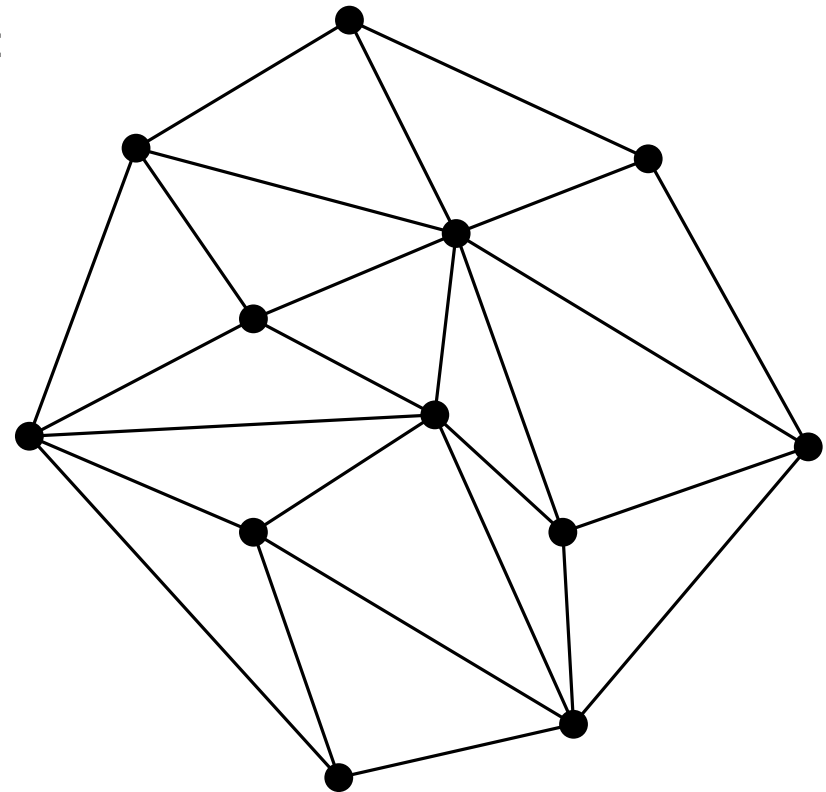
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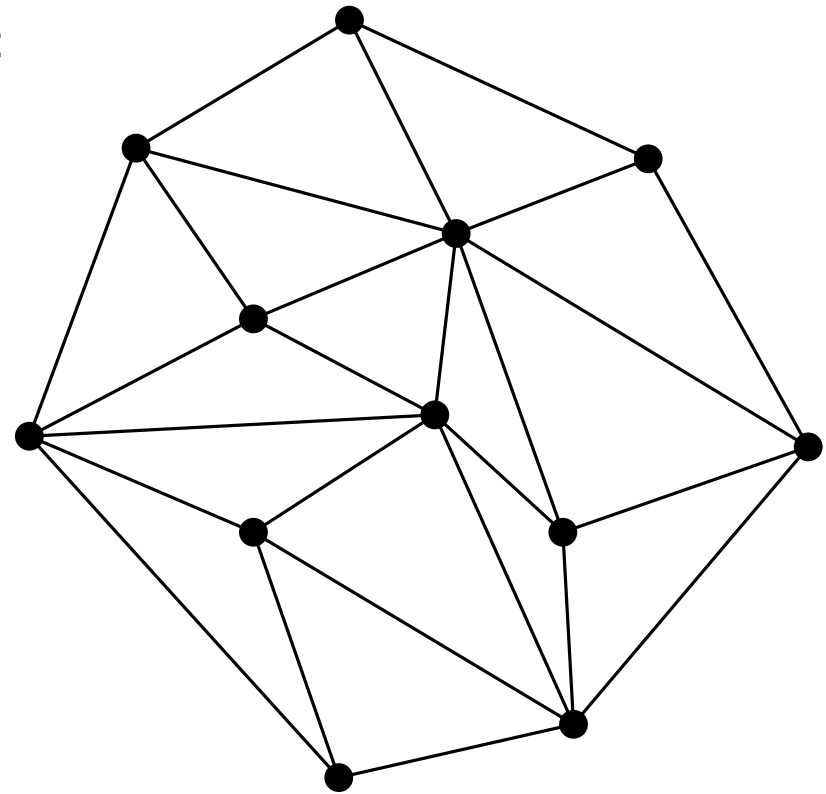
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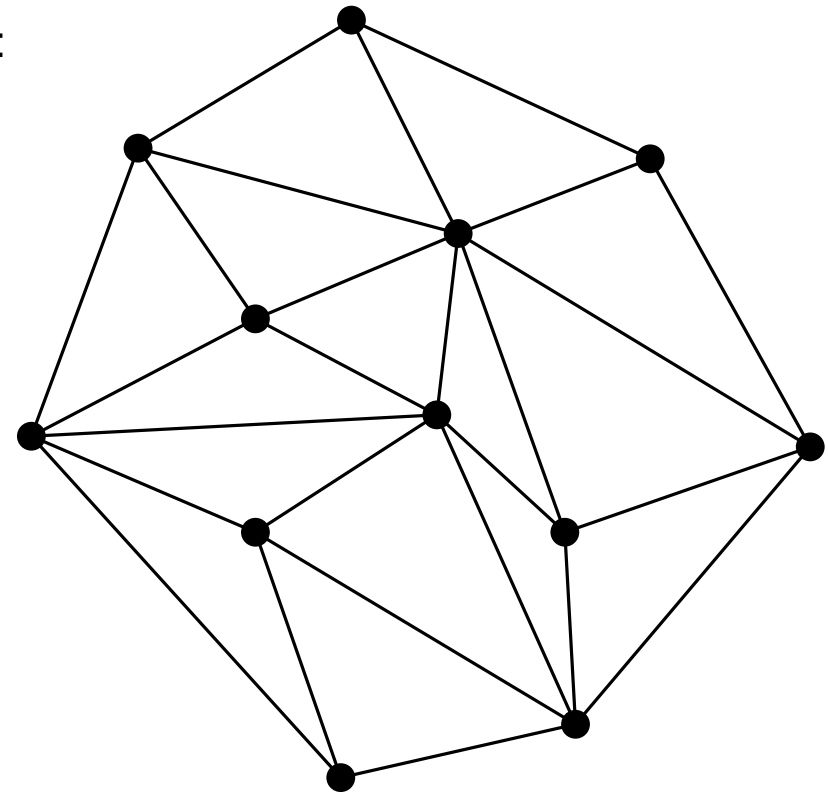
v	x	y	e
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Table of faces

t	e
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DCEL

e	v_B	v_E	f_L	f_R	e_P	e_N
-----	-------	-------	-------	-------	-------	-------



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ALGORITHMS

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1. Incremental algorithms

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1.1. Without sorting

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For each i , determine if p_i lies inside $ch(p_1, \dots, p_{i-1})$ (**point location step**)

- If p_i is external, compute the supporting lines from p_i to $ch(p_1, \dots, p_{i-1})$ and add all intermediate diagonals to the triangulation.
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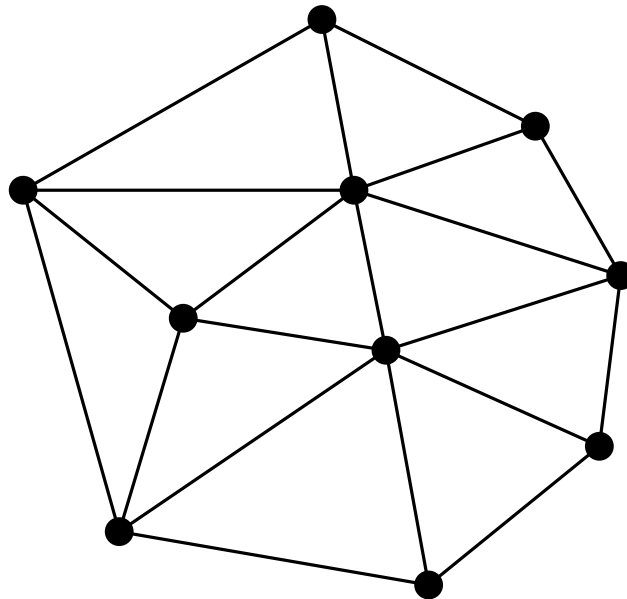
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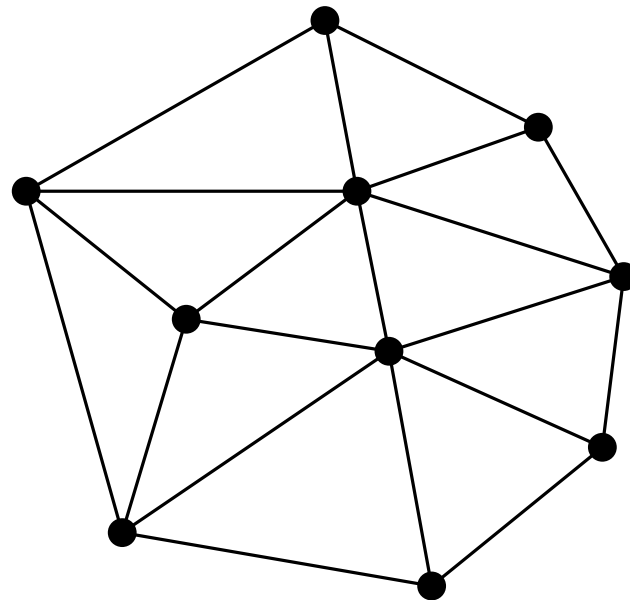
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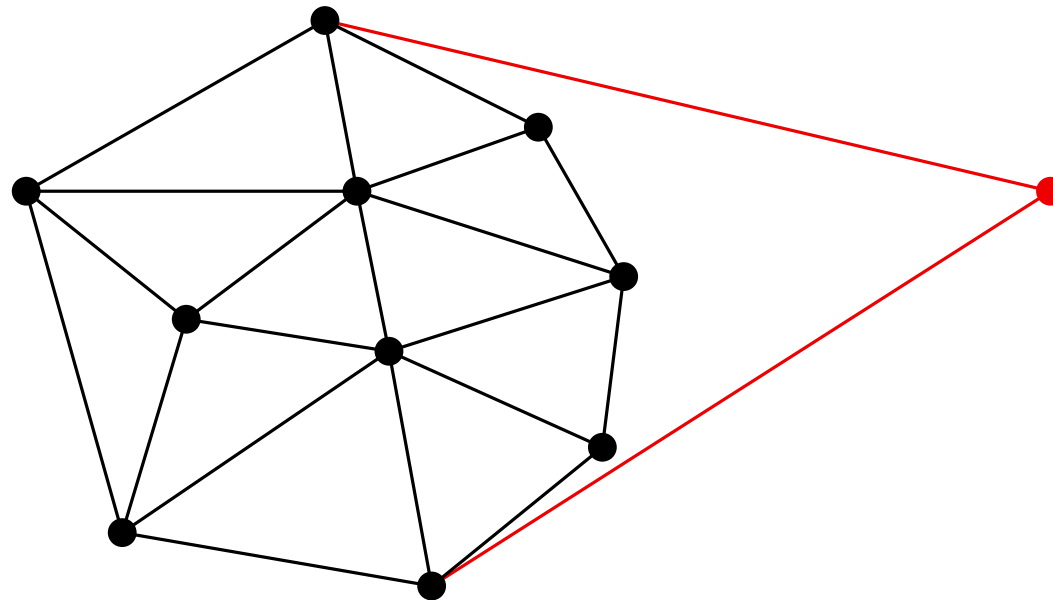
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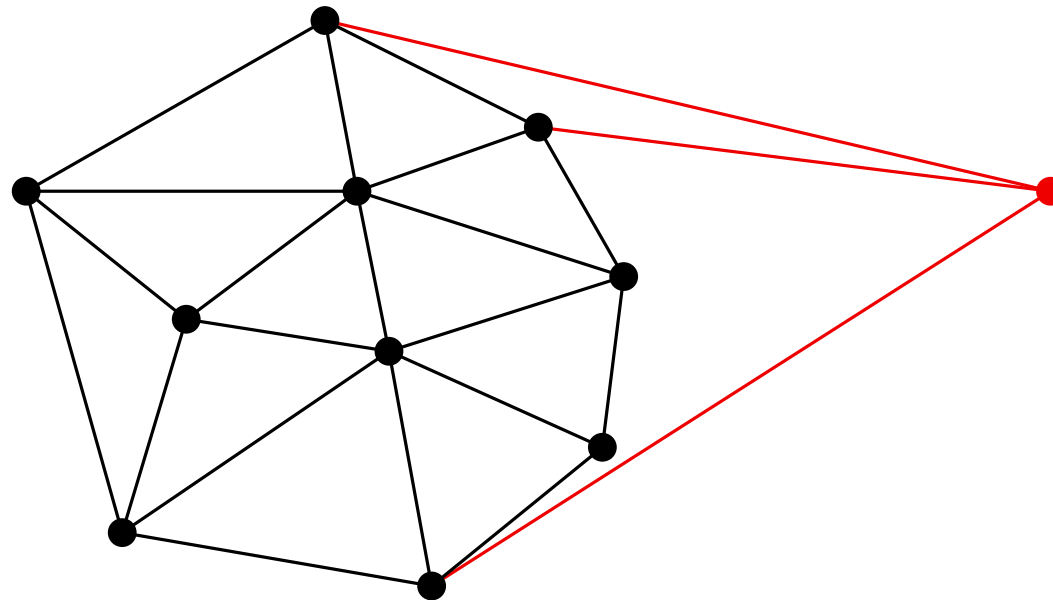
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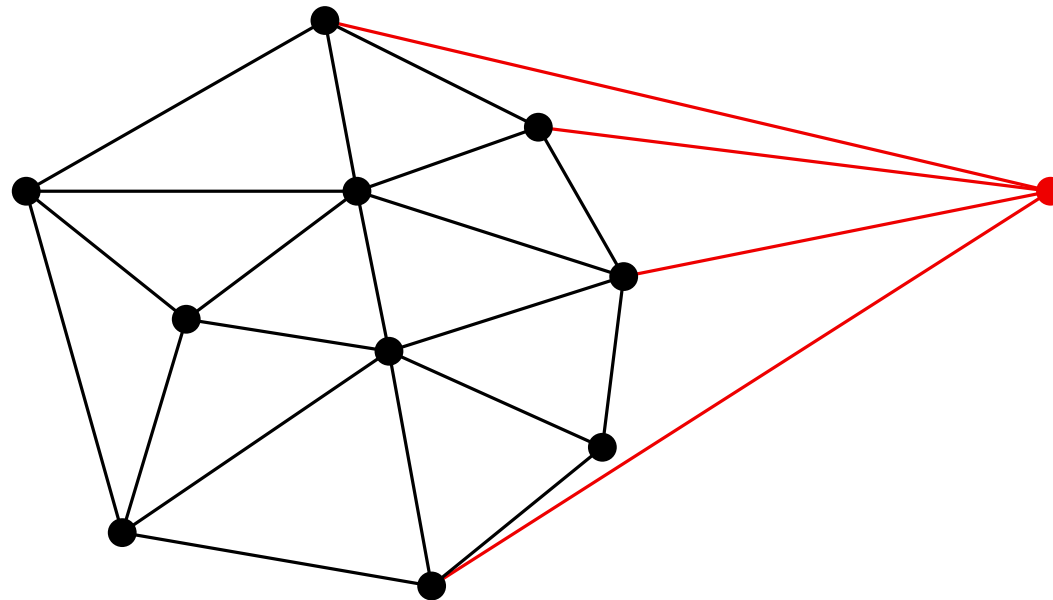
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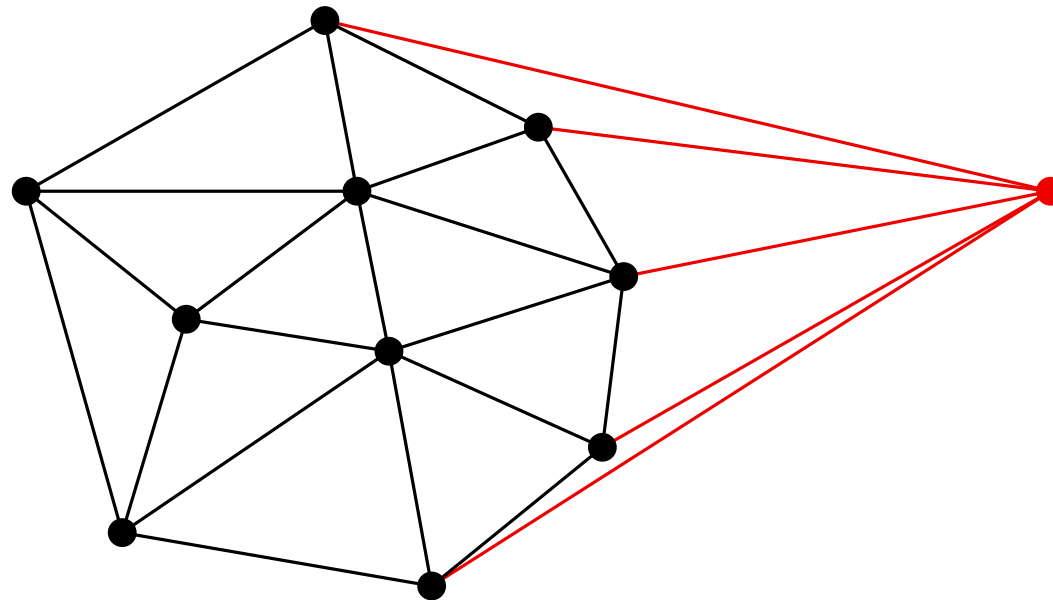
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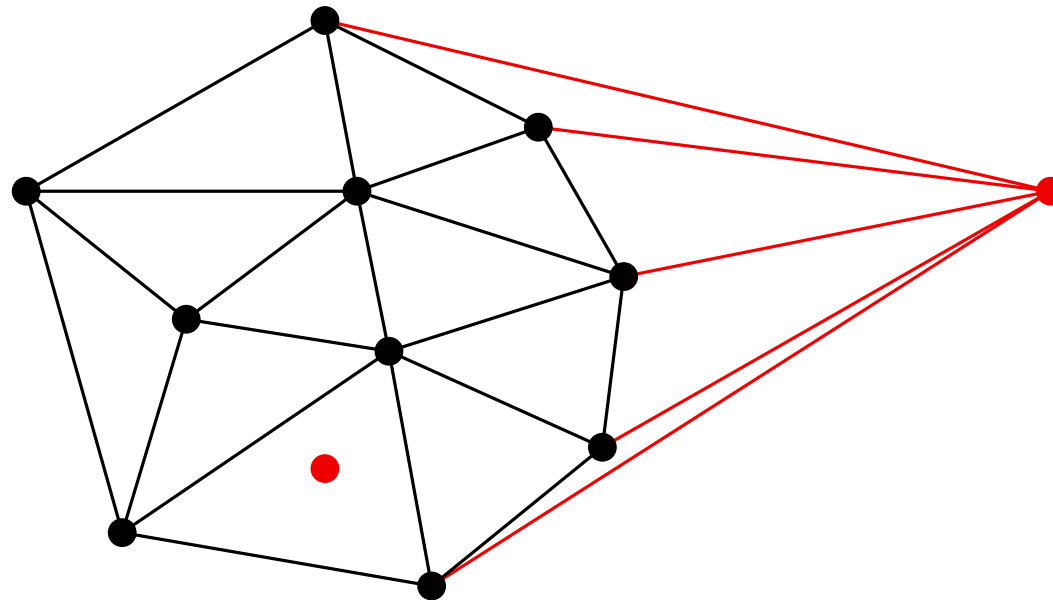
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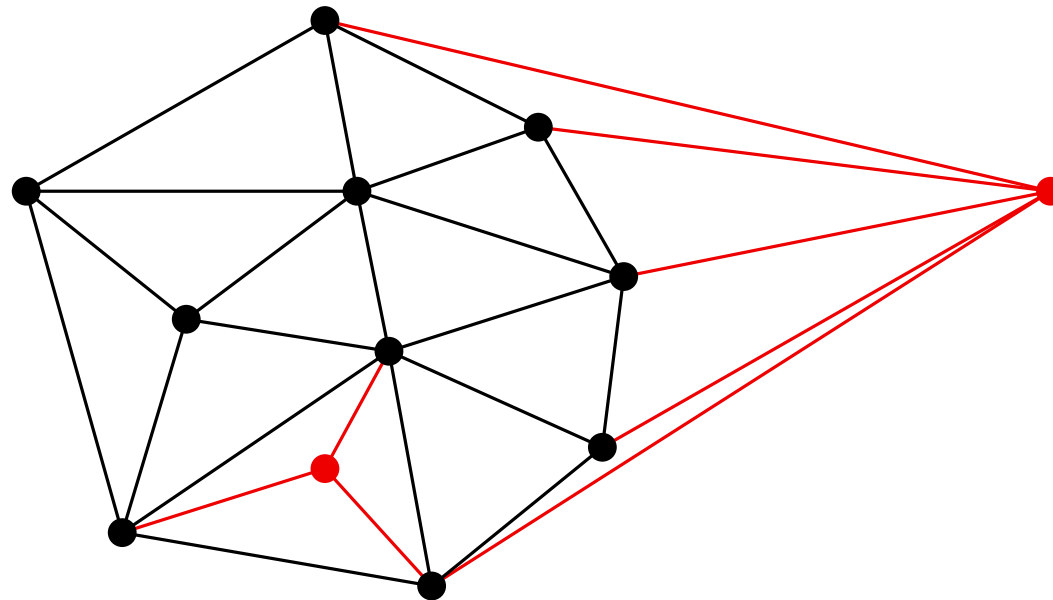
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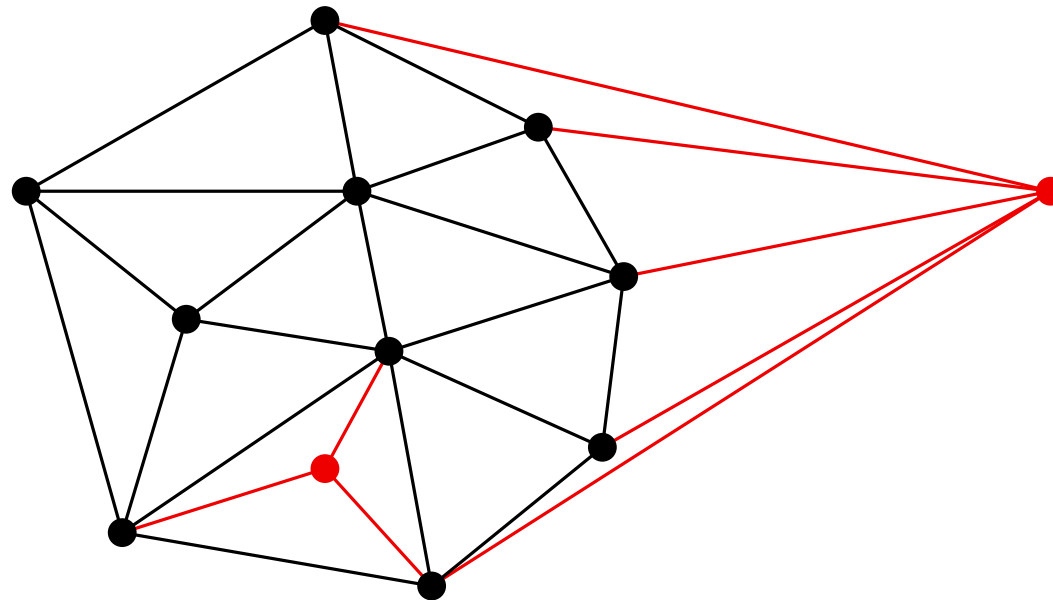
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Running time: $O(n^2)$

1.2. With sorting

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1. Incremental algorithms

1.1. Without sorting

Running time: $O(n^2)$

1.2. With sorting

First, sort the points in lexicographical order, and then insert points in that order. Now every inserted point is external!

Sort points lexicographically

For each i

- p_i must be external, so compute the supporting lines from p_i to $ch(p_1, \dots, p_{i-1})$ and add all intermediate diagonals to the triangulation.

Once the supporting lines are computed, the sorted order of the points along the convex hull allows to add the i diagonals in $O(i)$ time. Thus the *amortized* time of the insertion of all diagonals, for all points, is $O(n)$.

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1. Incremental algorithms

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Running time: $O(n^2)$

1.2. With sorting

Running time: $O(n \log n)$

1.3. Point location with hierarchical structure

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ALGORITHMS

1. Incremental algorithms

1.1. Without sorting

Running time: $O(n^2)$

1.2. With sorting

Running time: $O(n \log n)$

1.3. Point location with hierarchical structure

Use an auxiliary enclosing triangle so that all inserted points are internal.

To facilitate point location, use a hierarchy of triangles; each time a new point is added, a triangle gets subdivided into three children.

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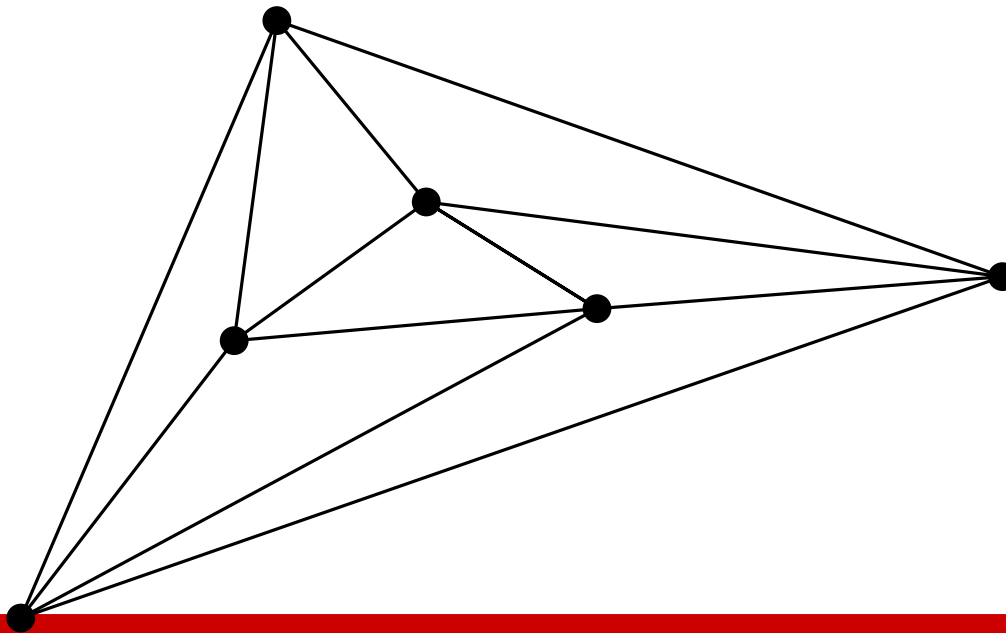
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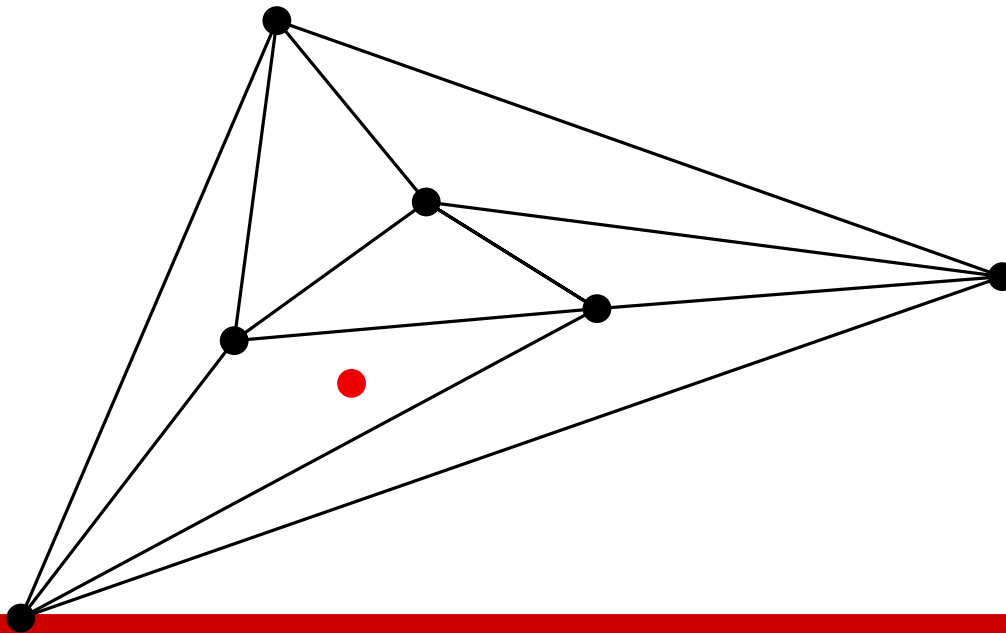
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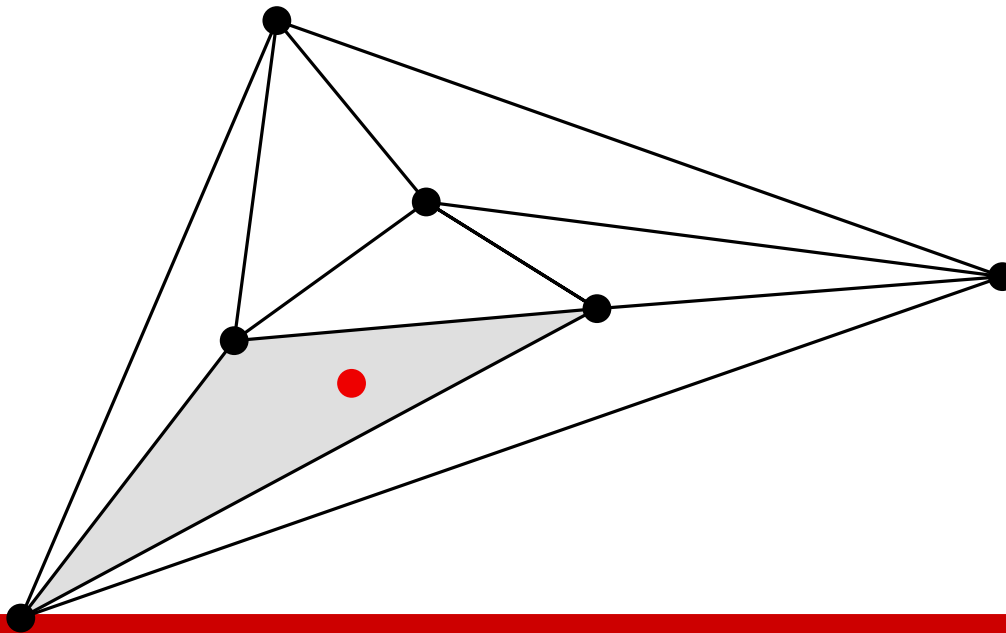
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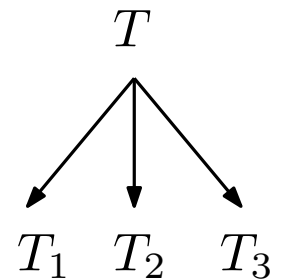
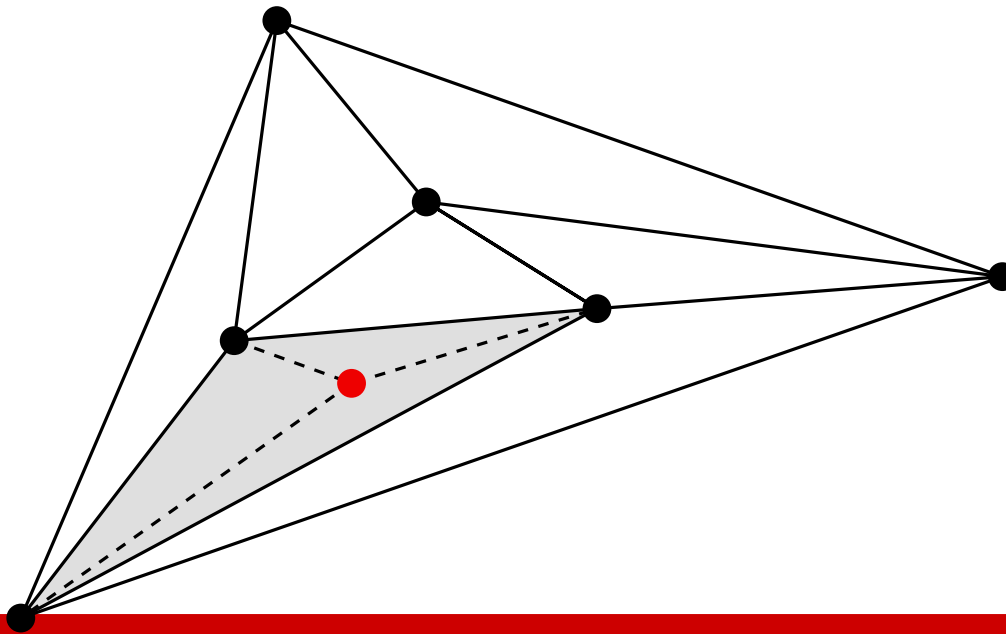
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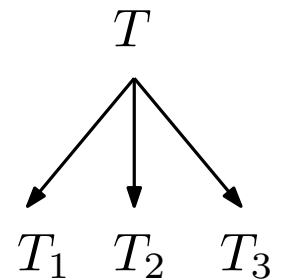
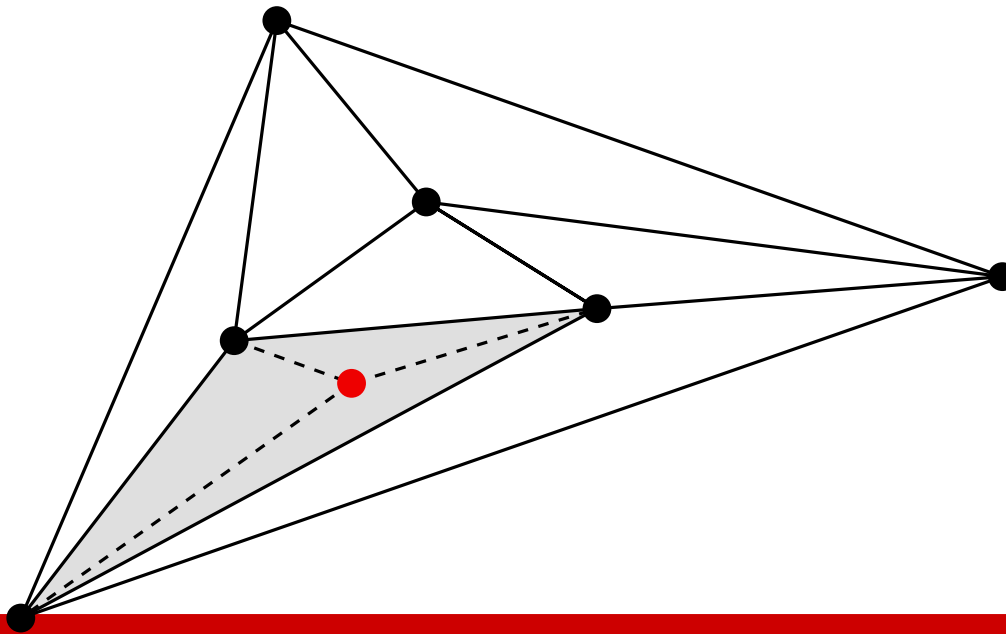
Running time: $O(n \log n)$

1.3. Point location with hierarchical structure

Running time: $O(n^2)$ worst case, $O(n \log n)$ if balanced

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1.4. Hierarchical, randomized

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1.4. Hierarchical, randomized

Same as 1.3, but insert points in random order. This results in a search tree that is expected to be *balanced*, thus have $O(\log n)$ height.

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1.5. Point location with auxiliary point(s)

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1.5. Point location with auxiliary point(s)

A fixed point p is used as a reference, and $P \cup \{p\}$ is enclosed in an auxiliary triangle.
When inserting each point p_i :

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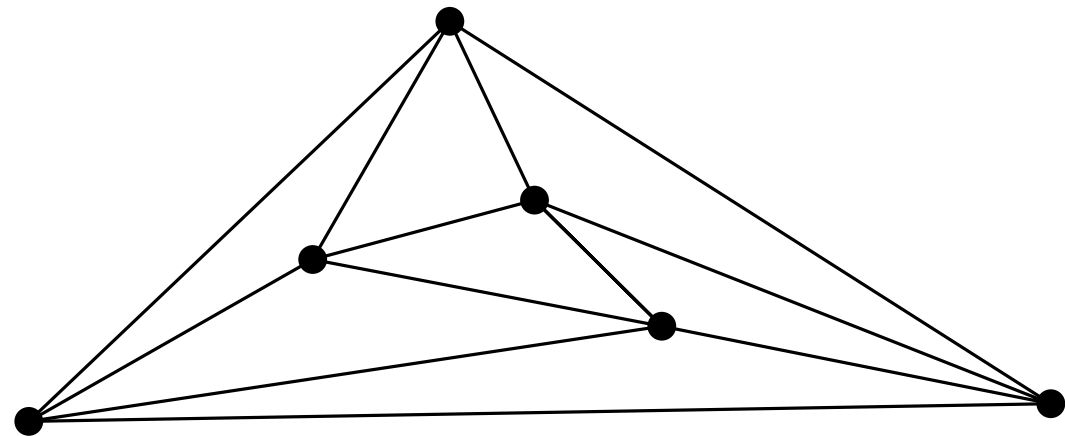
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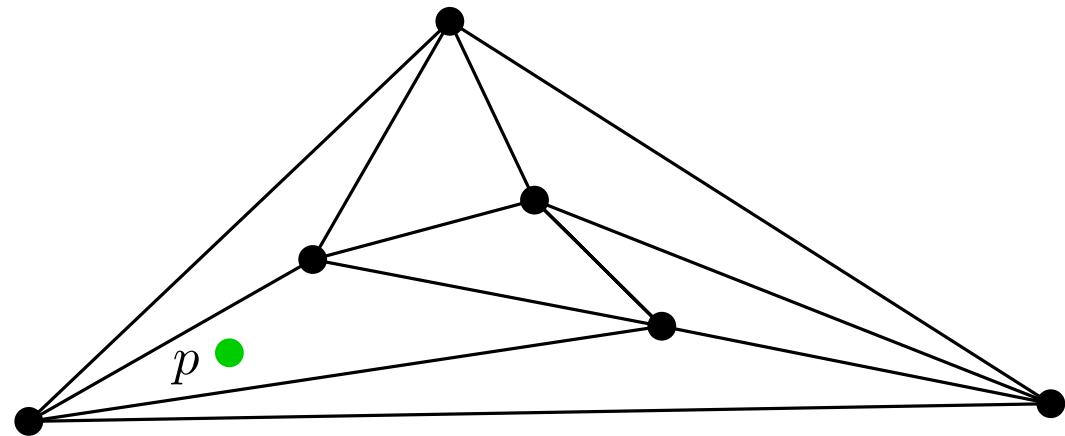
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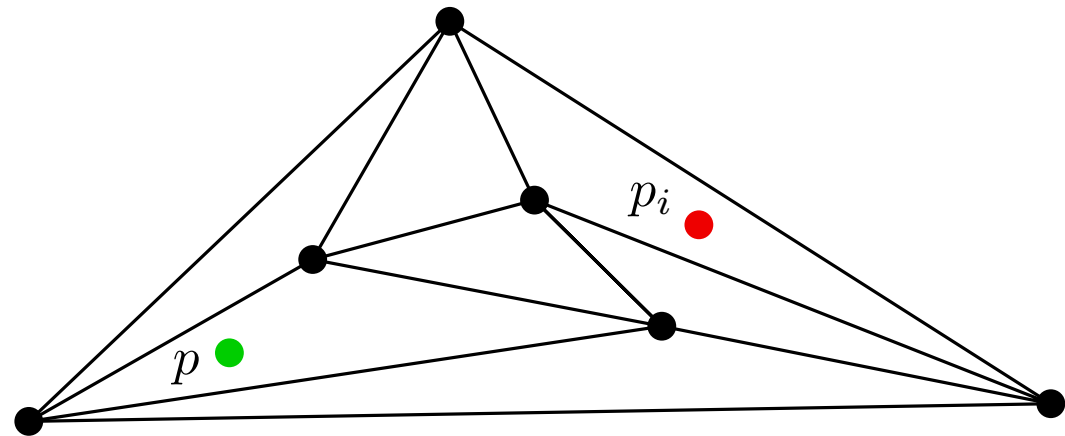
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Running time: $O(n^2)$ worst case, $O(n \log n)$ if balanced

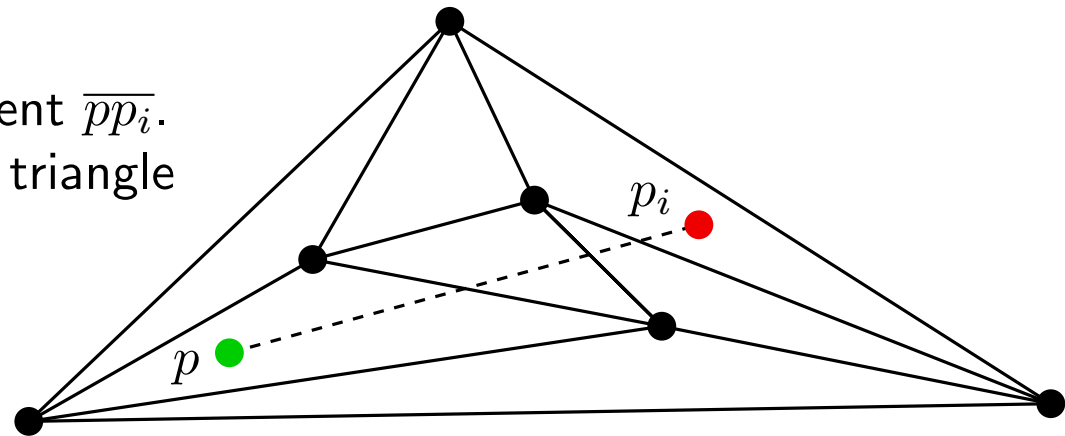
1.4. Hierarchical, randomized

Running time: $O(n \log n)$ expected

1.5. Point location with auxiliary point(s)

A fixed point p is used as a reference, and $P \cup \{p\}$ is enclosed in an auxiliary triangle. When inserting each point p_i :

- Scan triangles stabbed by the segment $\overline{pp_i}$.
- Update, if necessary, information of triangle containing p .



TRIANGULATING POINT SETS

ALGORITHMS

1. Incremental algorithms

1.1. Without sorting

Running time: $O(n^2)$

1.2. With sorting

Running time: $O(n \log n)$

1.3. Point location with hierarchical structure

Running time: $O(n^2)$ worst case, $O(n \log n)$ if balanced

1.4. Hierarchical, randomized

Running time: $O(n \log n)$ expected

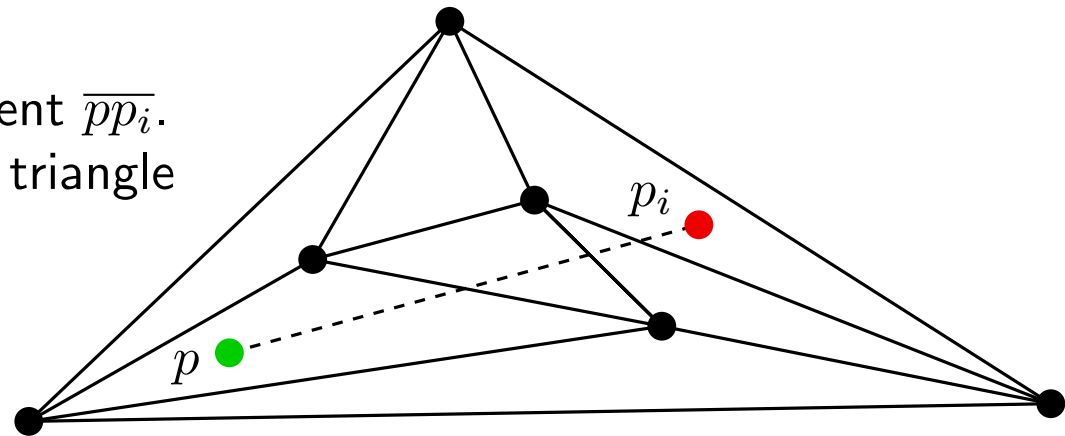
1.5. Point location with auxiliary point(s)

Running time: $O(n^2)$ worst case, $O(n^{3/2})$ expected

A fixed point p is used as a reference, and $P \cup \{p\}$ is enclosed in an auxiliary triangle.

When inserting each point p_i :

- Scan triangles stabbed by the segment $\overline{pp_i}$.
- Update, if necessary, information of triangle containing p .



TRIANGULATING POINT SETS

ALGORITHMS

1. Incremental algorithms

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Running time: $O(n^2)$

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TRIANGULATING POINT SETS

ALGORITHMS

TRIANGULATING POINT SETS

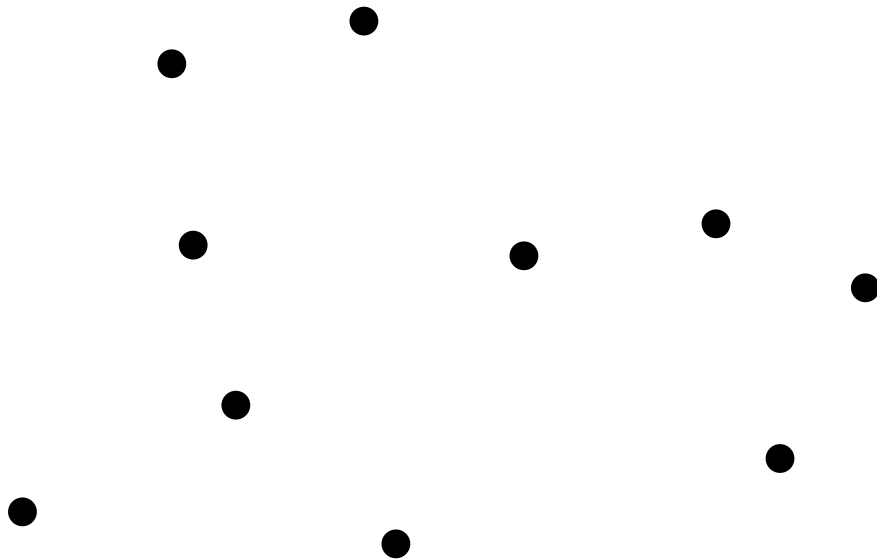
ALGORITHMS

2. Graham's algorithm

TRIANGULATING POINT SETS

ALGORITHMS

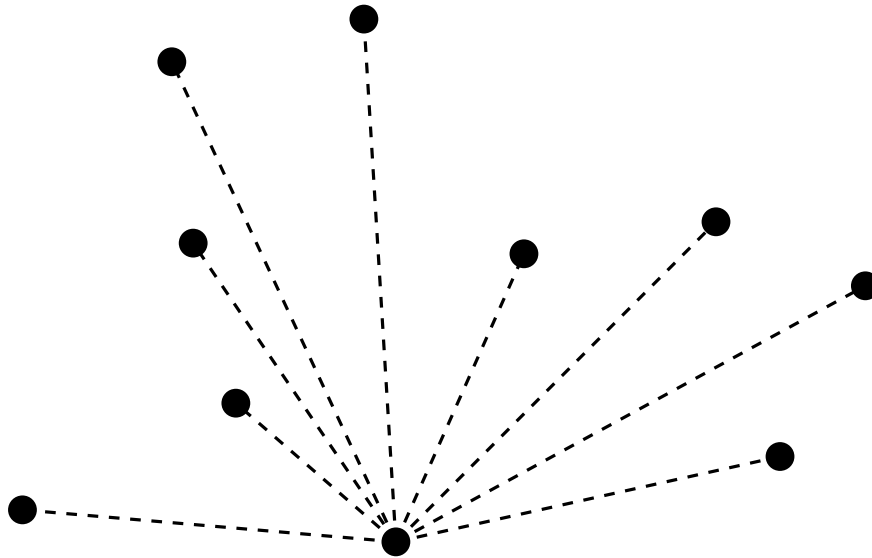
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

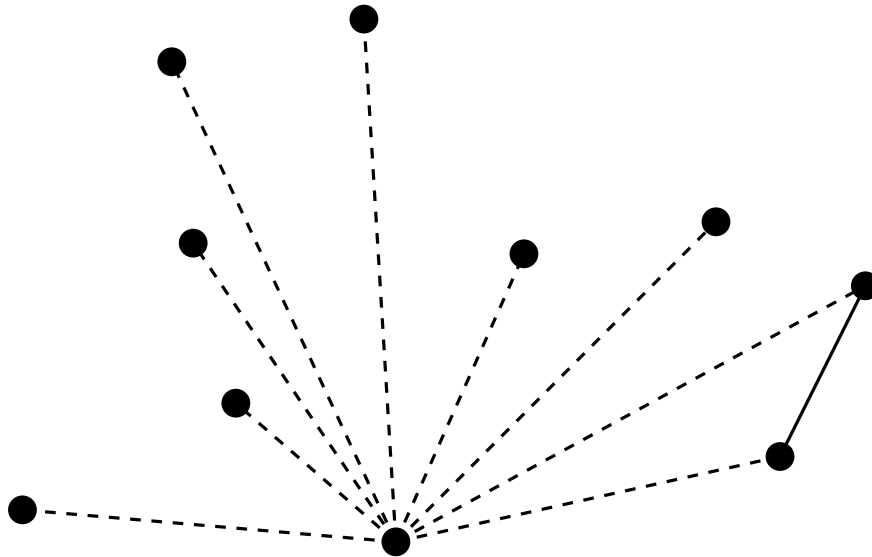
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

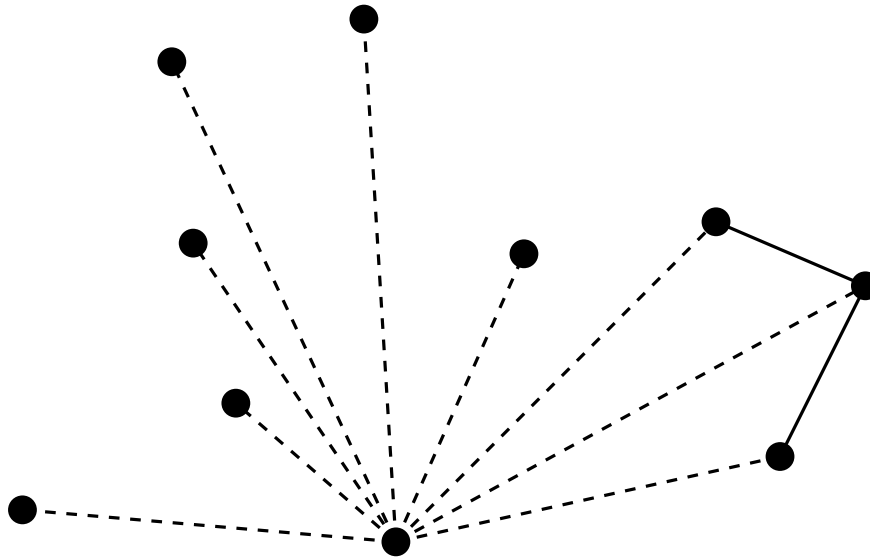
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

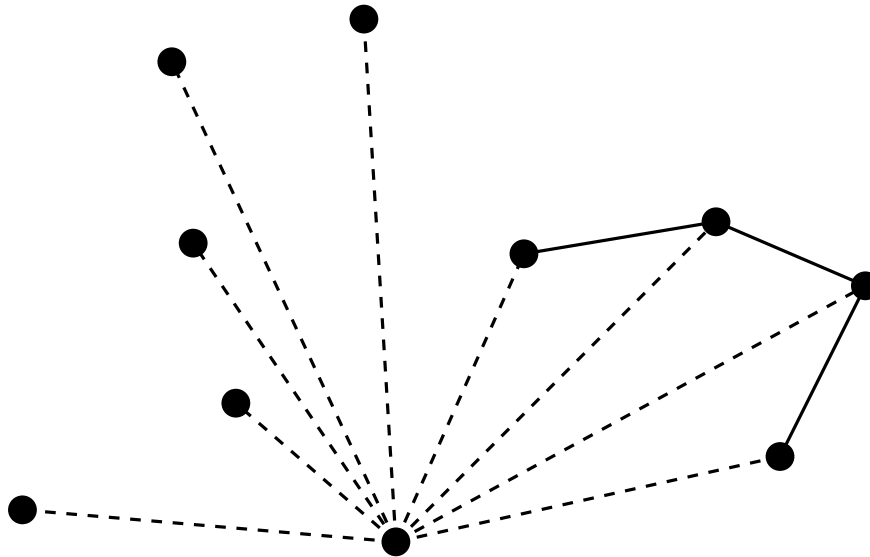
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

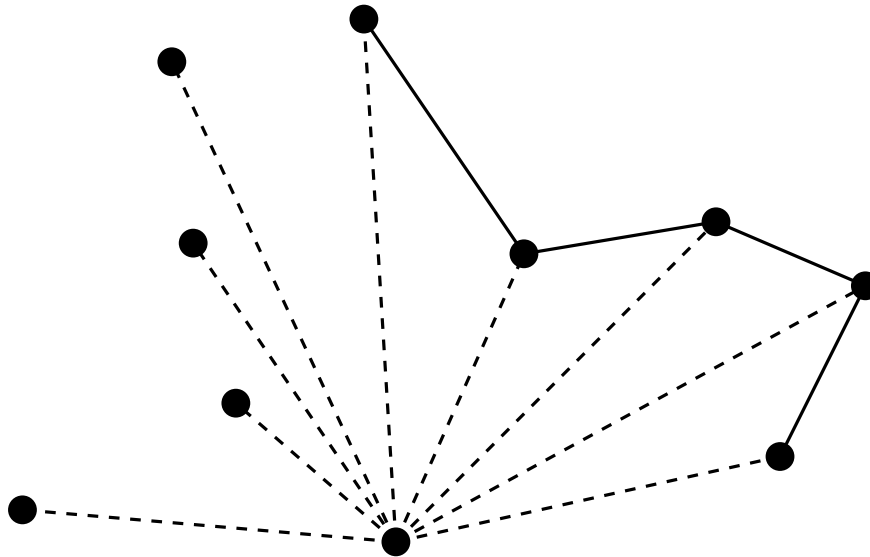
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

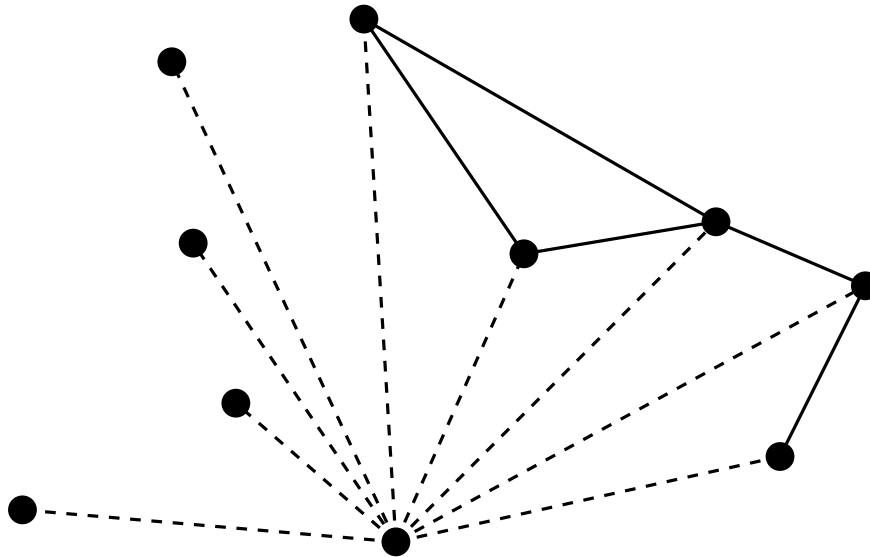
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

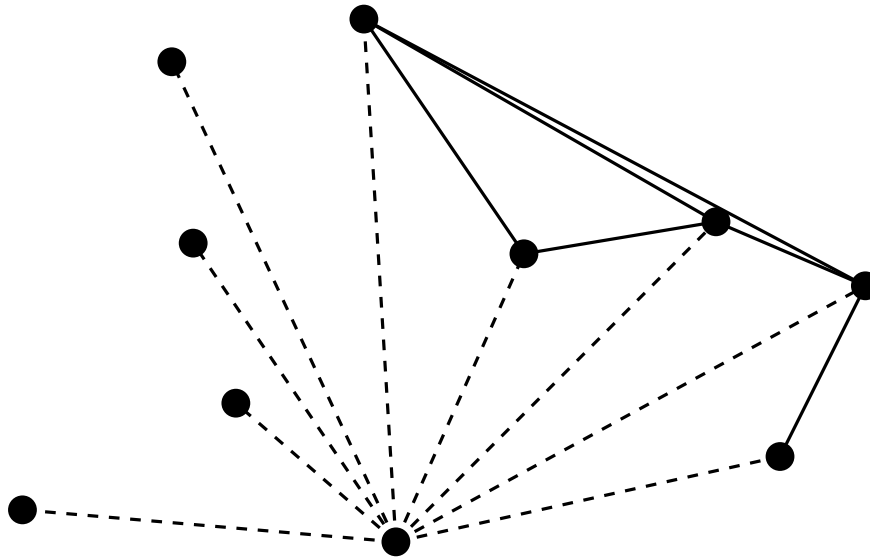
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

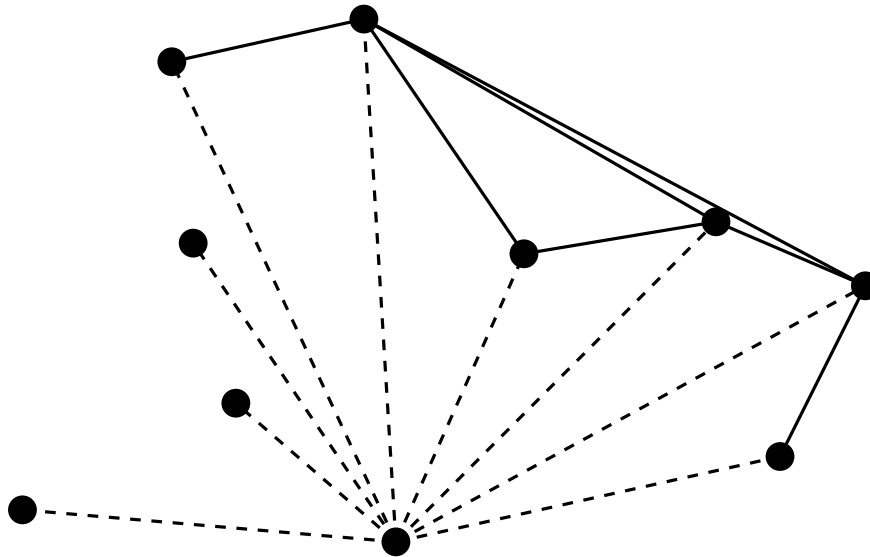
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

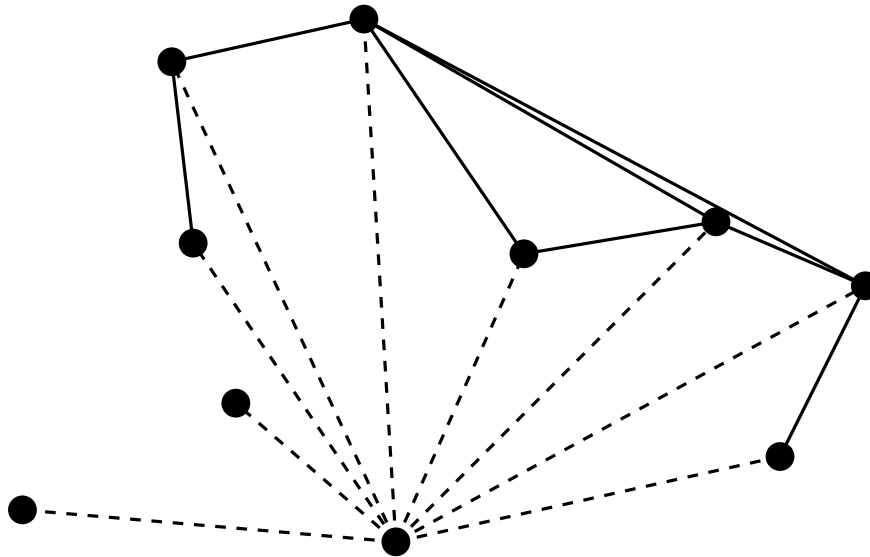
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

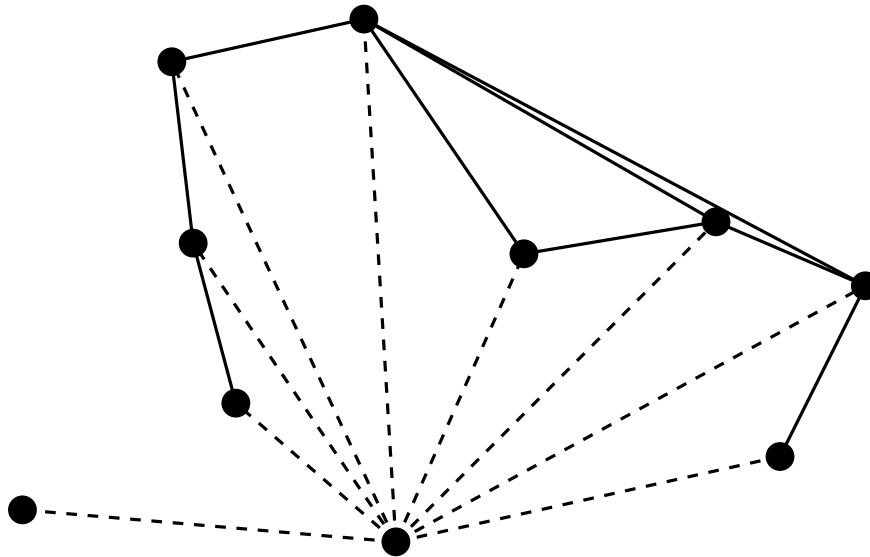
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

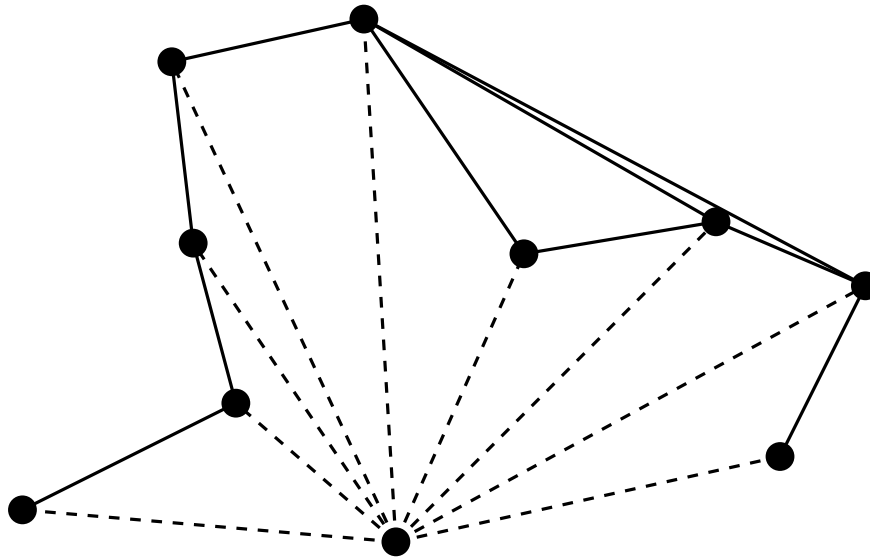
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

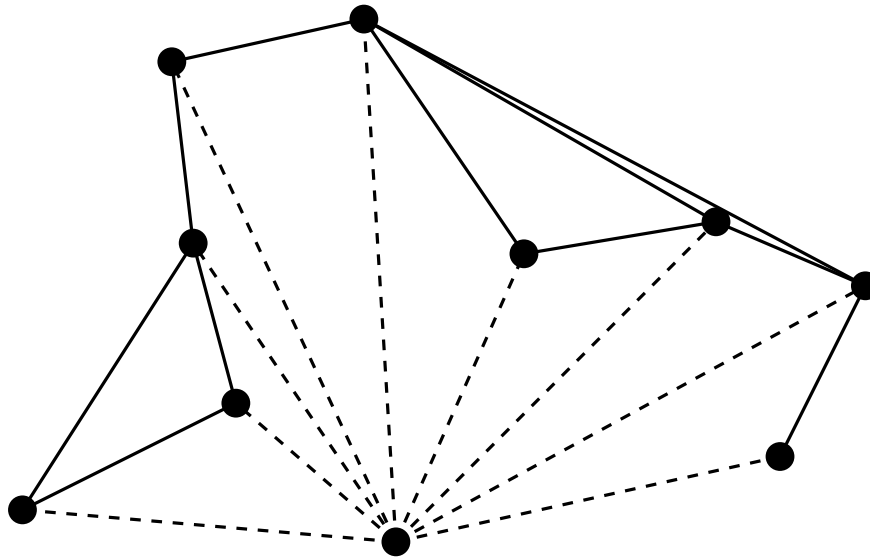
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

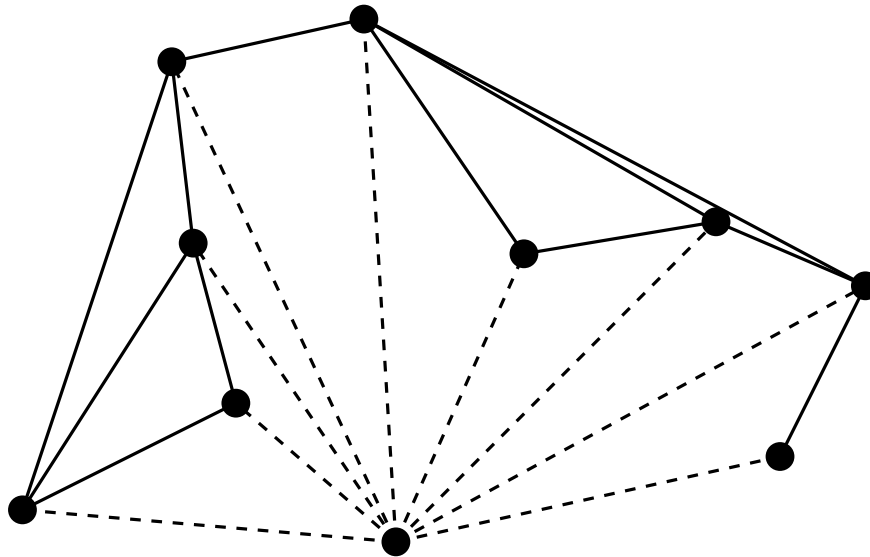
2. Graham's algorithm



TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

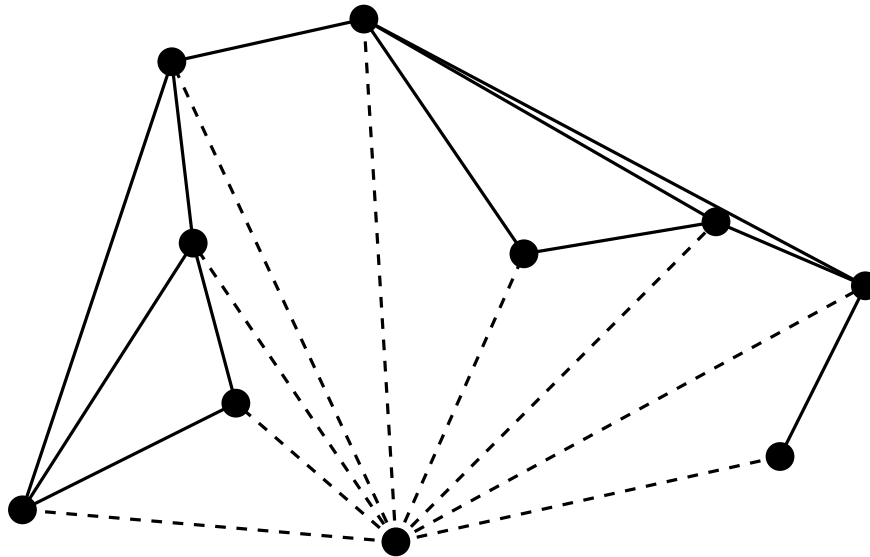


TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$



TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$

3. Divide and conquer

TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$

3. Divide and conquer

Initialization

Sort the points by abscissa

Advance

- Partition: divide the points into roughly two vertically separated halves
- Recursion: recursively triangulate each half
- Fusion: compute the external common tangents and triangulate the intermediate space

TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$

3. Divide and conquer

Running time: $O(n \log n)$

Initialization

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TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$

3. Divide and conquer

Running time: $O(n \log n)$

LOWER BOUND

TRIANGULATING POINT SETS

ALGORITHMS

2. Graham's algorithm

Running time: $O(n \log n)$

3. Divide and conquer

Running time: $O(n \log n)$

LOWER BOUND

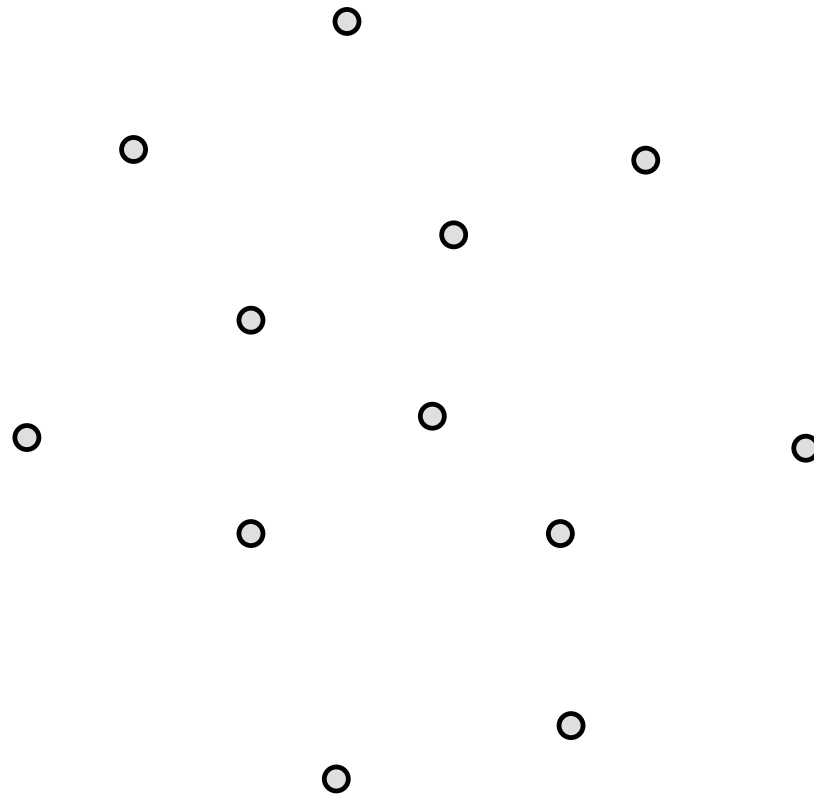
This problem has an $\Omega(n \log n)$ lower bound, since the convex hull of the set of points can be trivially obtained in $O(n)$ time from the triangulation.

TRIANGULATING POINT SETS

Quality of a triangulation

TRIANGULATING POINT SETS

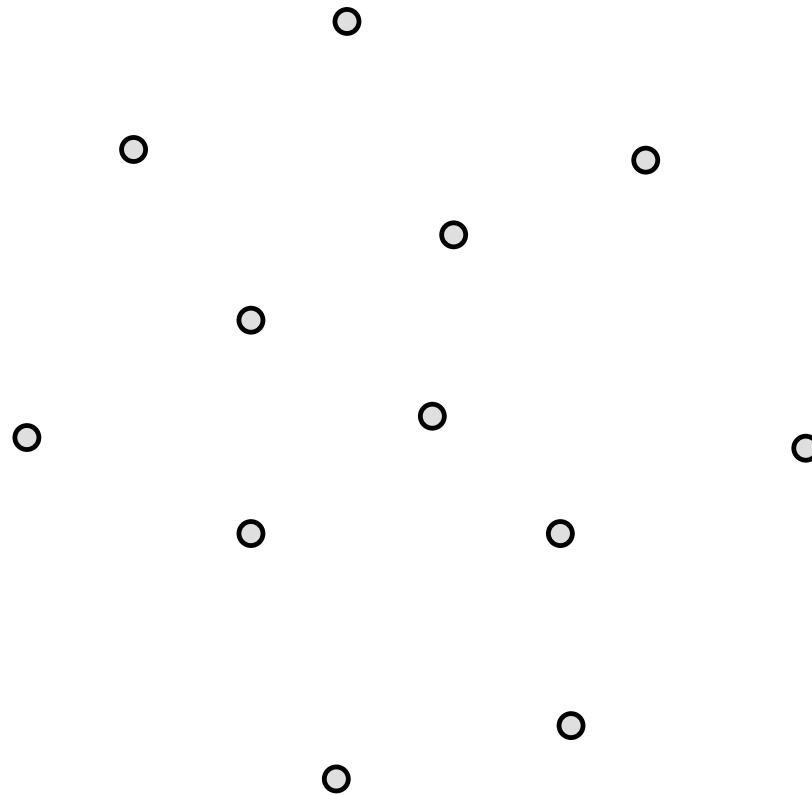
Quality of a triangulation



TRIANGULATING POINT SETS

Quality of a triangulation

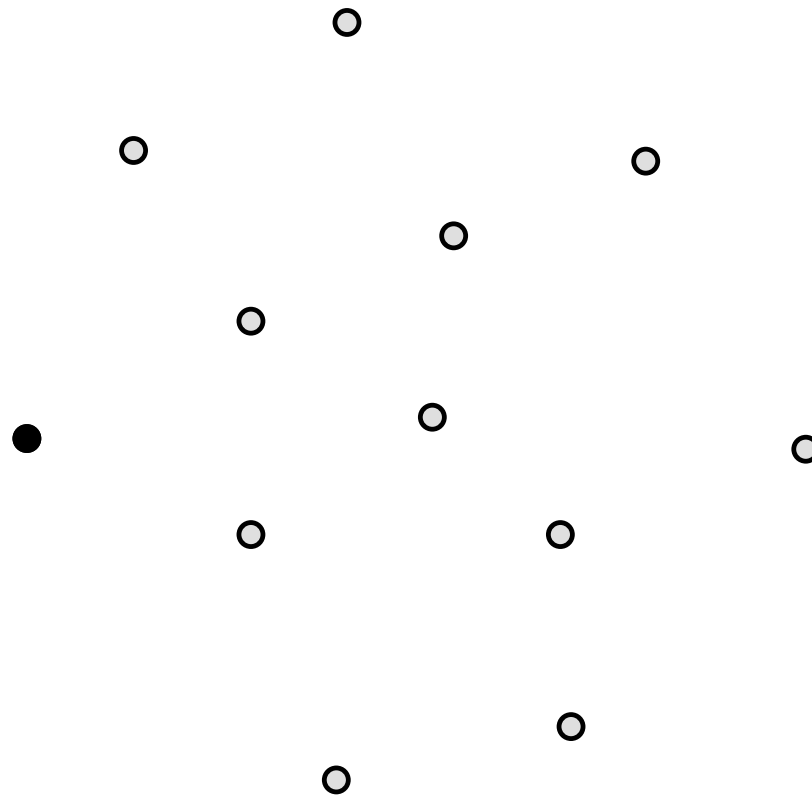
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

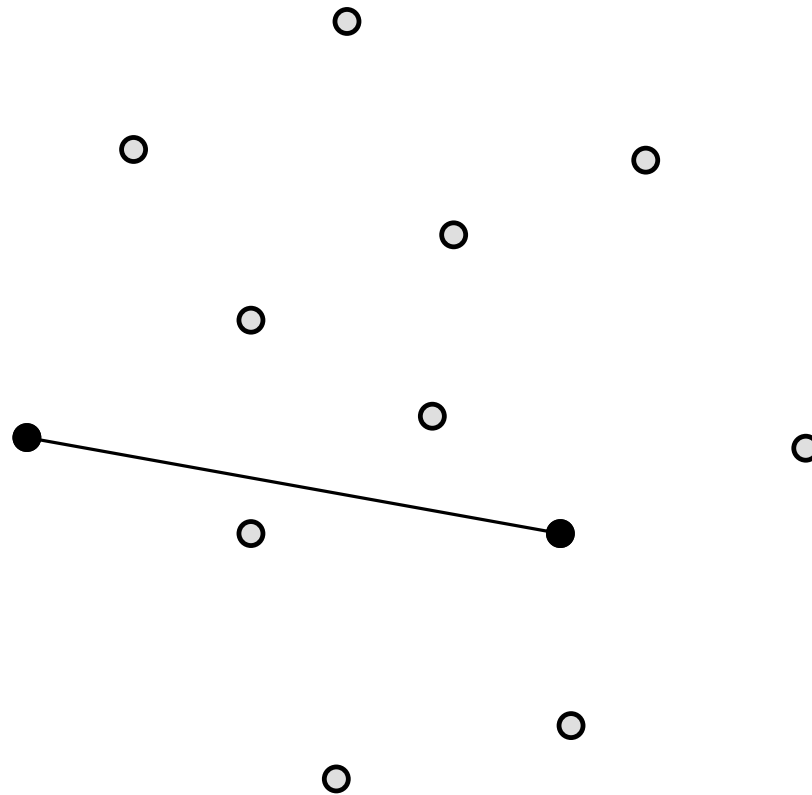
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

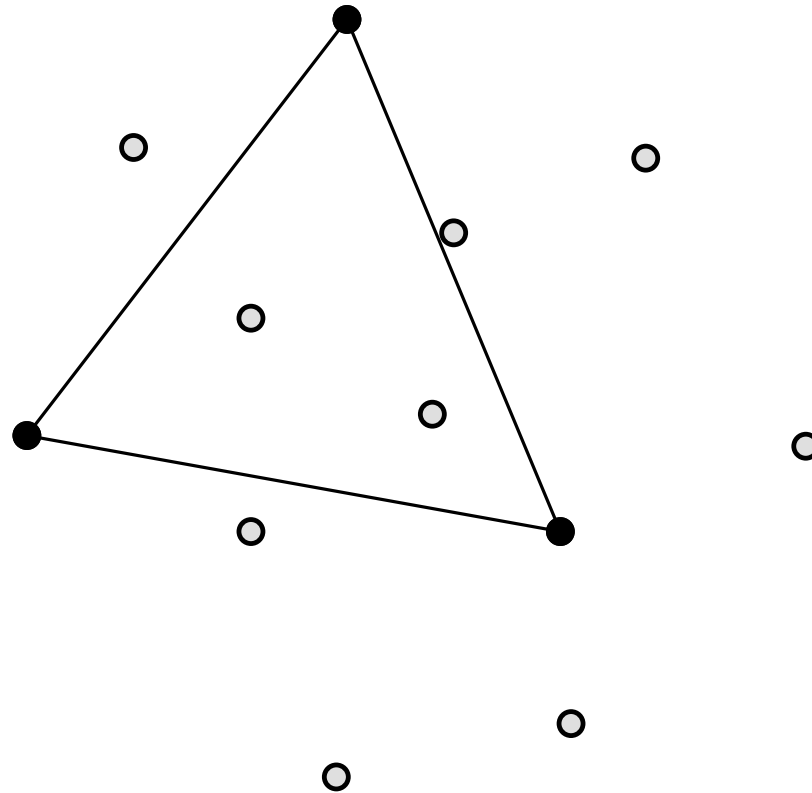
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

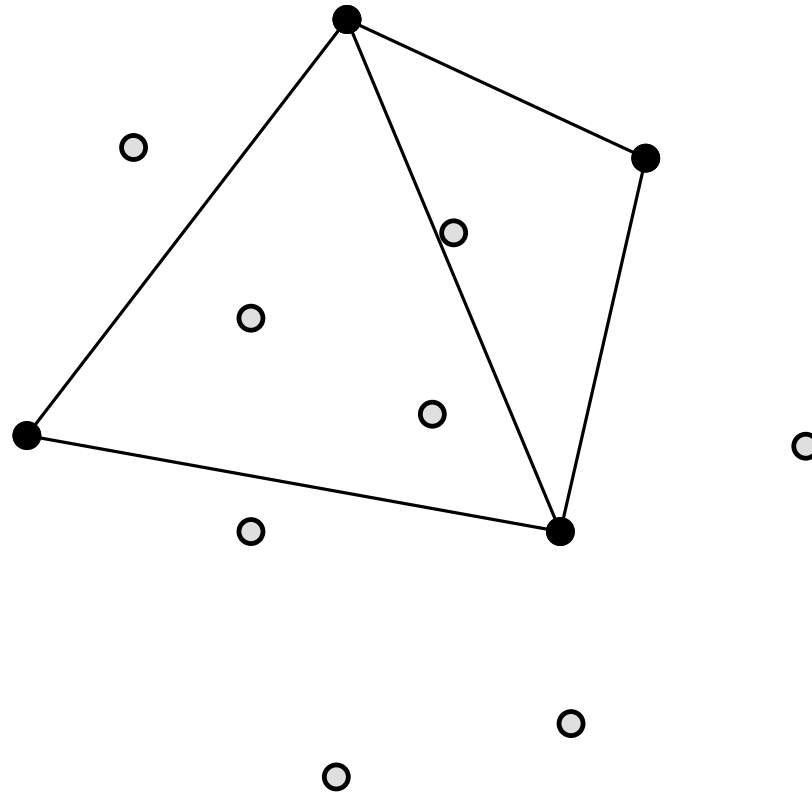
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

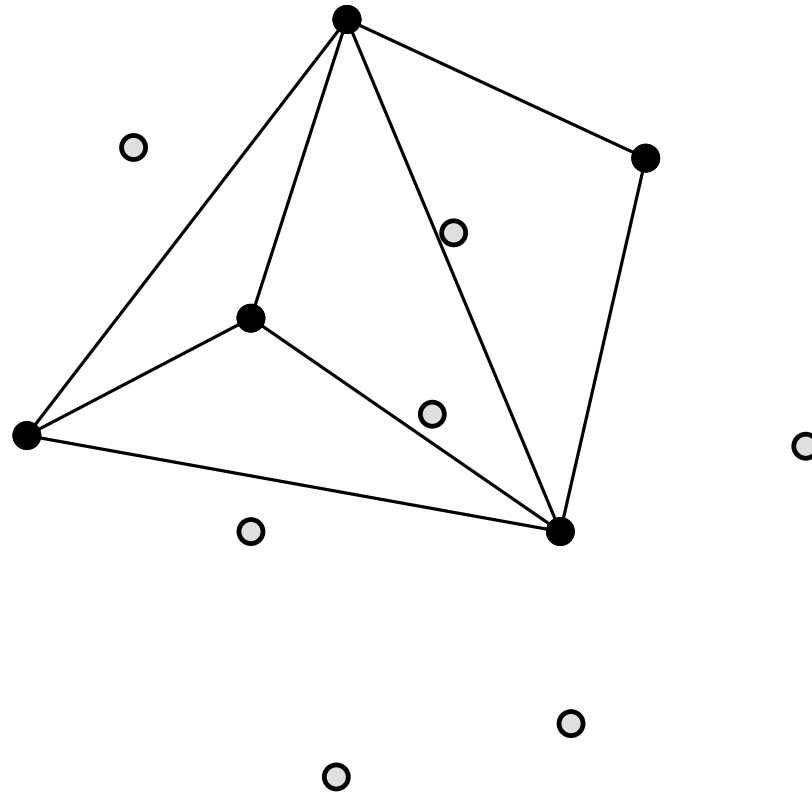
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

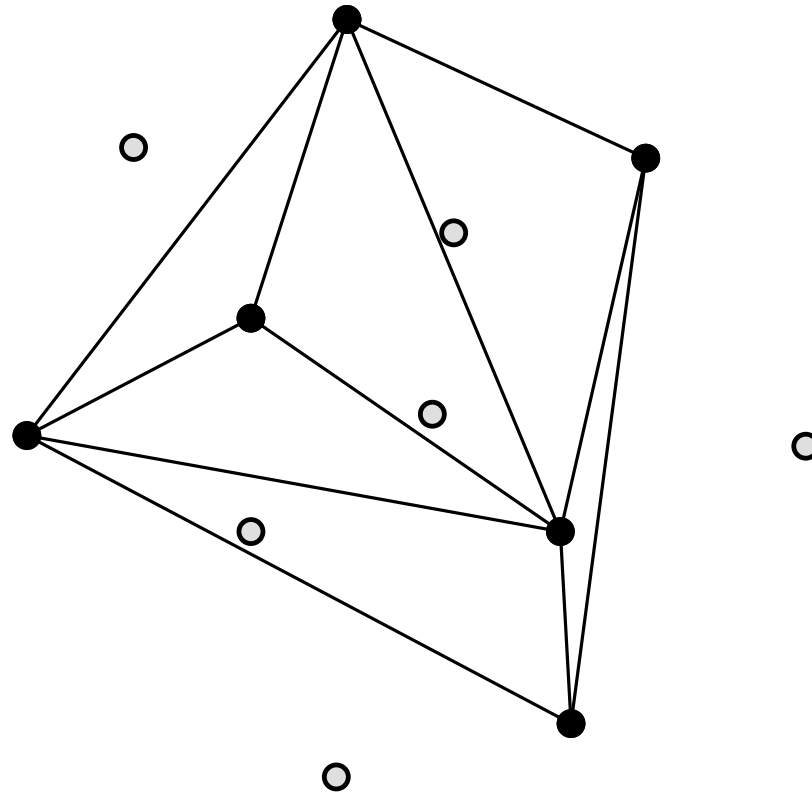
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

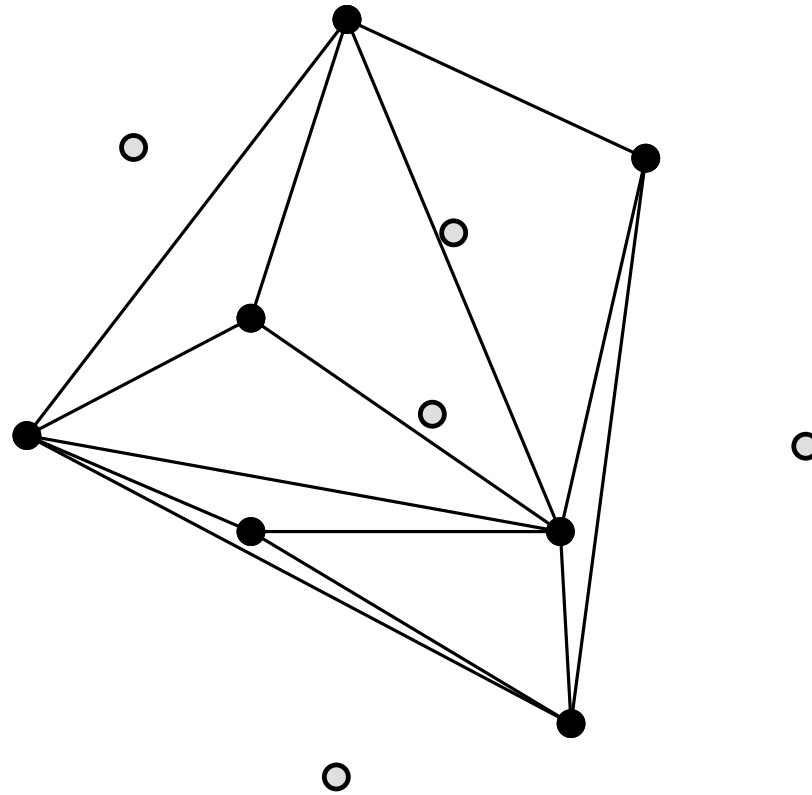
Incremental, without sorting



TRIANGULATING POINT SETS

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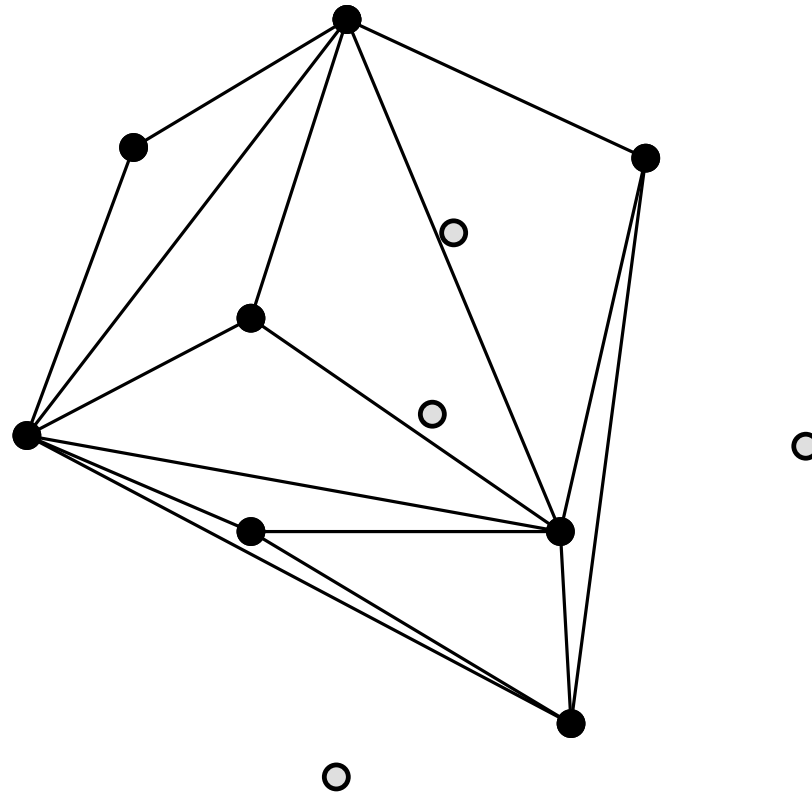
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

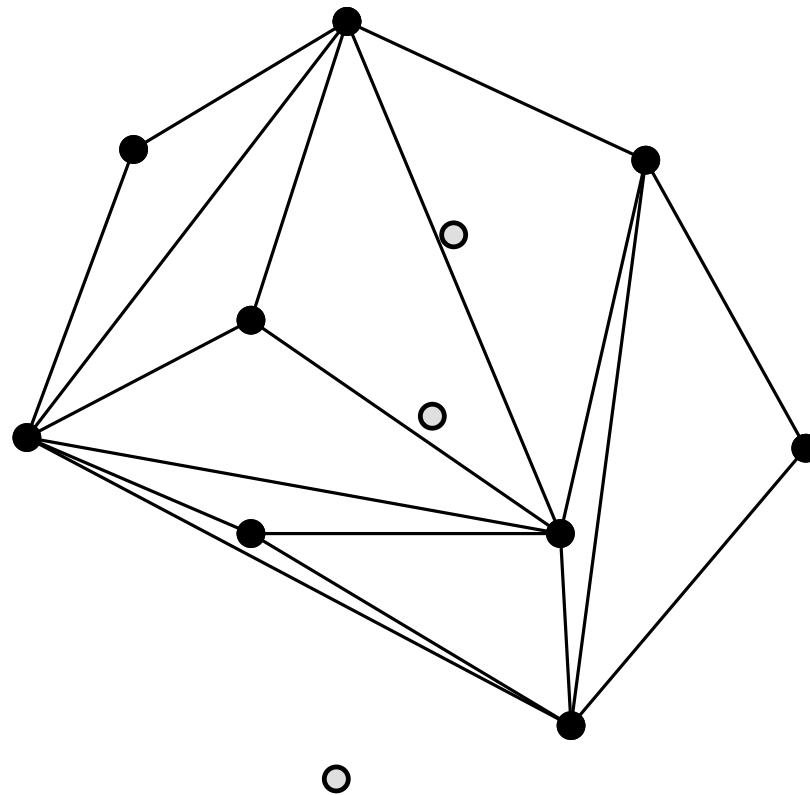
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

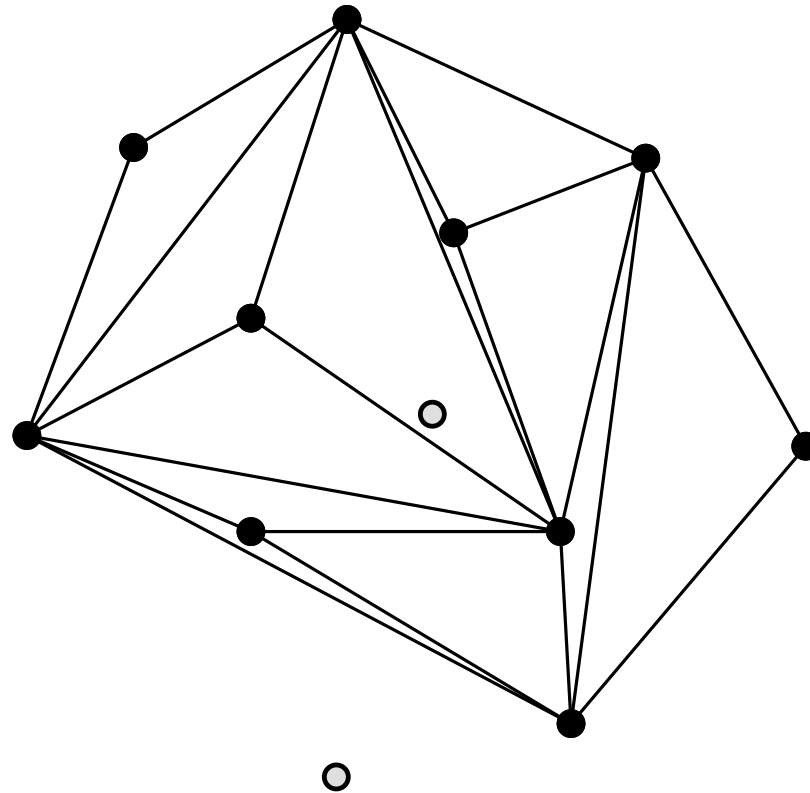
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

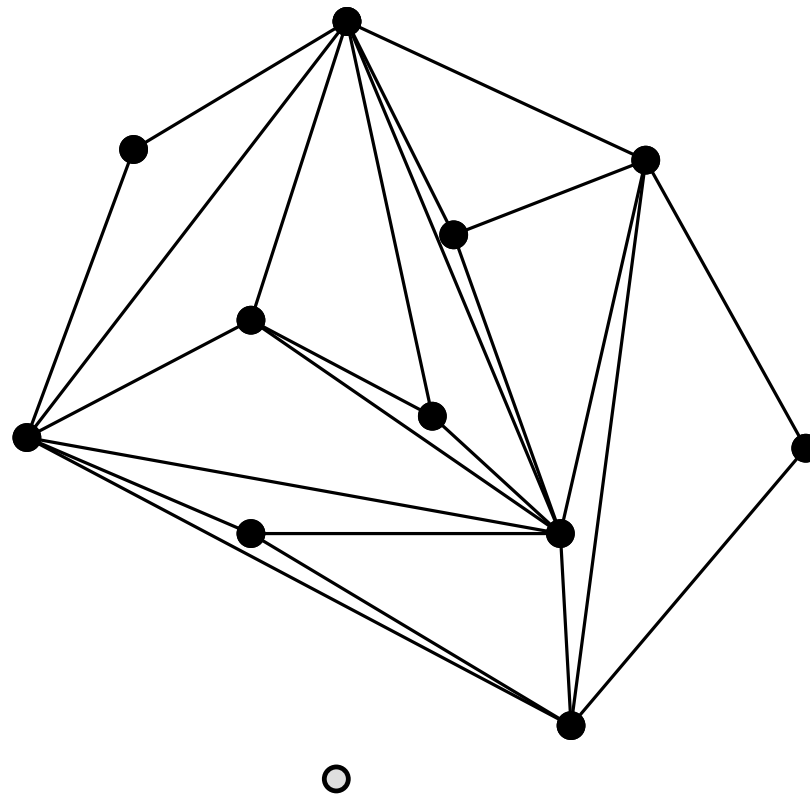
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

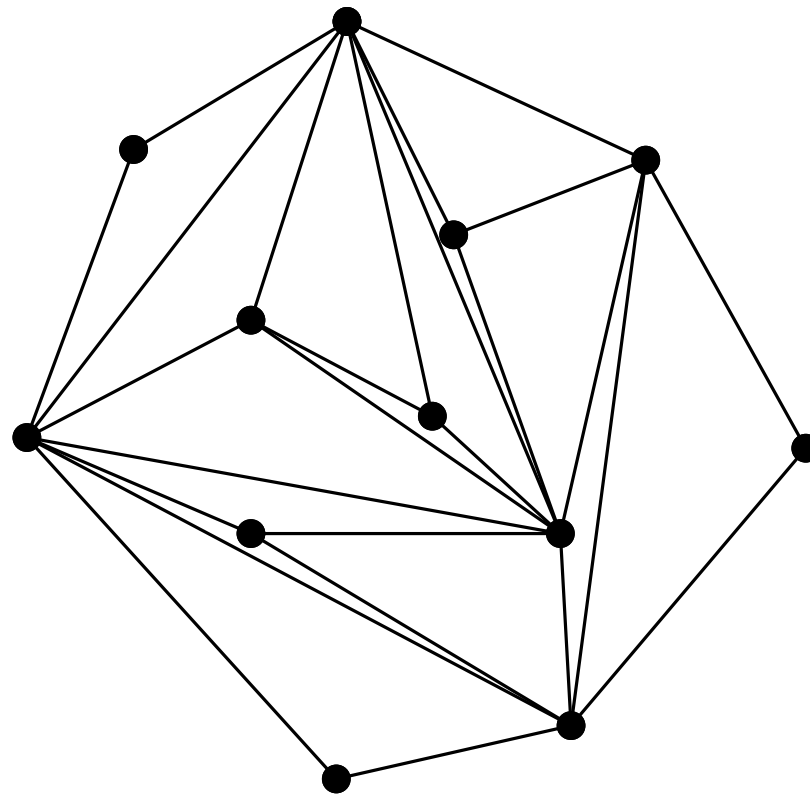
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

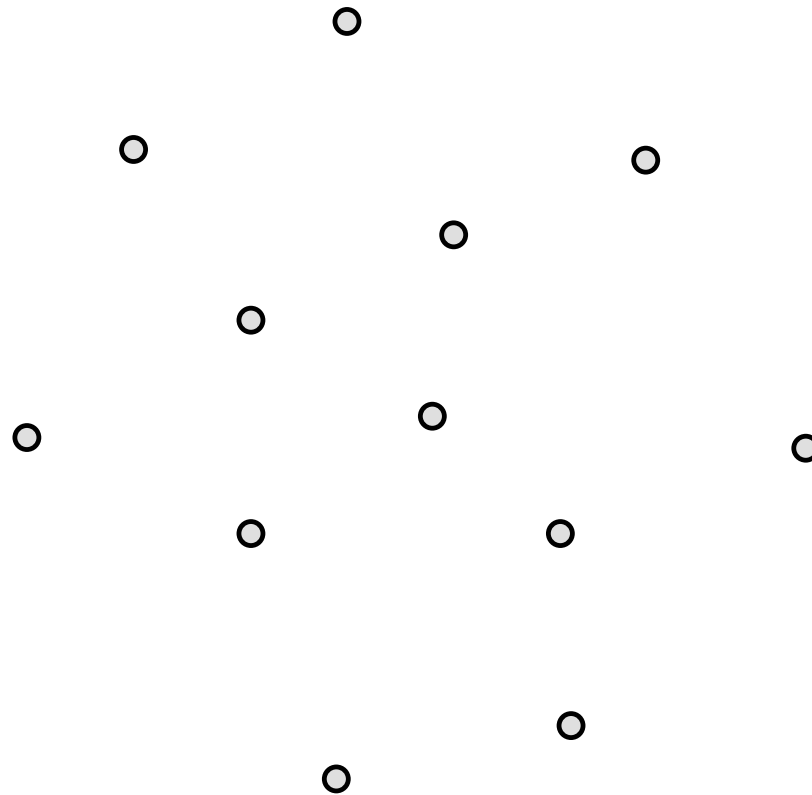
Incremental, without sorting



TRIANGULATING POINT SETS

Quality of a triangulation

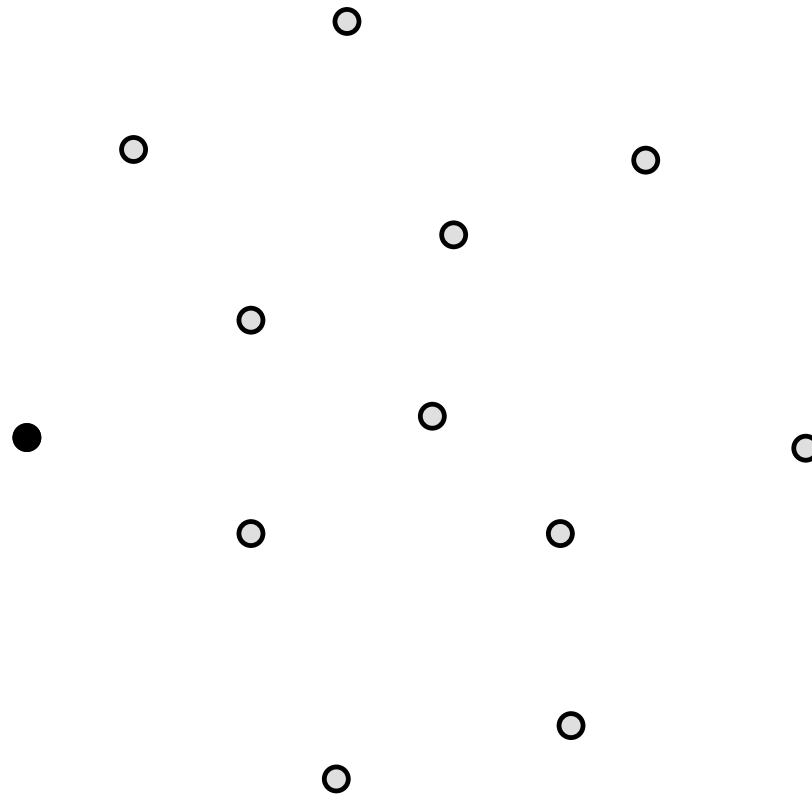
Incremental, with sorting



TRIANGULATING POINT SETS

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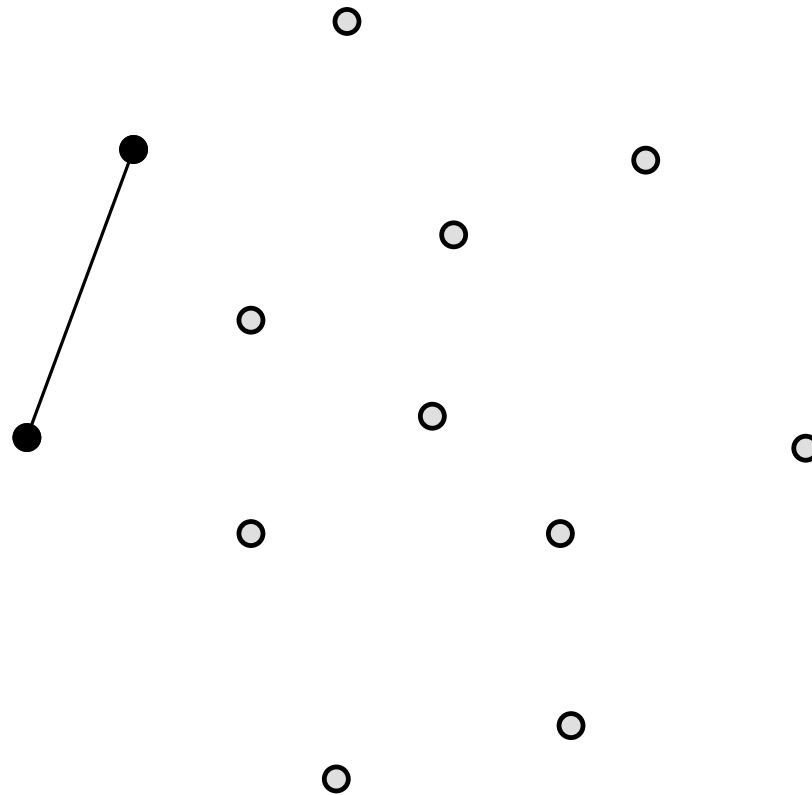
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

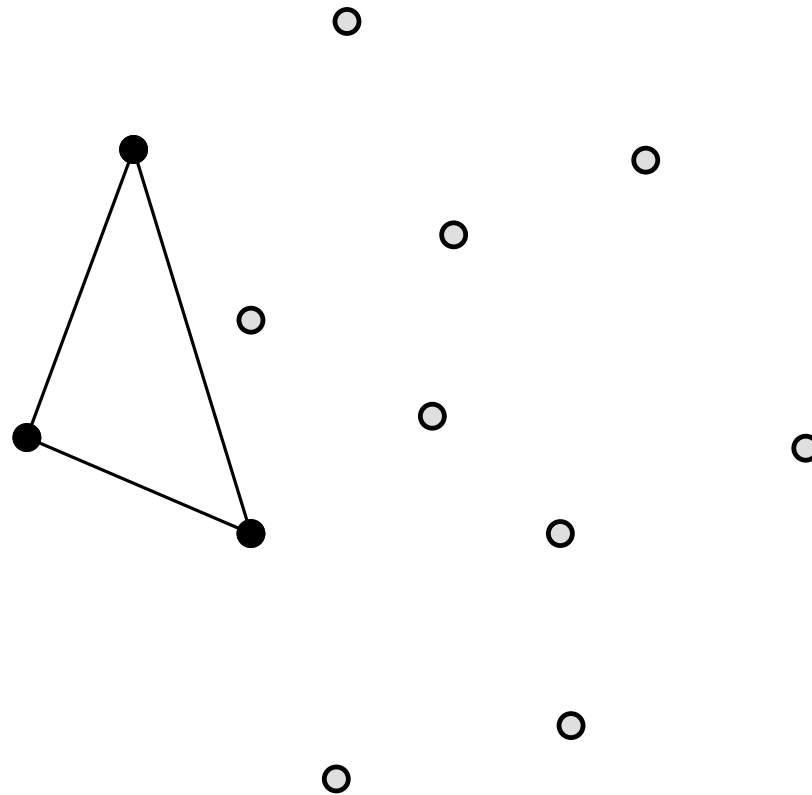
Incremental, with sorting



TRIANGULATING POINT SETS

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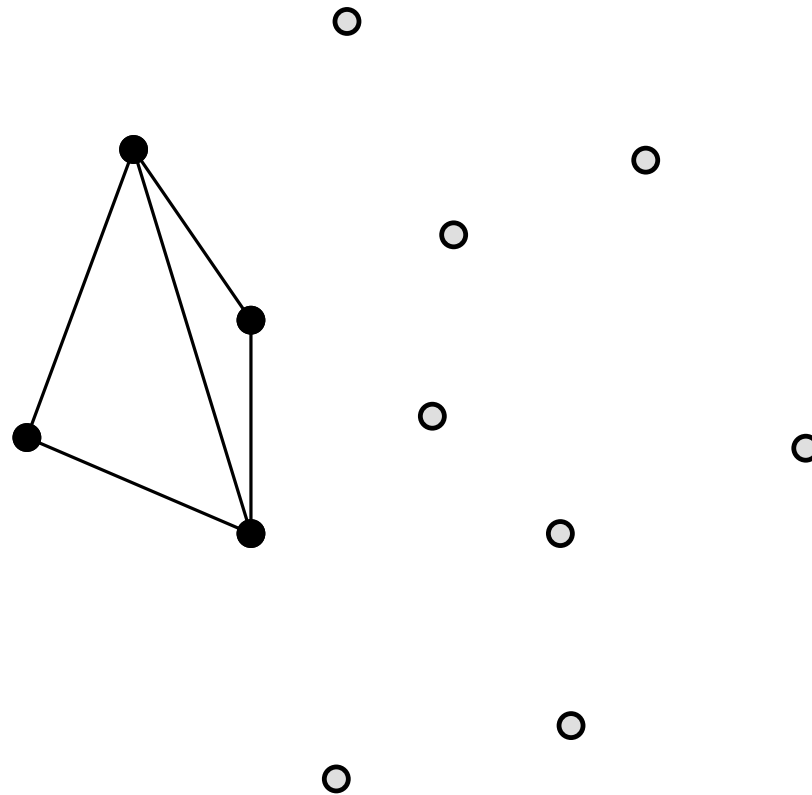
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

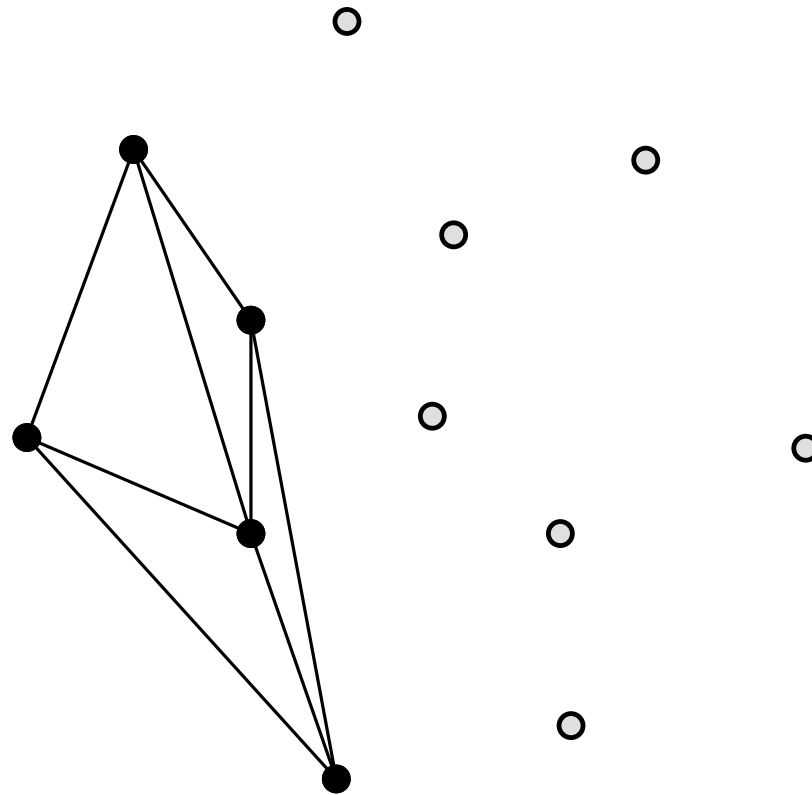
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

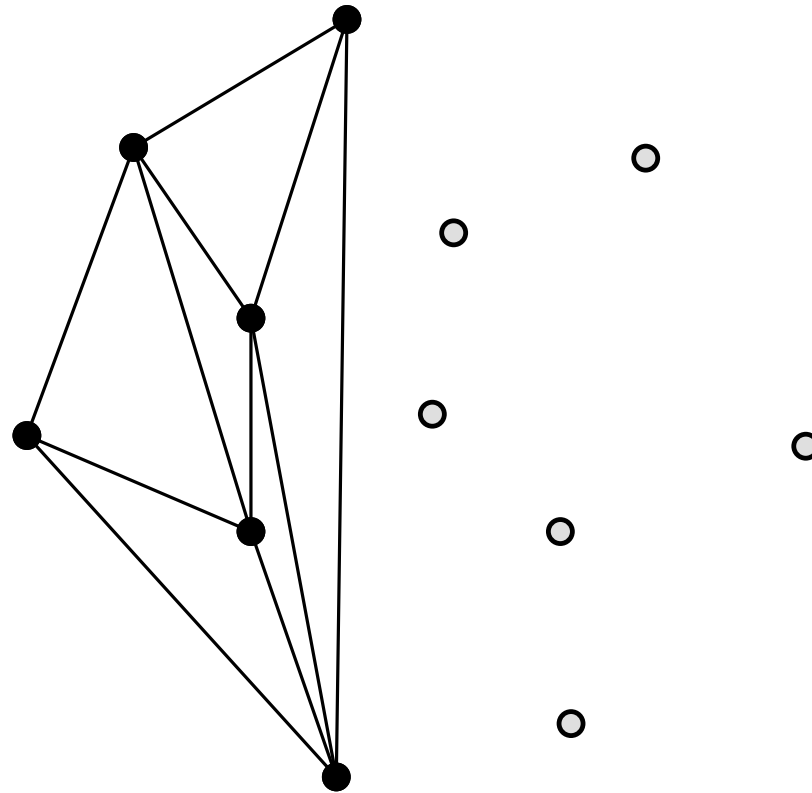
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

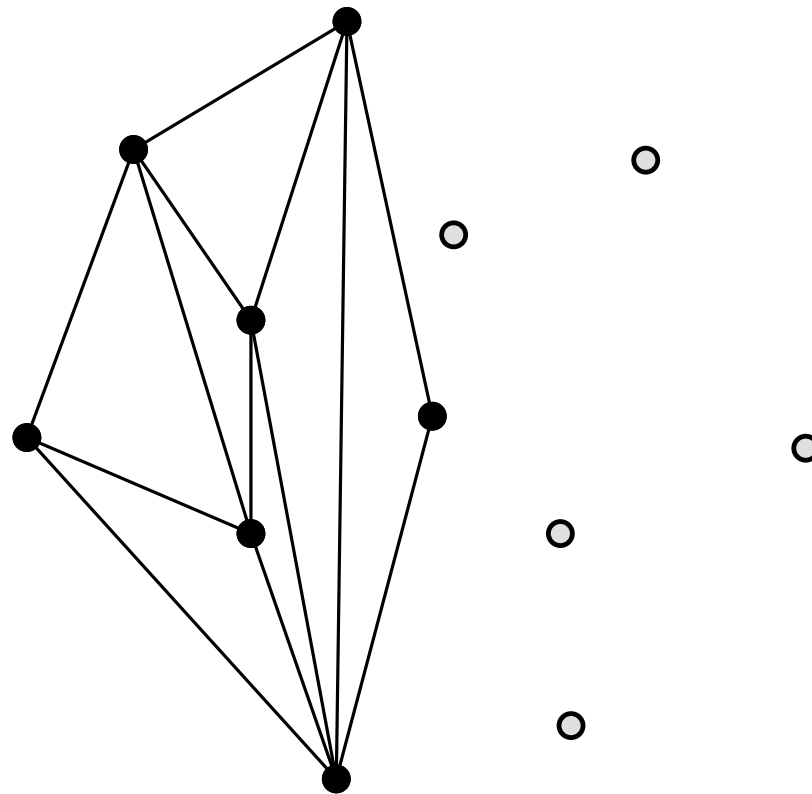
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

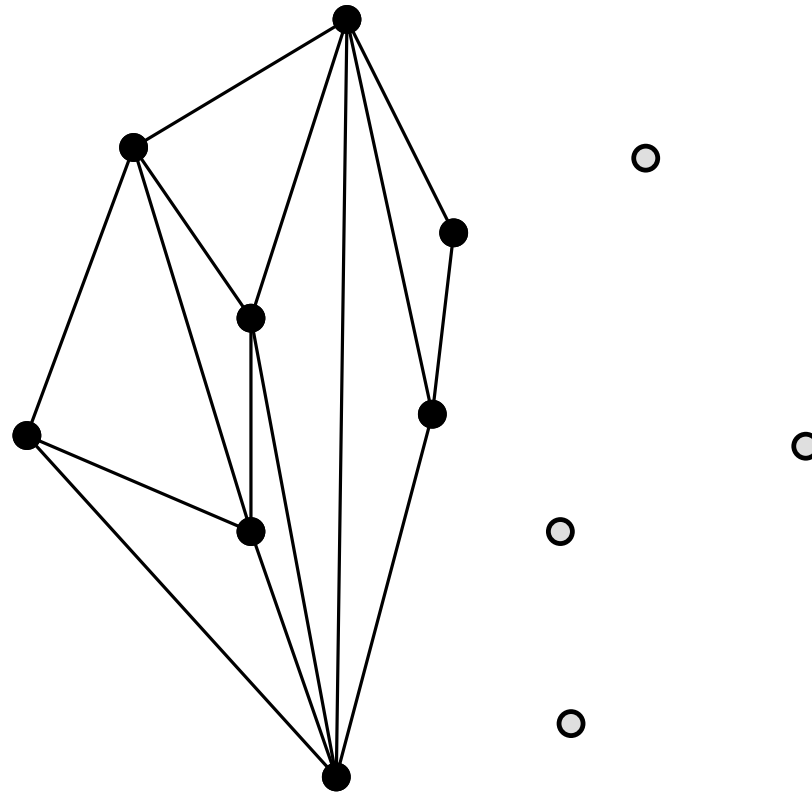
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

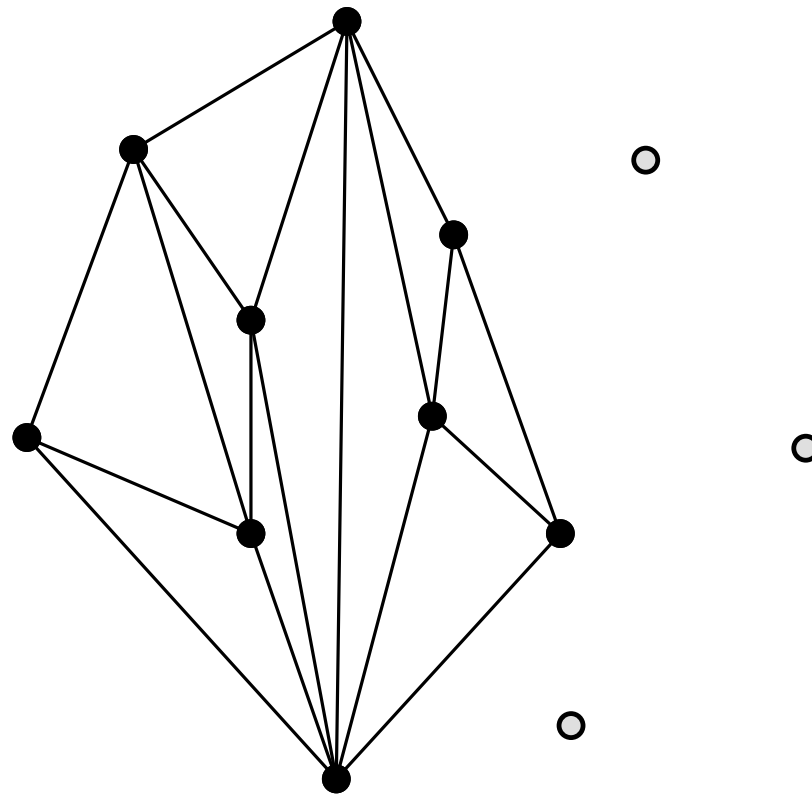
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

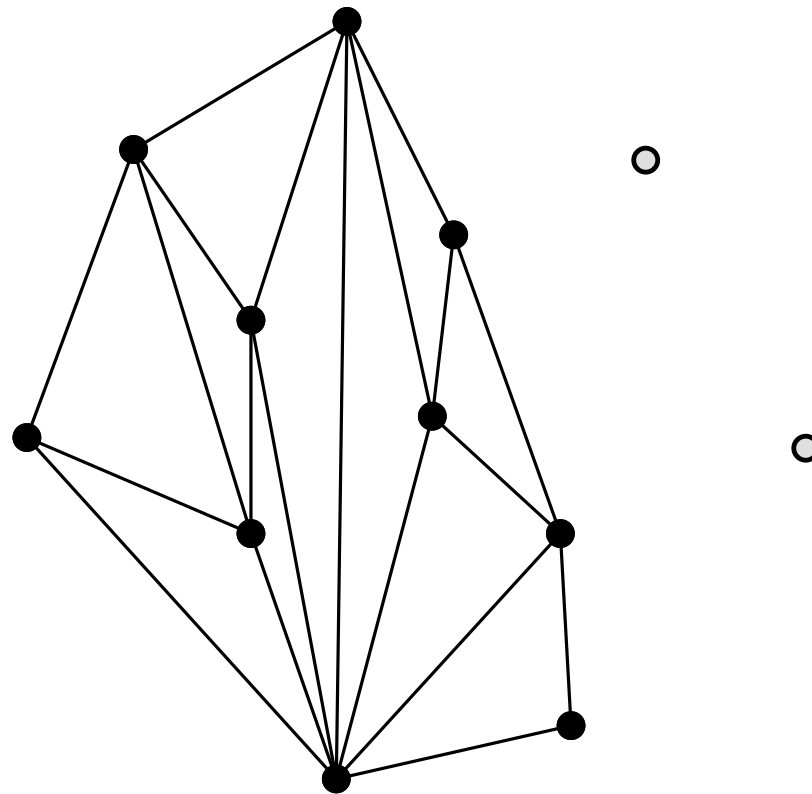
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

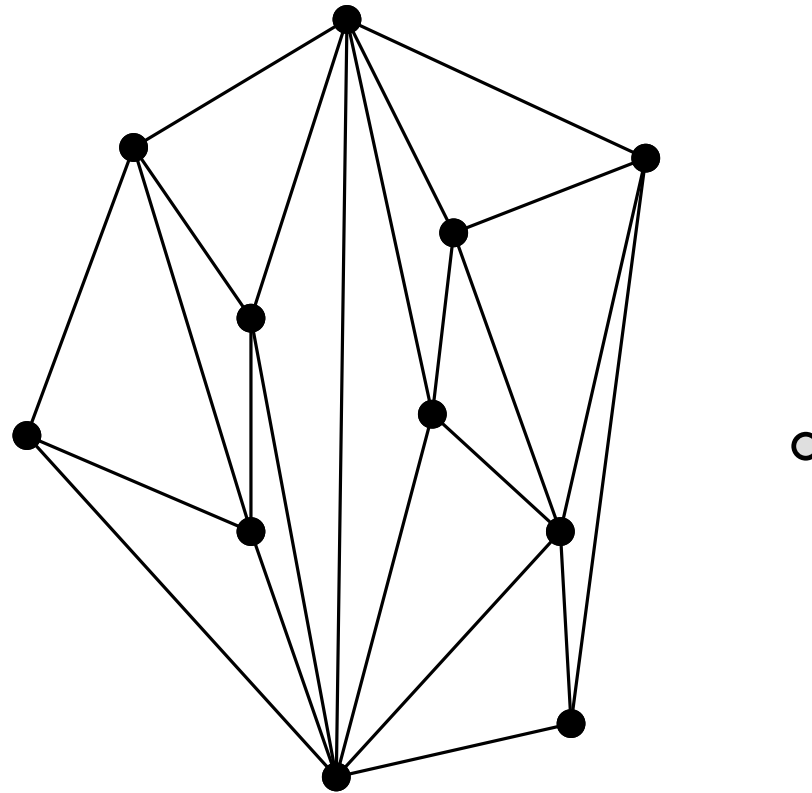
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

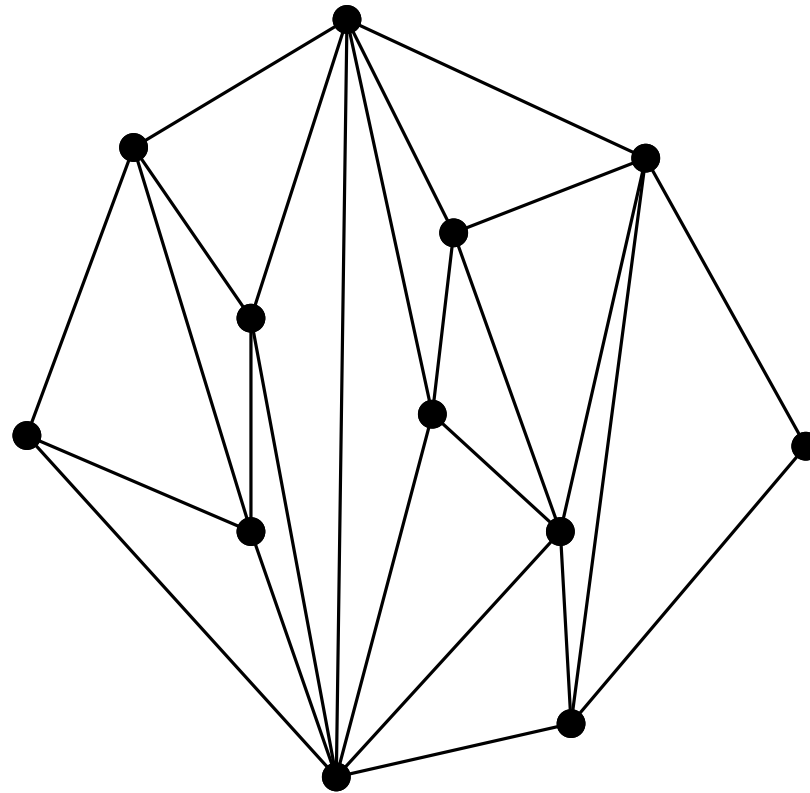
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

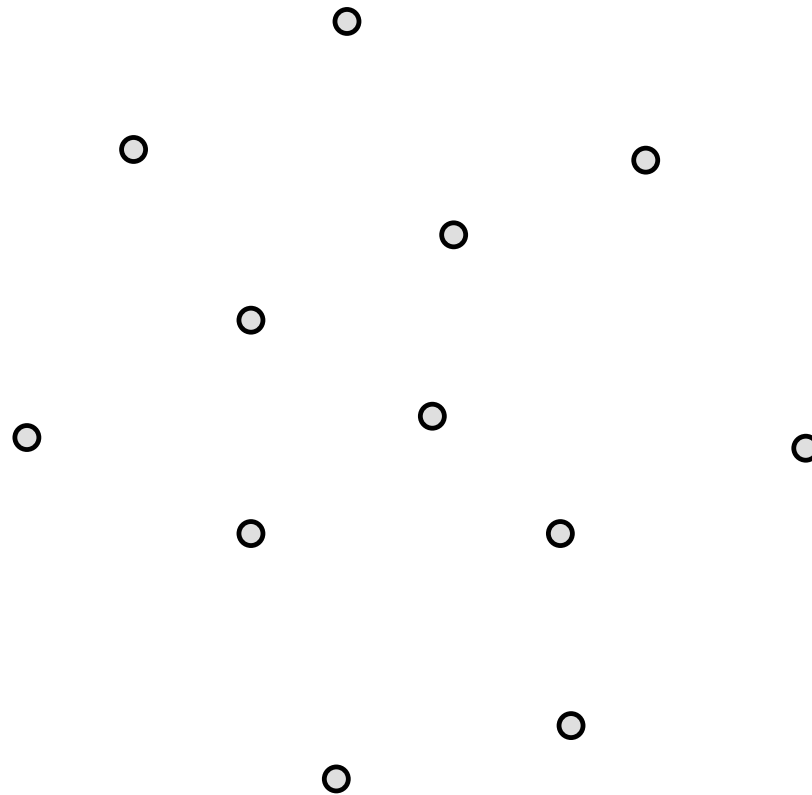
Incremental, with sorting



TRIANGULATING POINT SETS

Quality of a triangulation

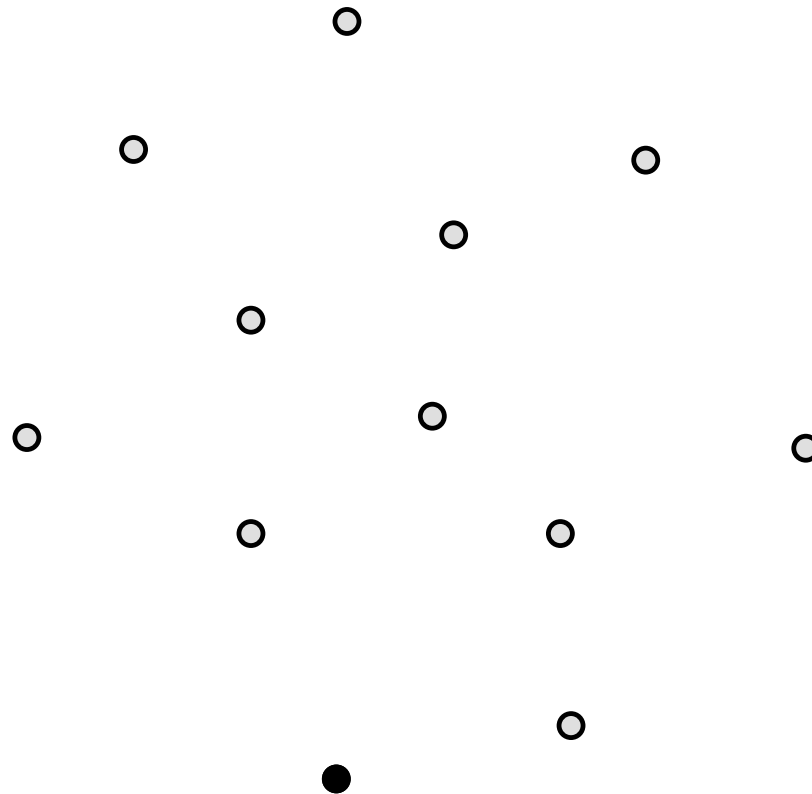
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

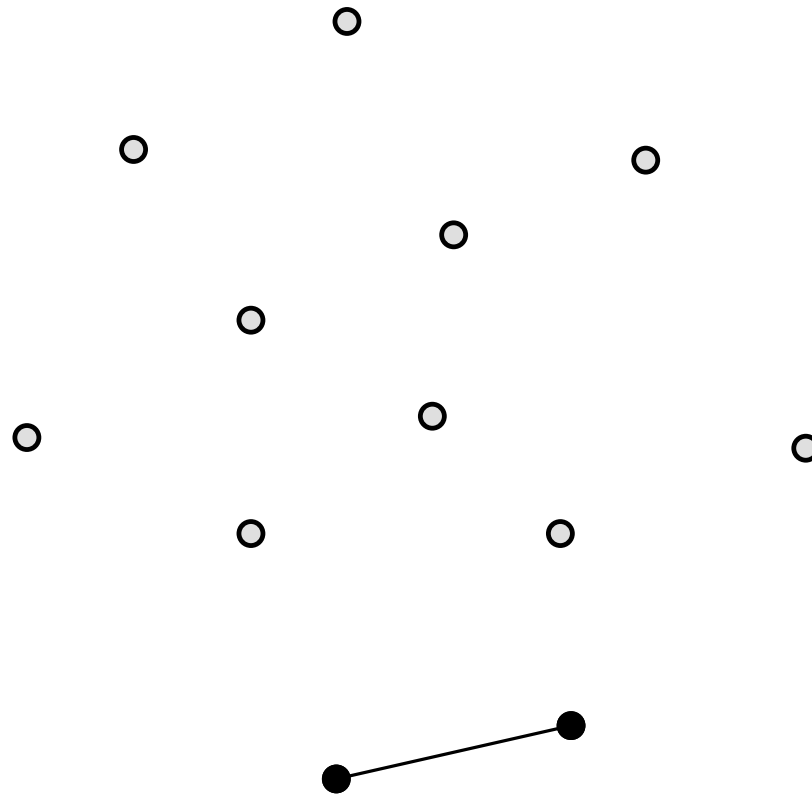
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

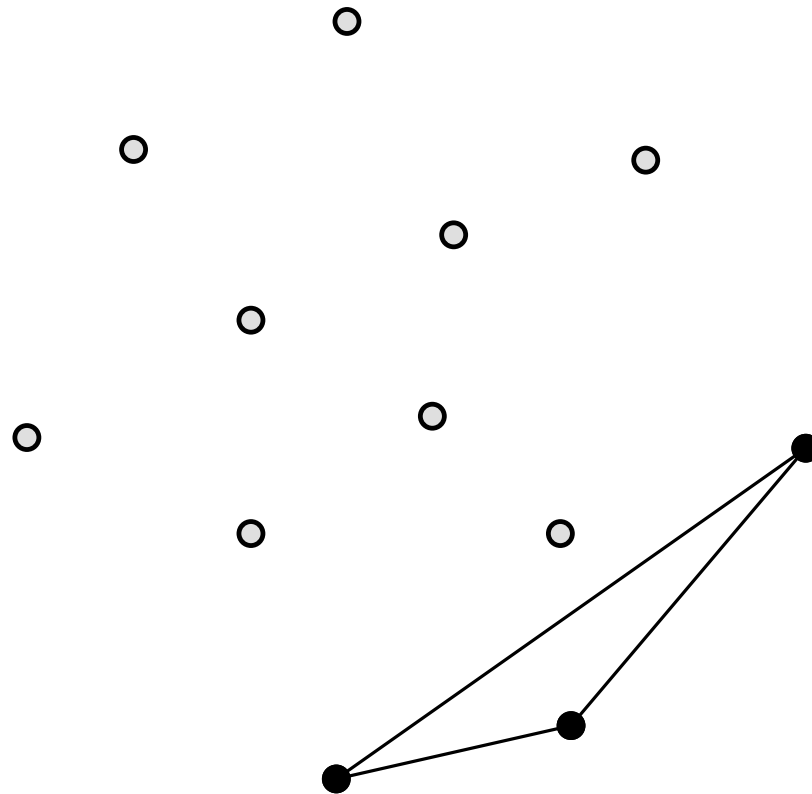
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

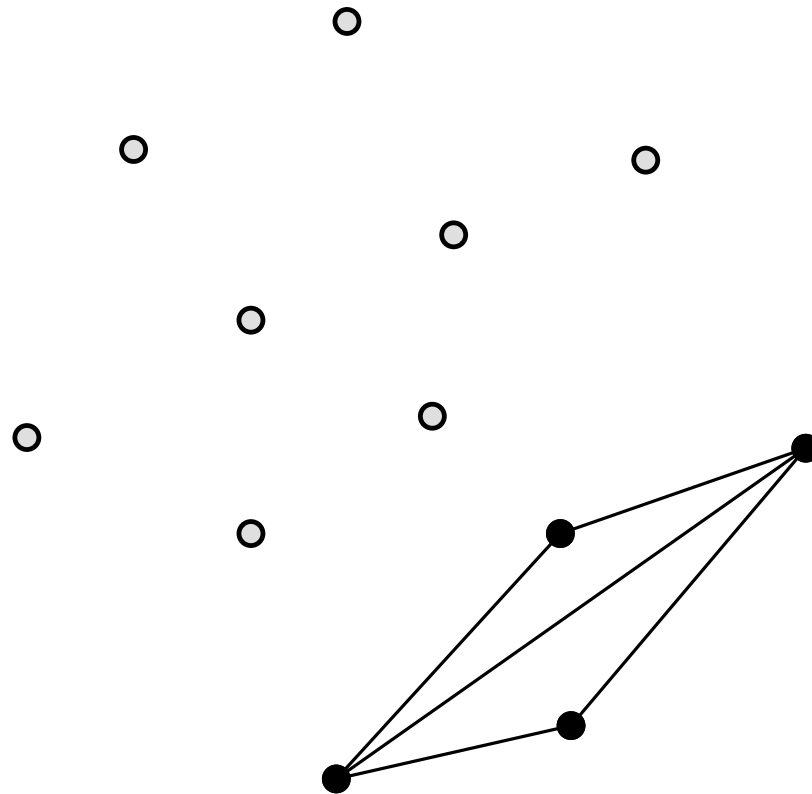
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

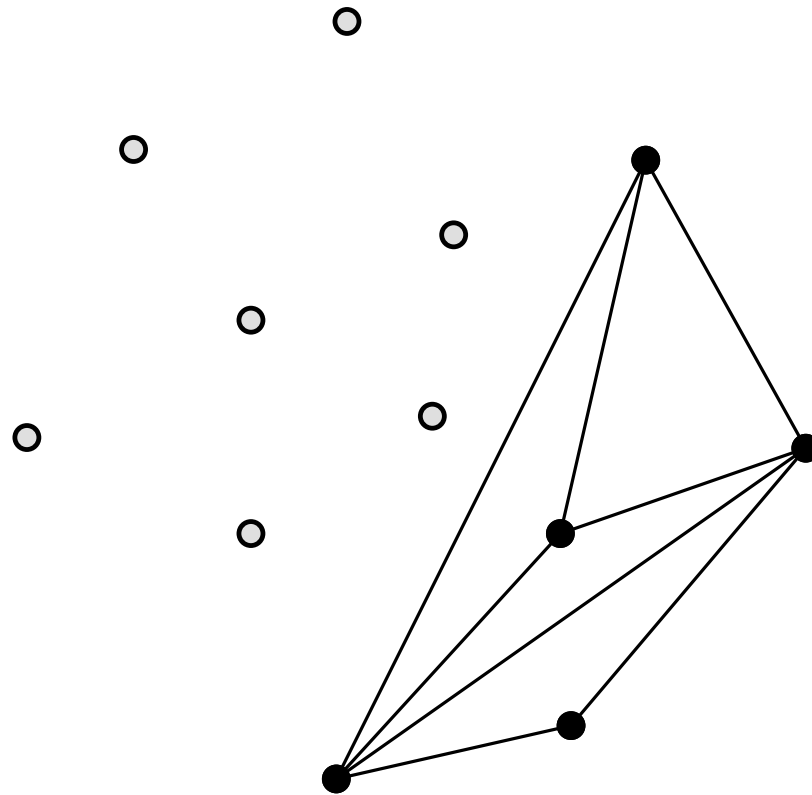
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

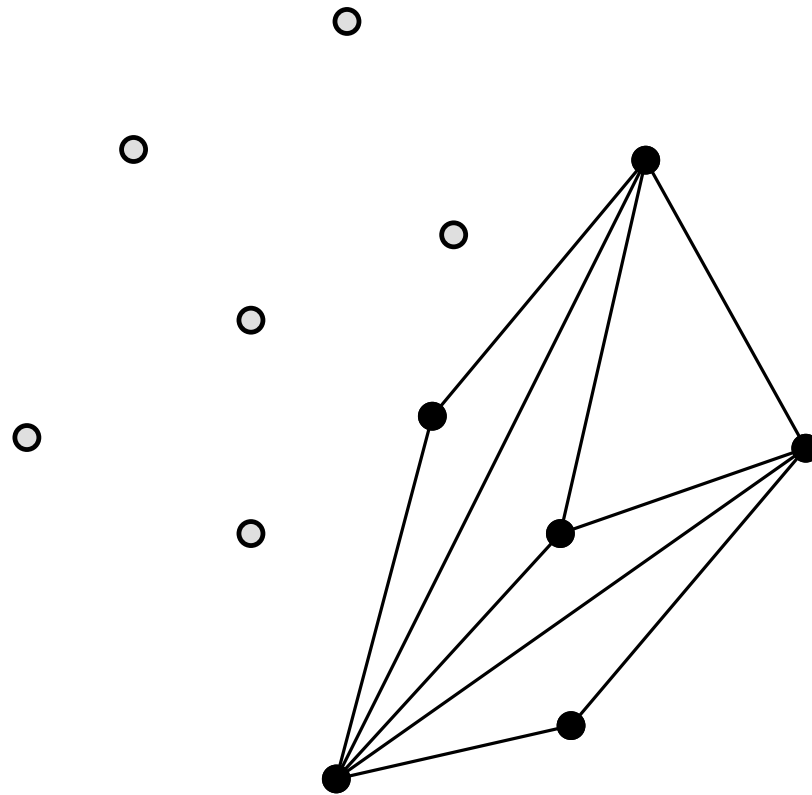
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

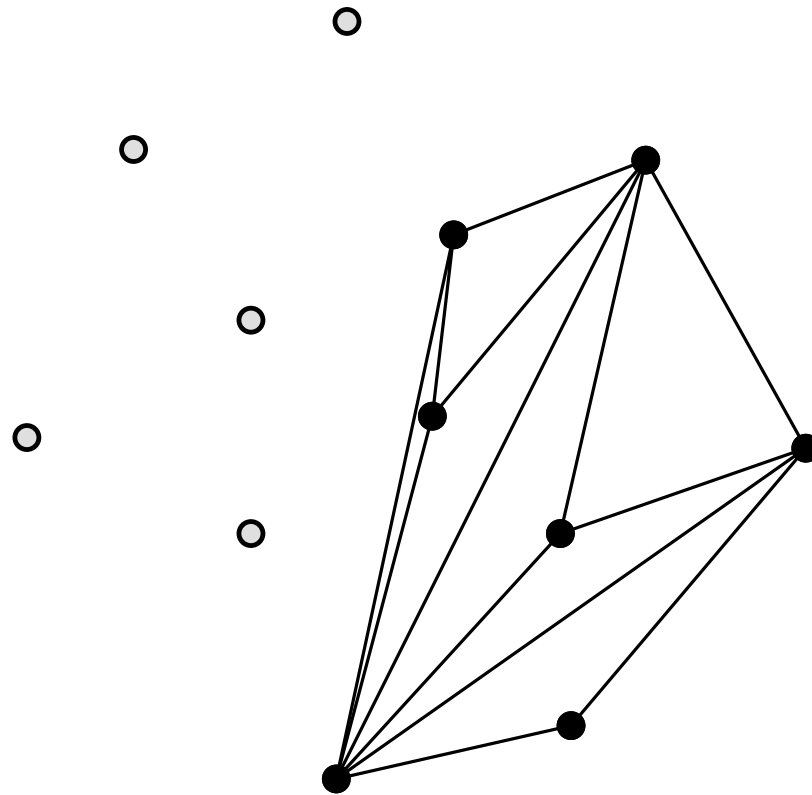
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

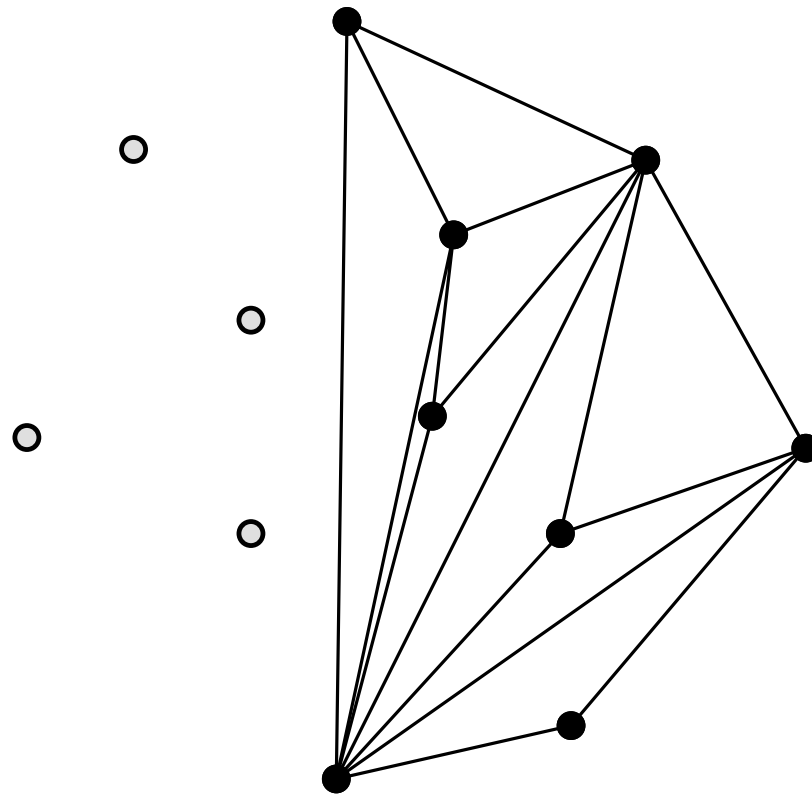
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

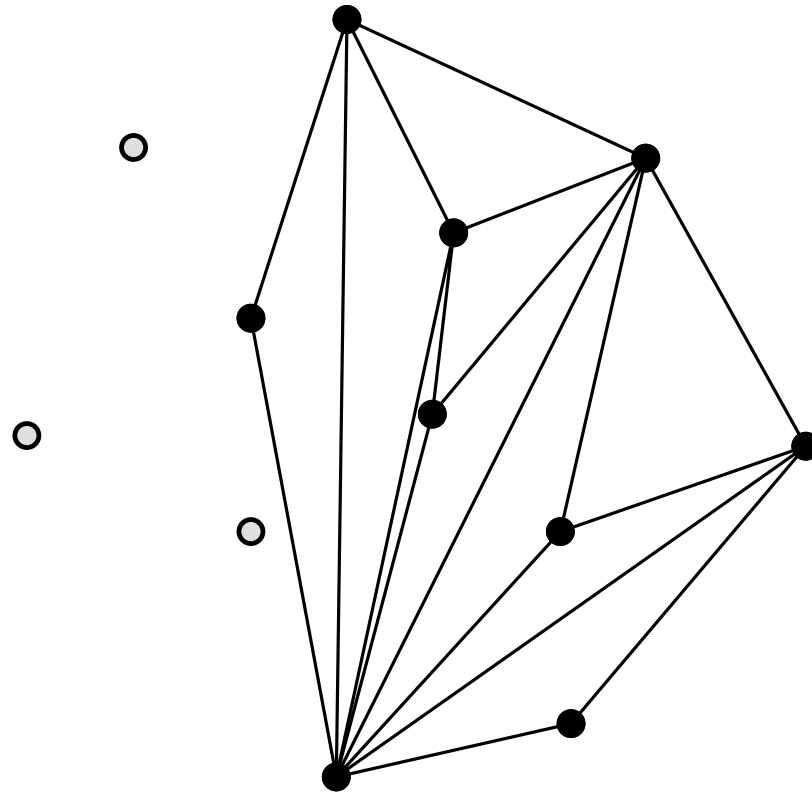
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

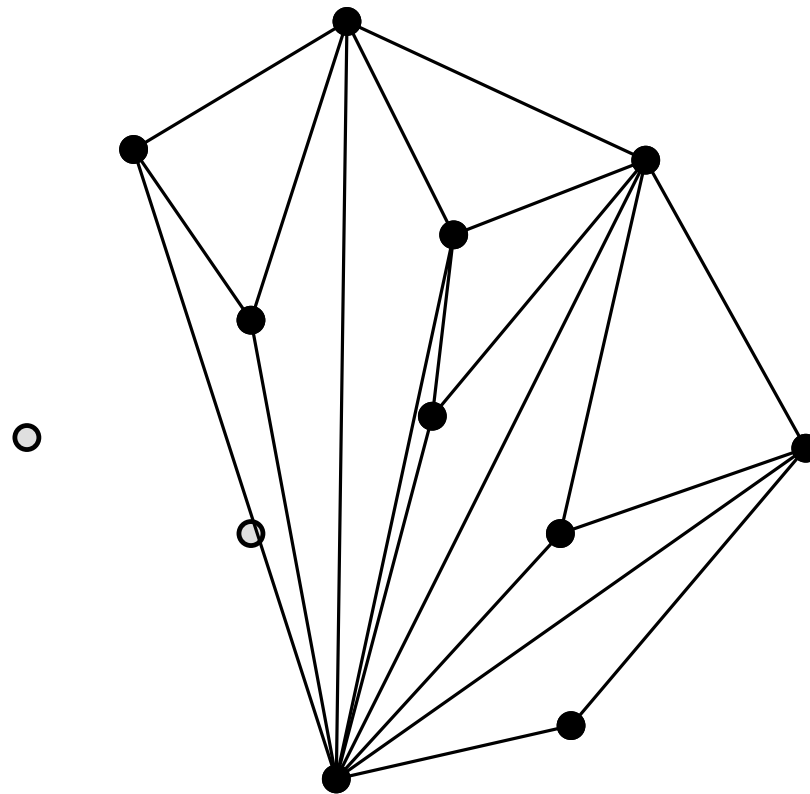
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

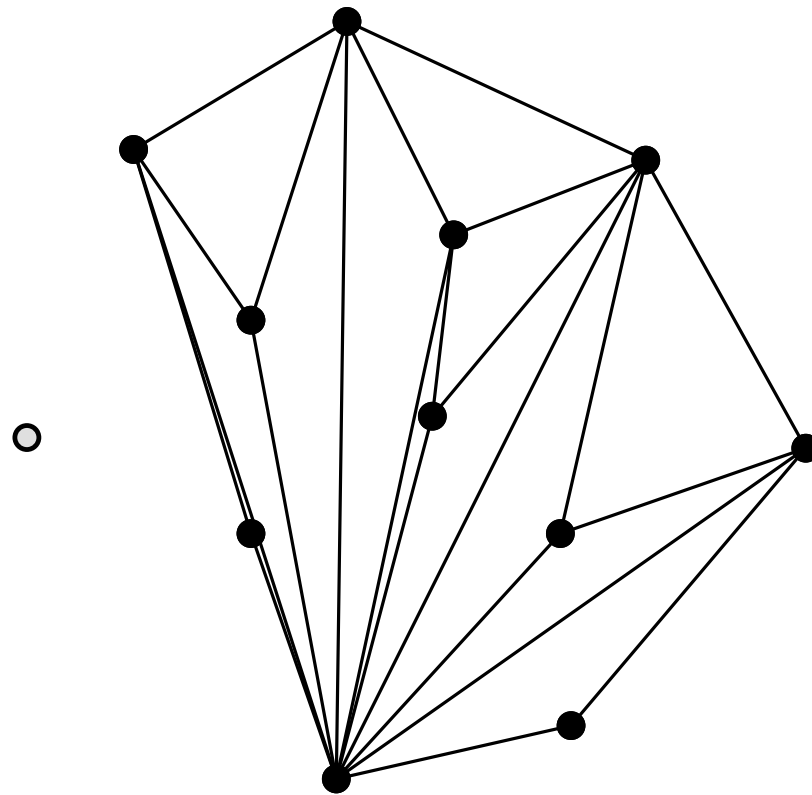
Graham's



TRIANGULATING POINT SETS

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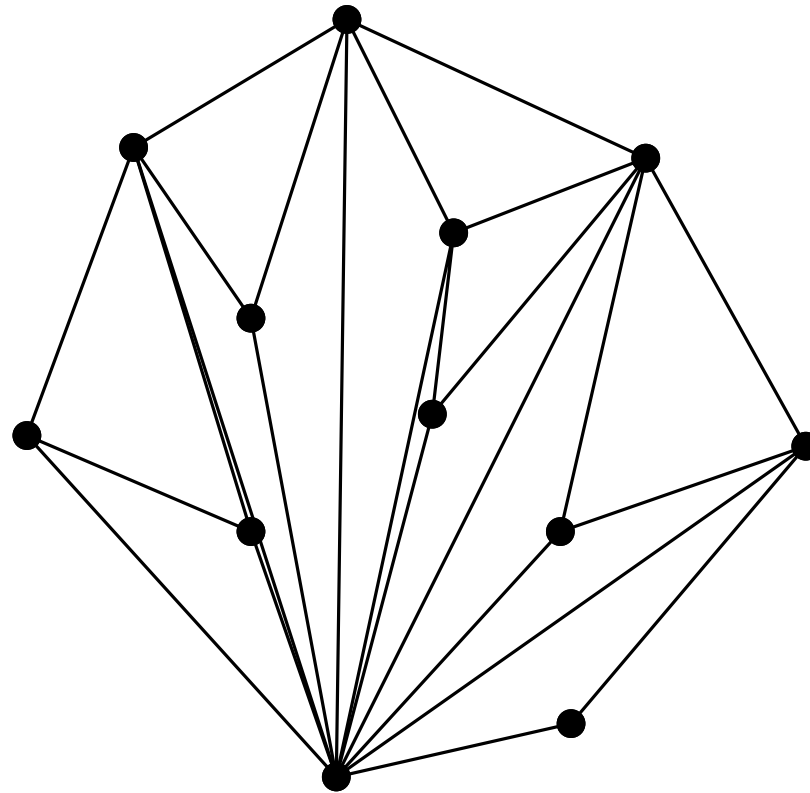
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

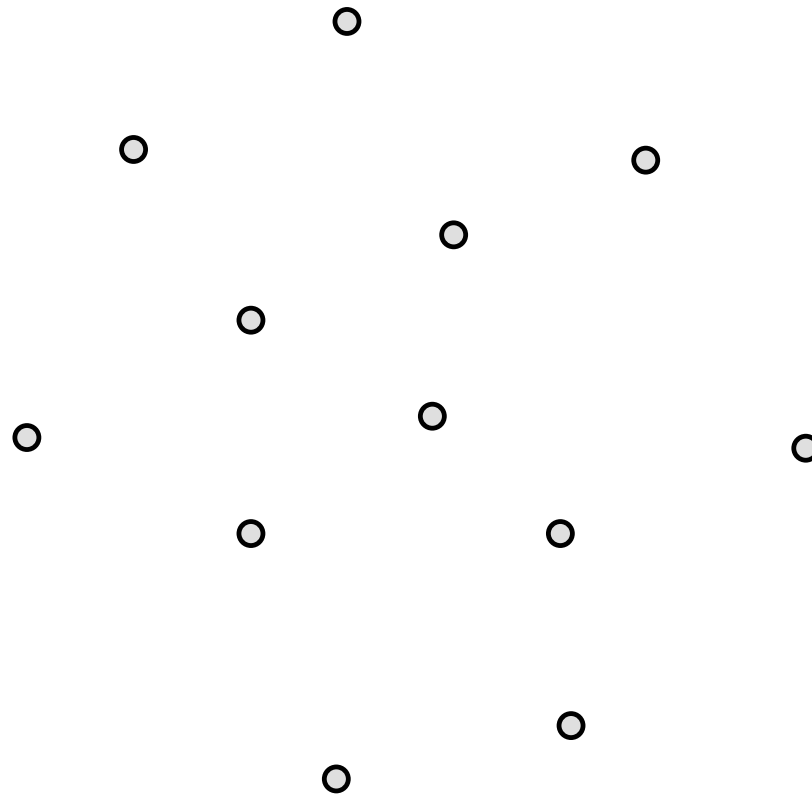
Graham's



TRIANGULATING POINT SETS

Quality of a triangulation

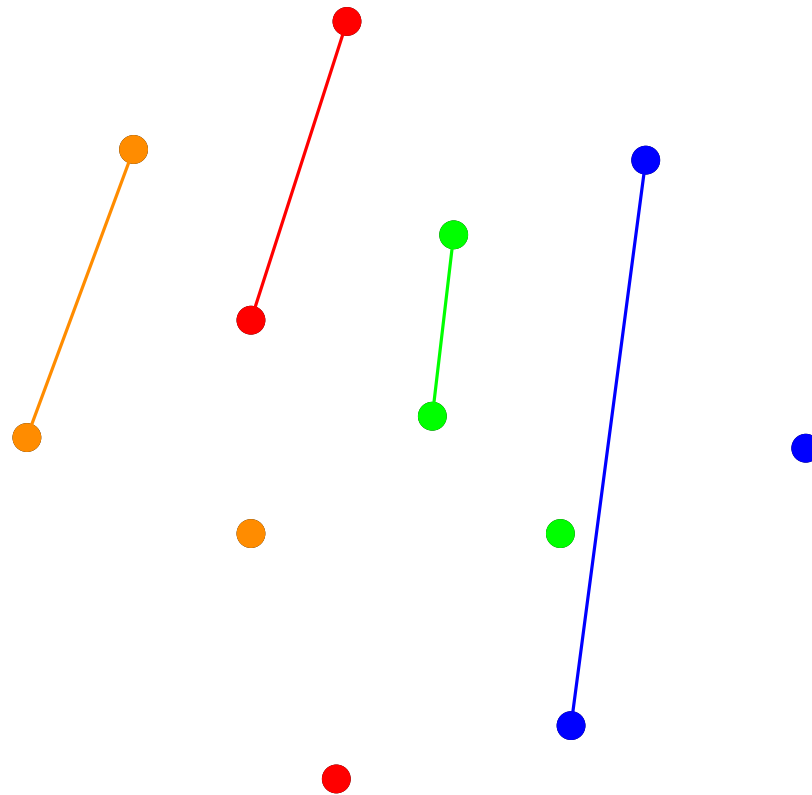
Divide and conquer



TRIANGULATING POINT SETS

Quality of a triangulation

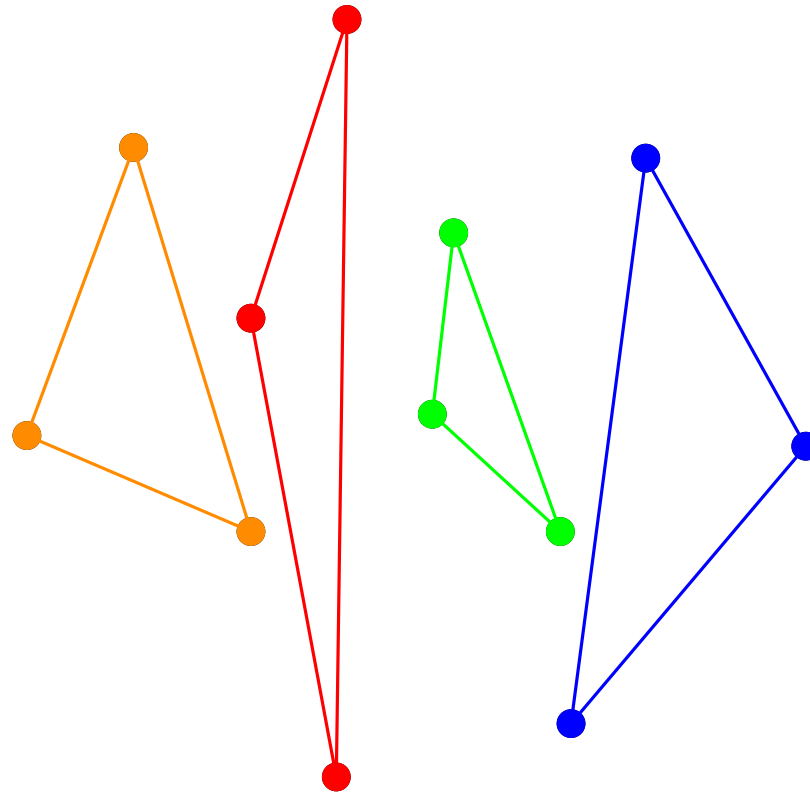
Divide and conquer



TRIANGULATING POINT SETS

Quality of a triangulation

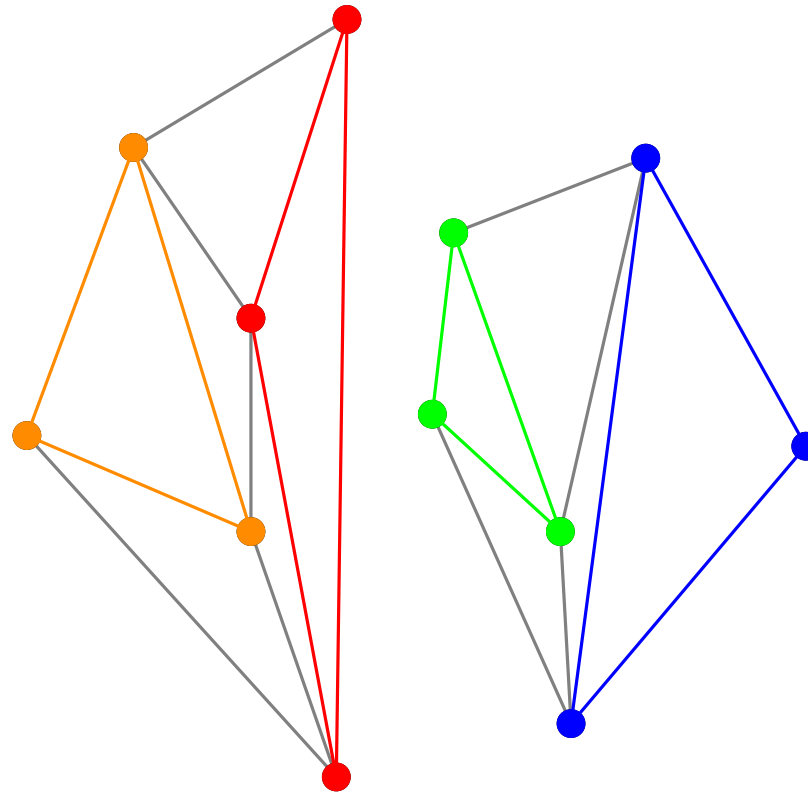
Divide and conquer



TRIANGULATING POINT SETS

Quality of a triangulation

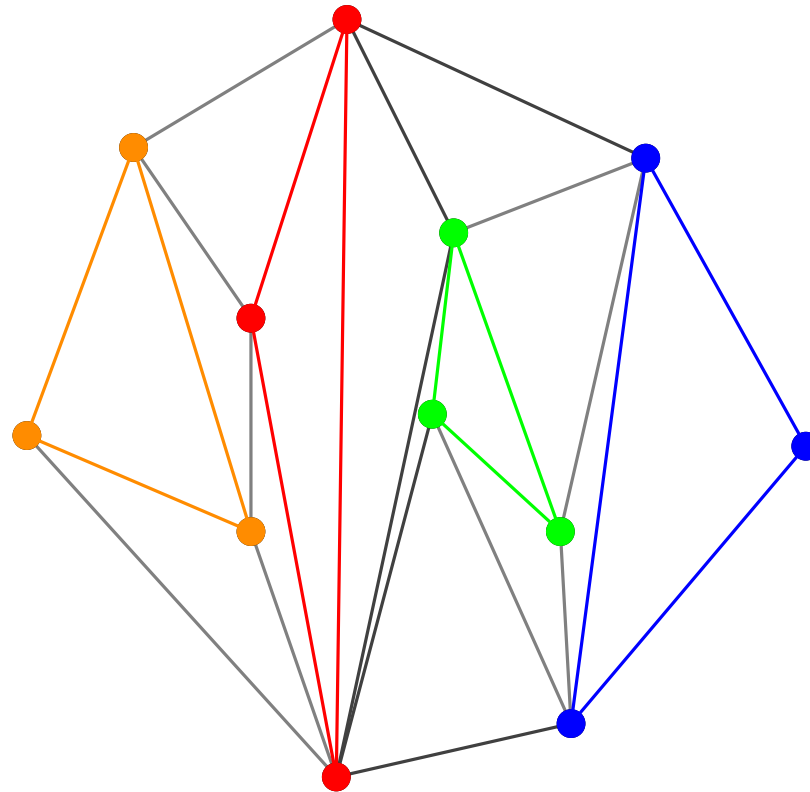
Divide and conquer



TRIANGULATING POINT SETS

Quality of a triangulation

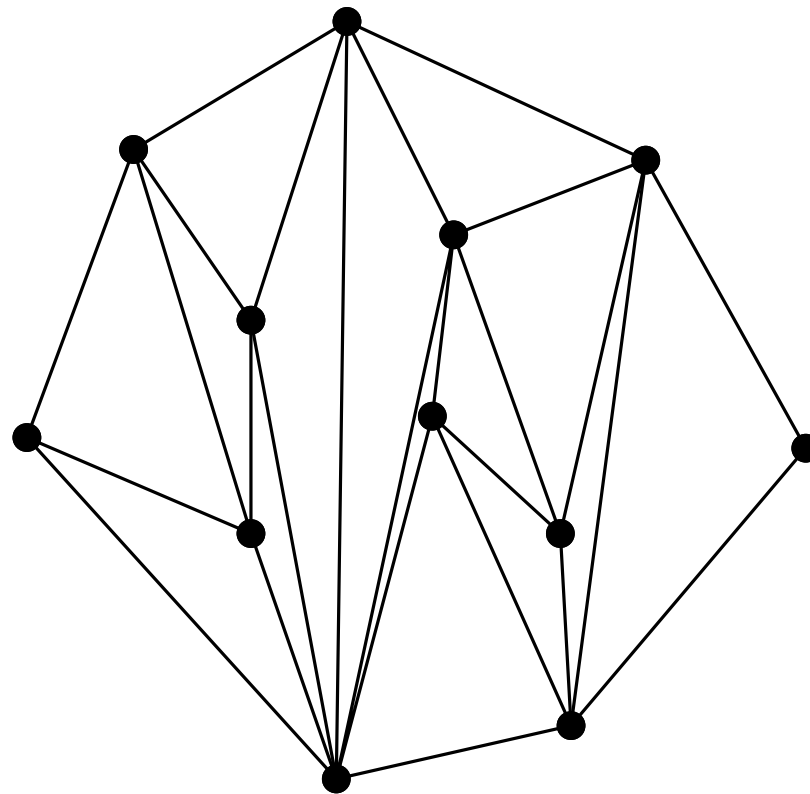
Divide and conquer



TRIANGULATING POINT SETS

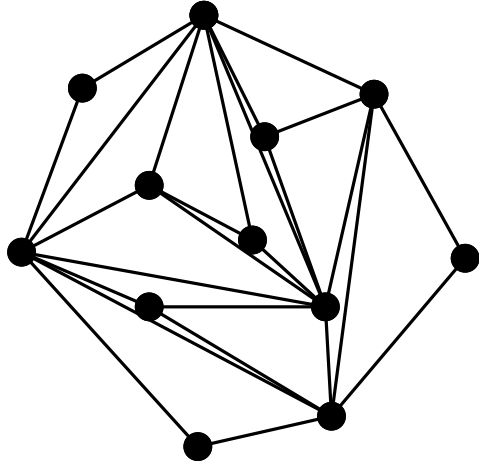
Quality of a triangulation

Divide and conquer

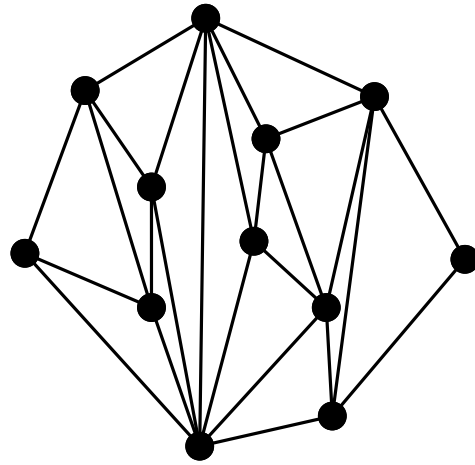


TRIANGULATING POINT SETS

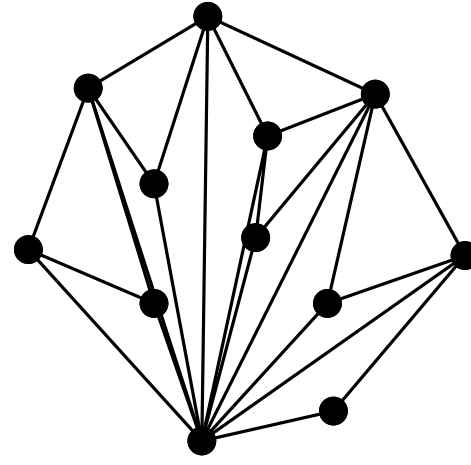
Quality of a triangulation



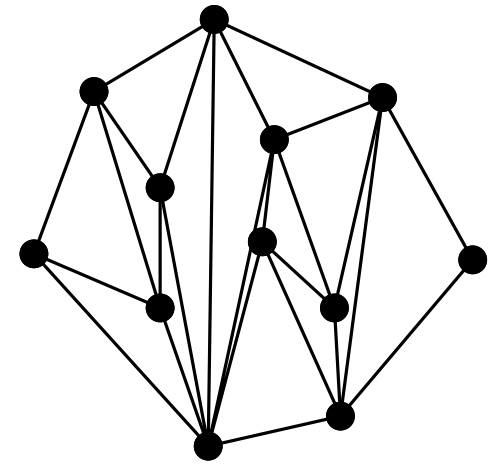
Incremental, unsorted



Incremental, sorted



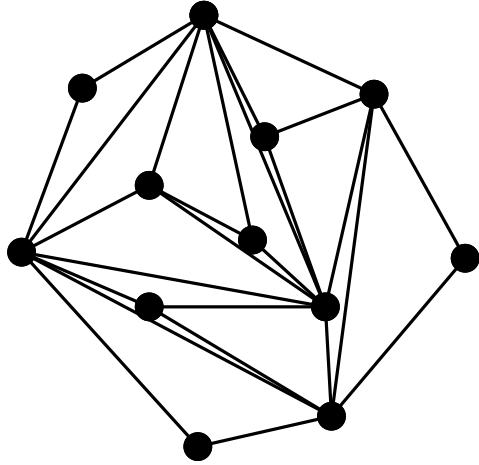
Graham's scan



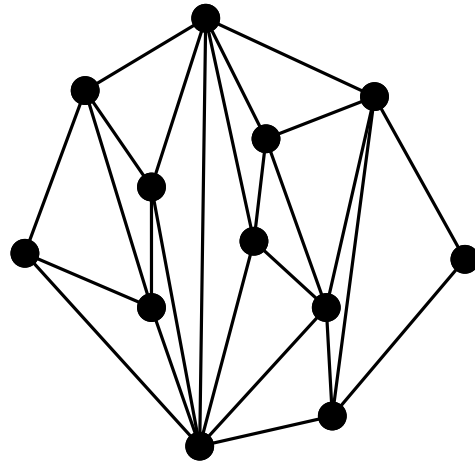
Divide and conquer

TRIANGULATING POINT SETS

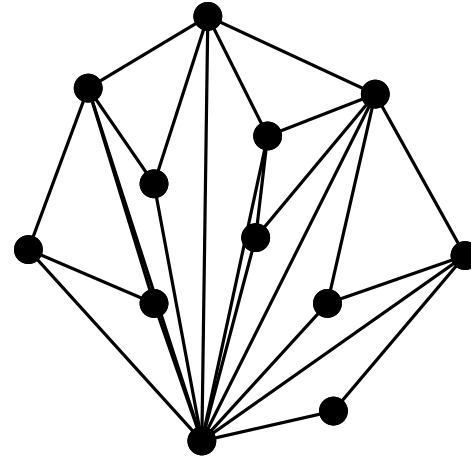
Quality of a triangulation



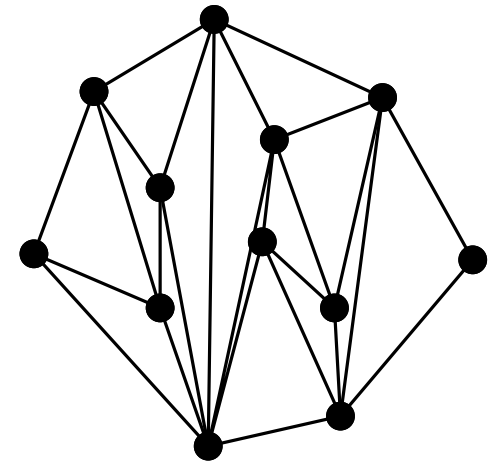
Incremental, unsorted



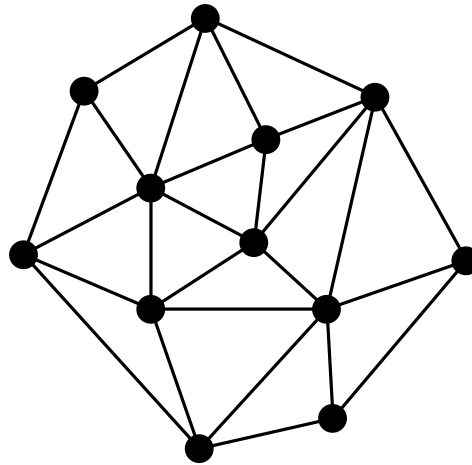
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Graham's scan

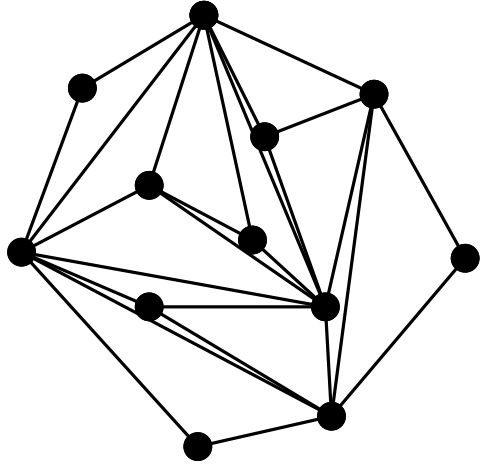


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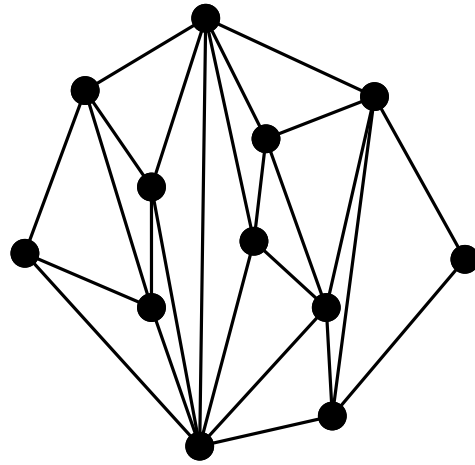


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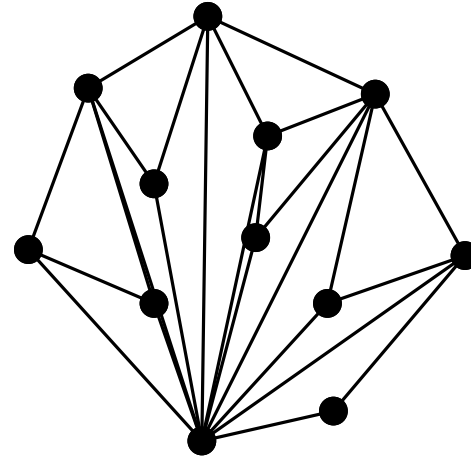
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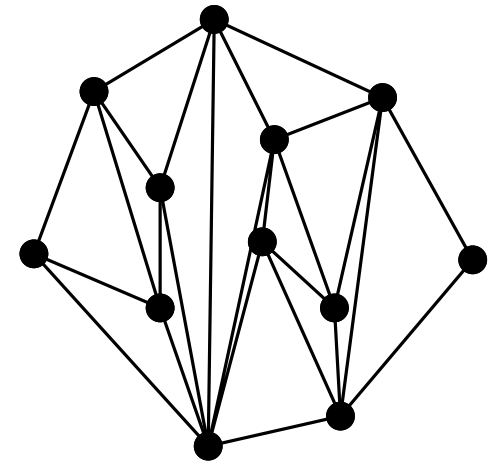
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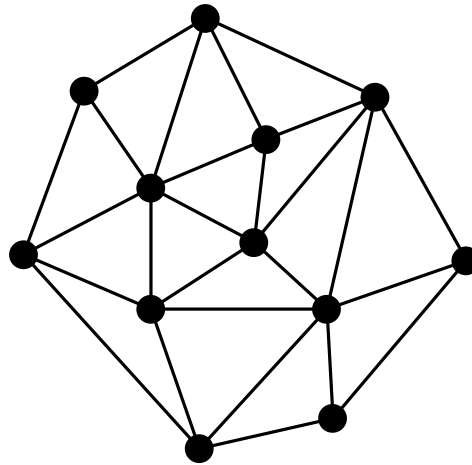
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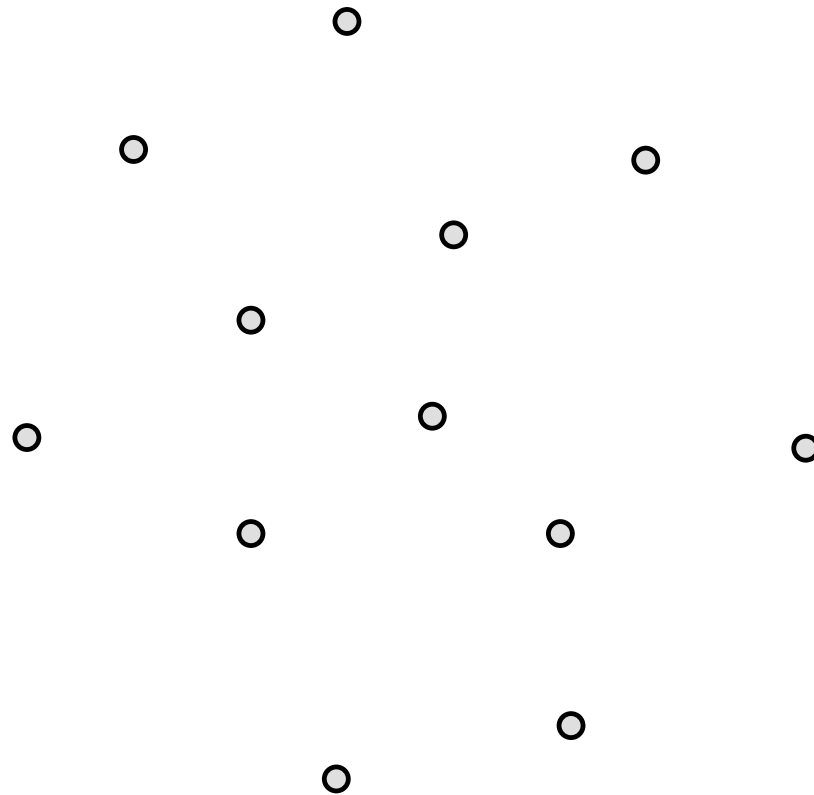


Delaunay triangulation

TRIANGULATING POINT SETS

Quality of a triangulation

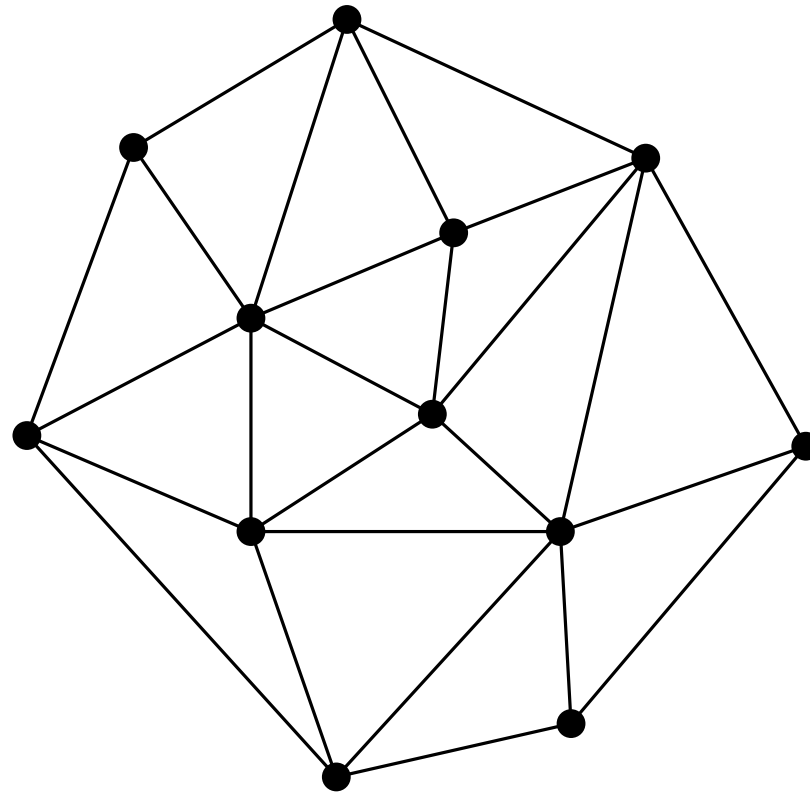
Delaunay



TRIANGULATING POINT SETS

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Delaunay



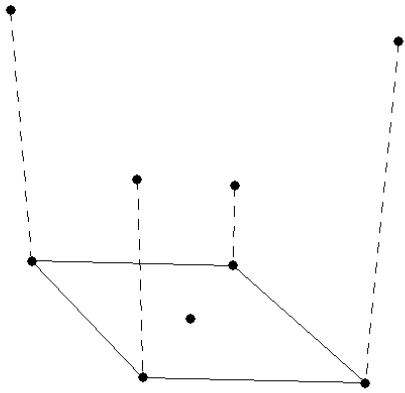
Delaunay Triangulation

DELAUNAY TRIANGULATION

A TOOL FOR INTERPOLATION

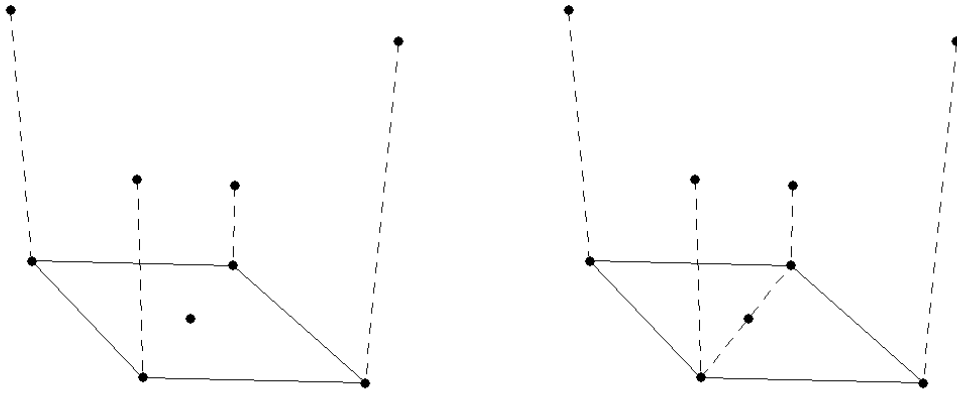
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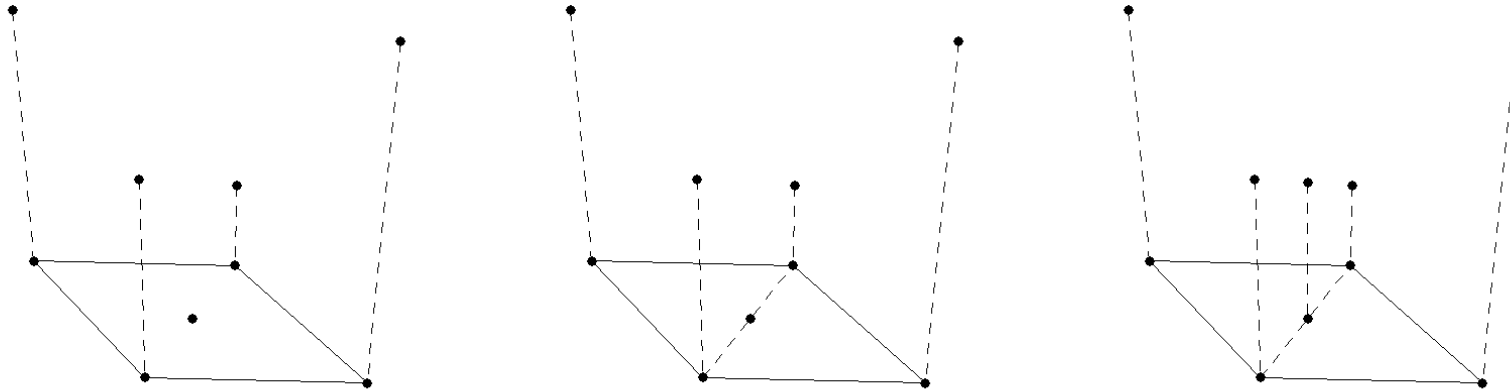
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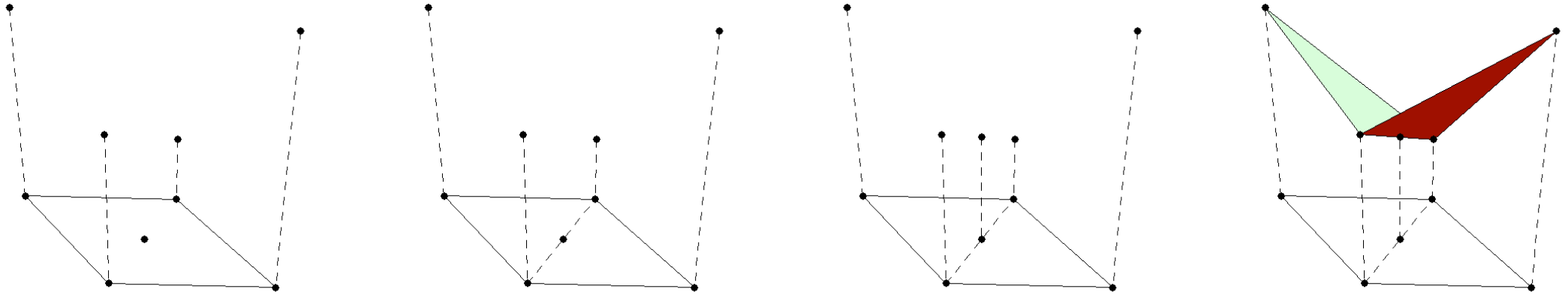
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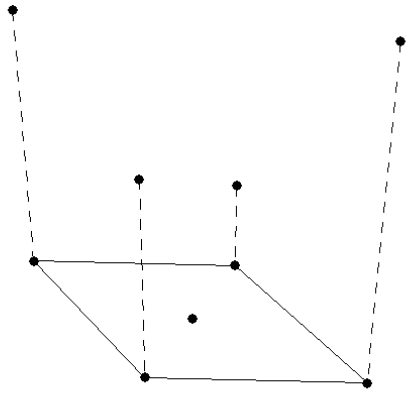
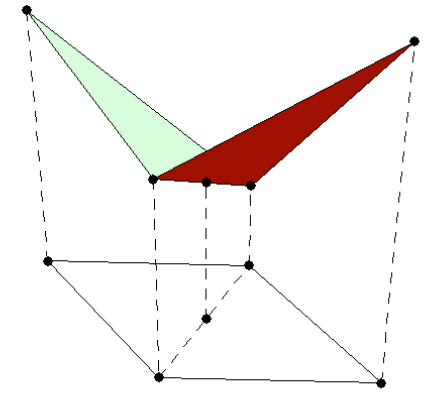
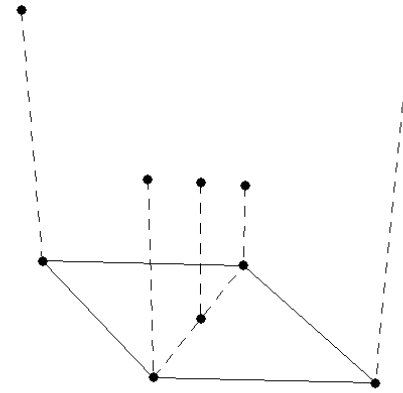
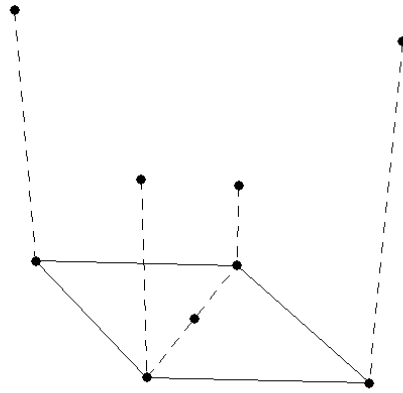
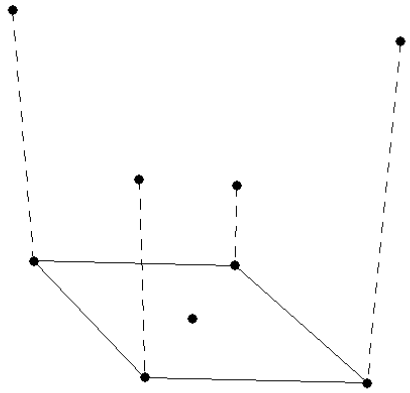
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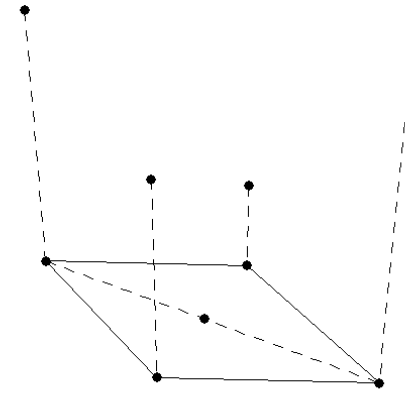
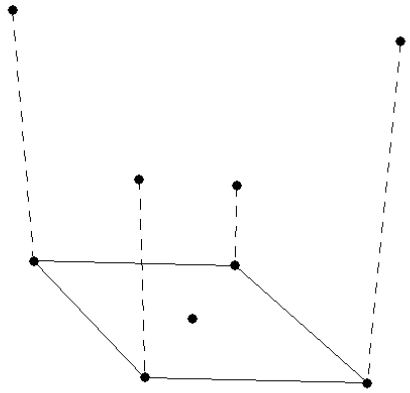
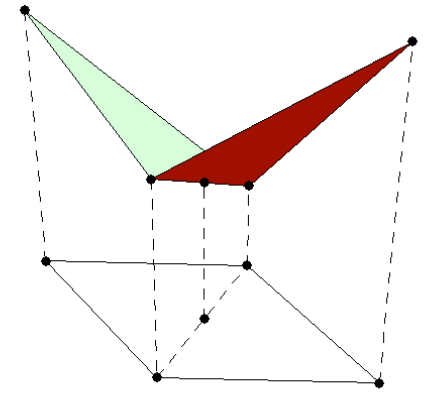
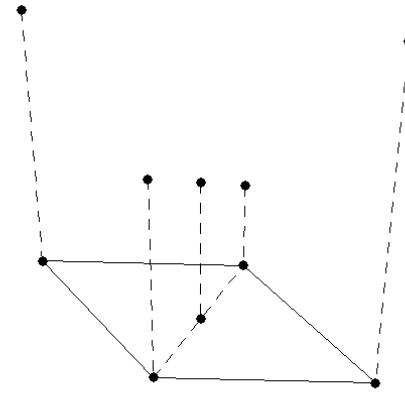
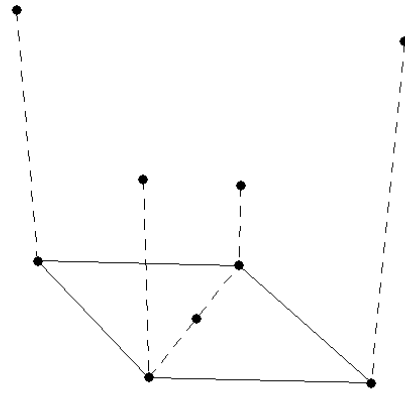
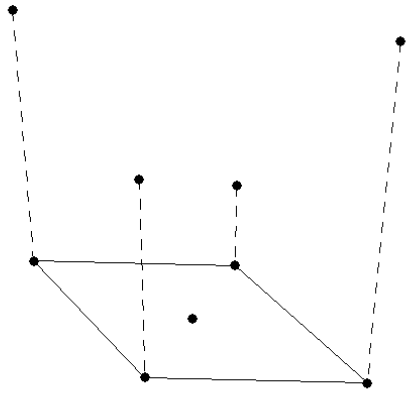
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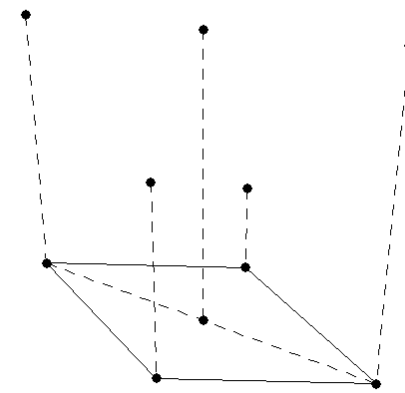
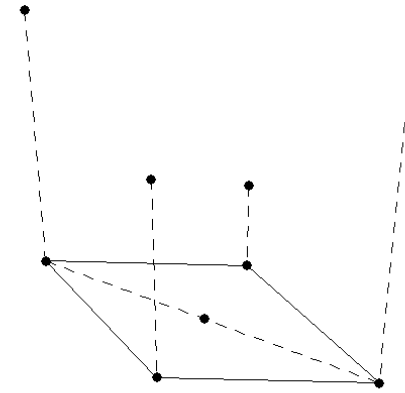
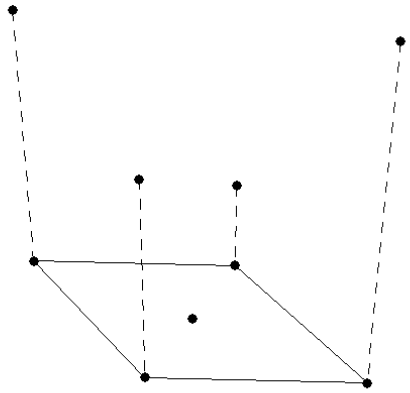
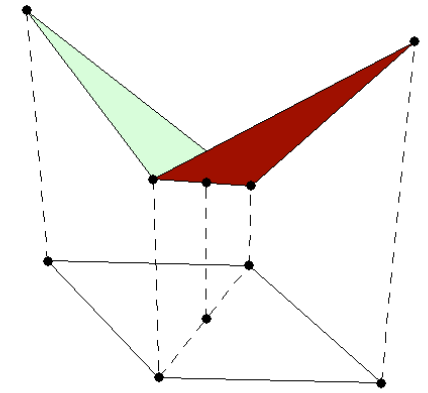
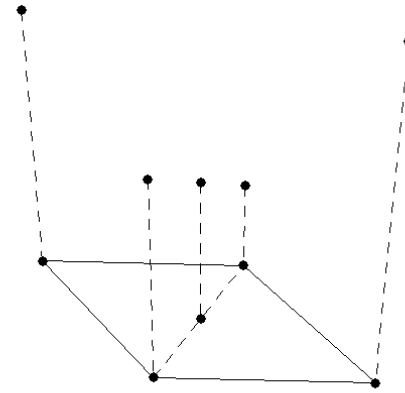
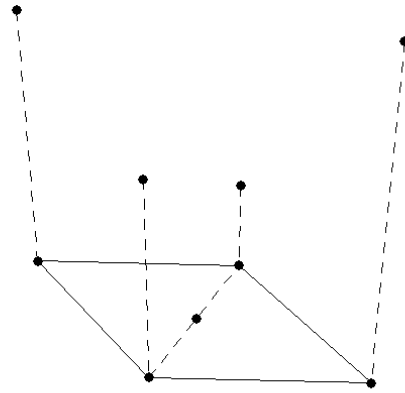
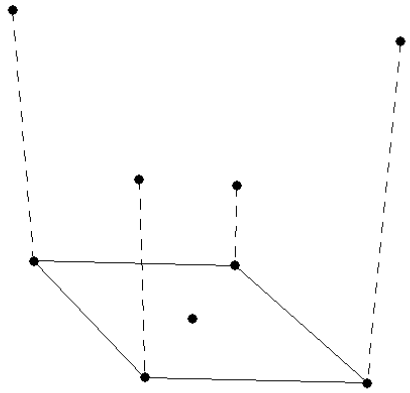
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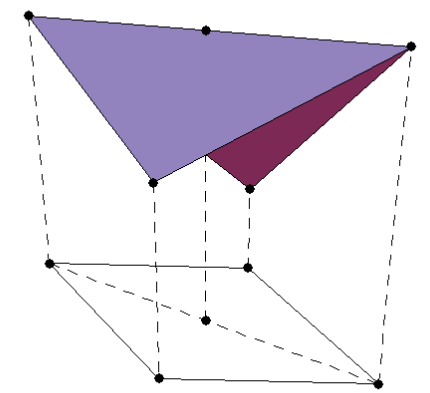
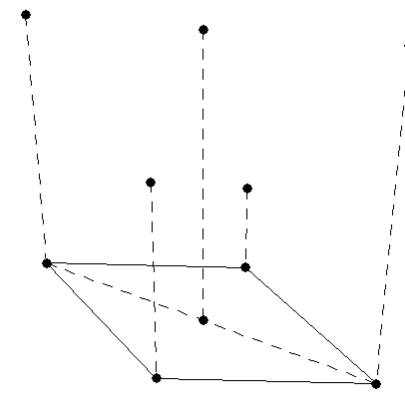
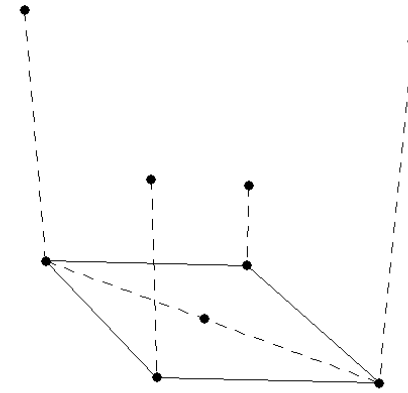
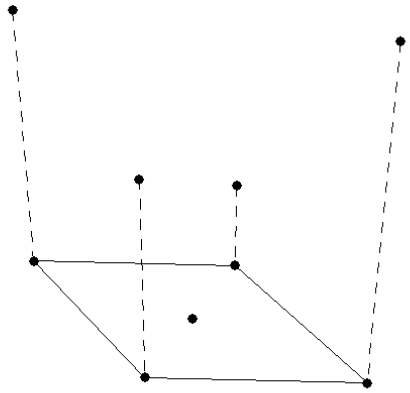
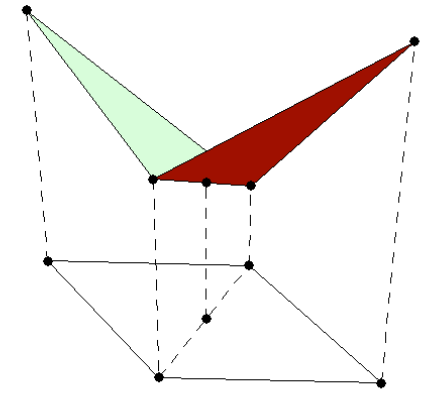
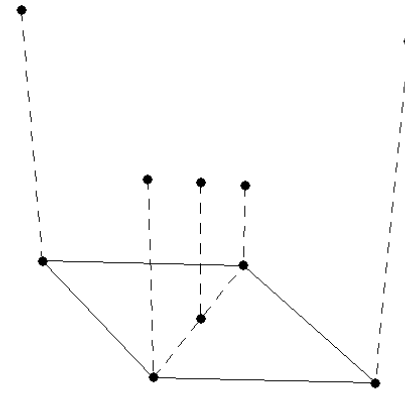
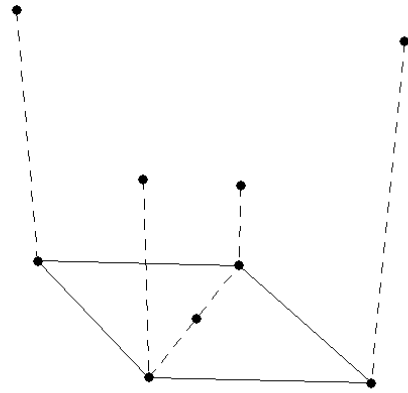
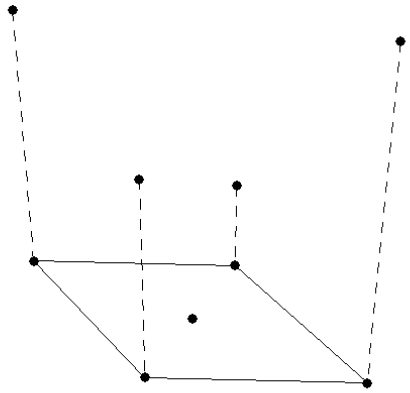
DELAUNAY TRIANGULATION

A TOOL FOR INTERPOLATION



DELAUNAY TRIANGULATION

A TOOL FOR INTERPOLATION



DELAUNAY TRIANGULATION

A TOOL FOR INTERPOLATION

Motivation

Triangulations give a natural way to interpolate! For many interpolation applications, it makes more sense to interpolate from points that are nearby than from points that are far away

DELAUNAY TRIANGULATION

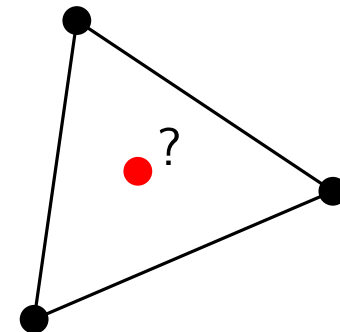
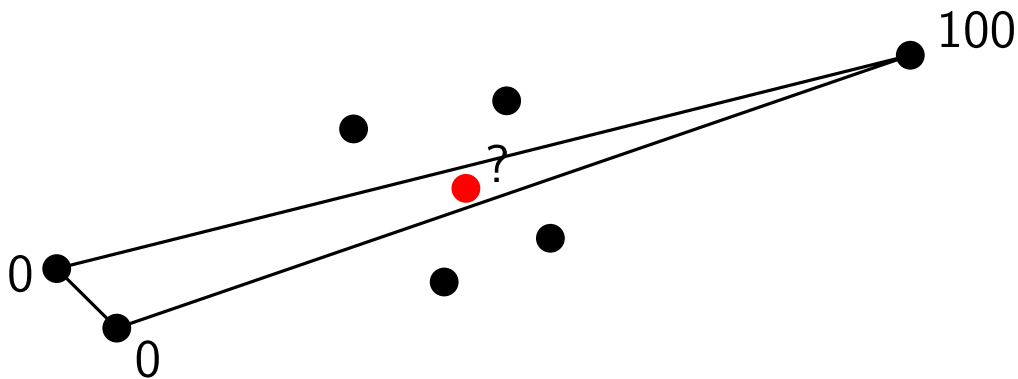
A TOOL FOR INTERPOLATION

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Triangulations give a natural way to interpolate! For many interpolation applications, it makes more sense to interpolate from points that are nearby than from points that are far away

To do that, the shape of the triangles matters:

- Long and skinny triangles tend to produce interpolation from points that are far
- Equilateral-like triangles tend to give interpolations from points closer, and well distributed



DELAUNAY TRIANGULATION

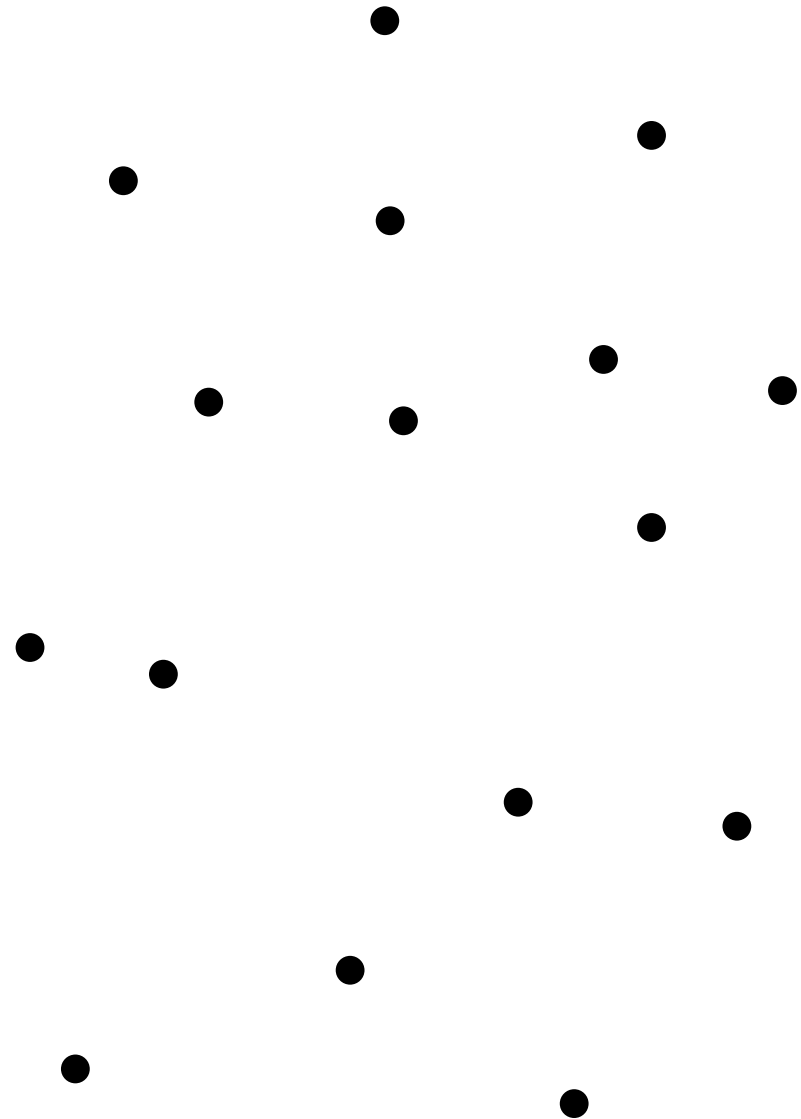
DEFINITION AND PROPERTIES

DELAUNAY TRIANGULATION

DEFINITION AND PROPERTIES

Definition

Given a set P with n points in the plane...



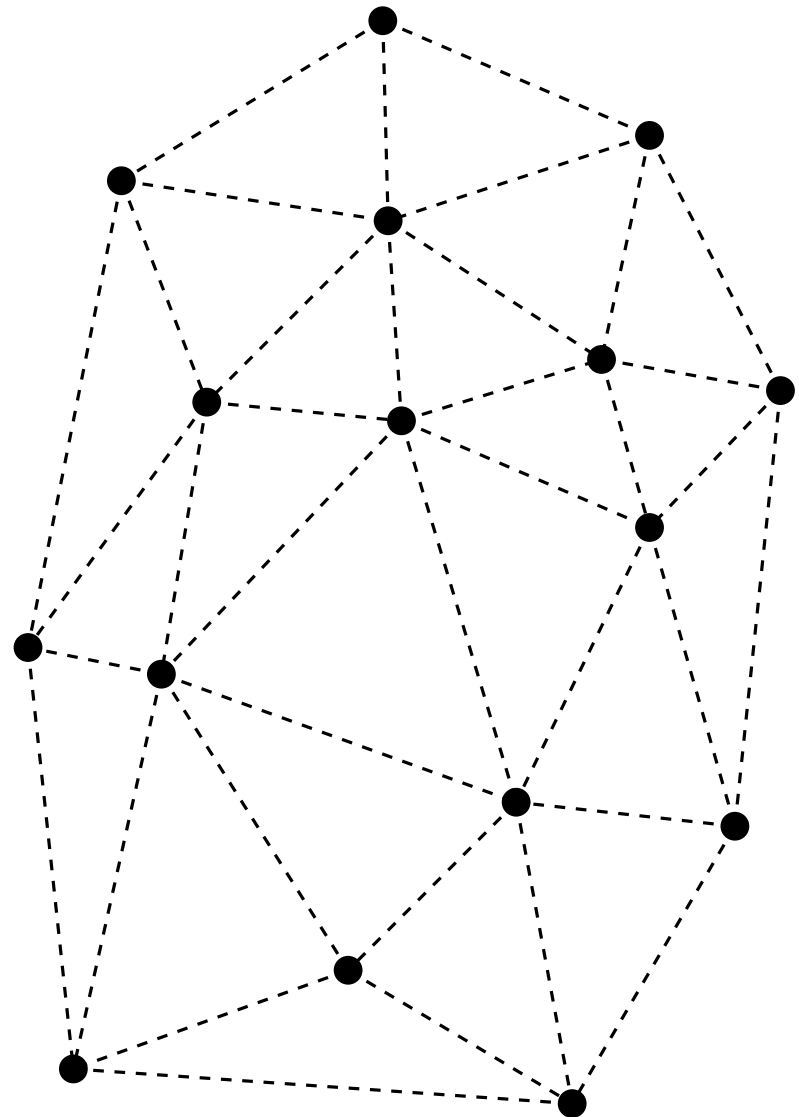
DELAUNAY TRIANGULATION

DEFINITION AND PROPERTIES

Definition

Given a set P of n points in the plane, the **Delaunay triangulation** of P , $Del(P)$, is a triangulation where all triangles are *Delaunay triangles*.

Three points p_i, p_j, p_k form a *Delaunay triangle* (in general, are vertices of a face) if and only if the circle through them does not contain any point of P in its interior.



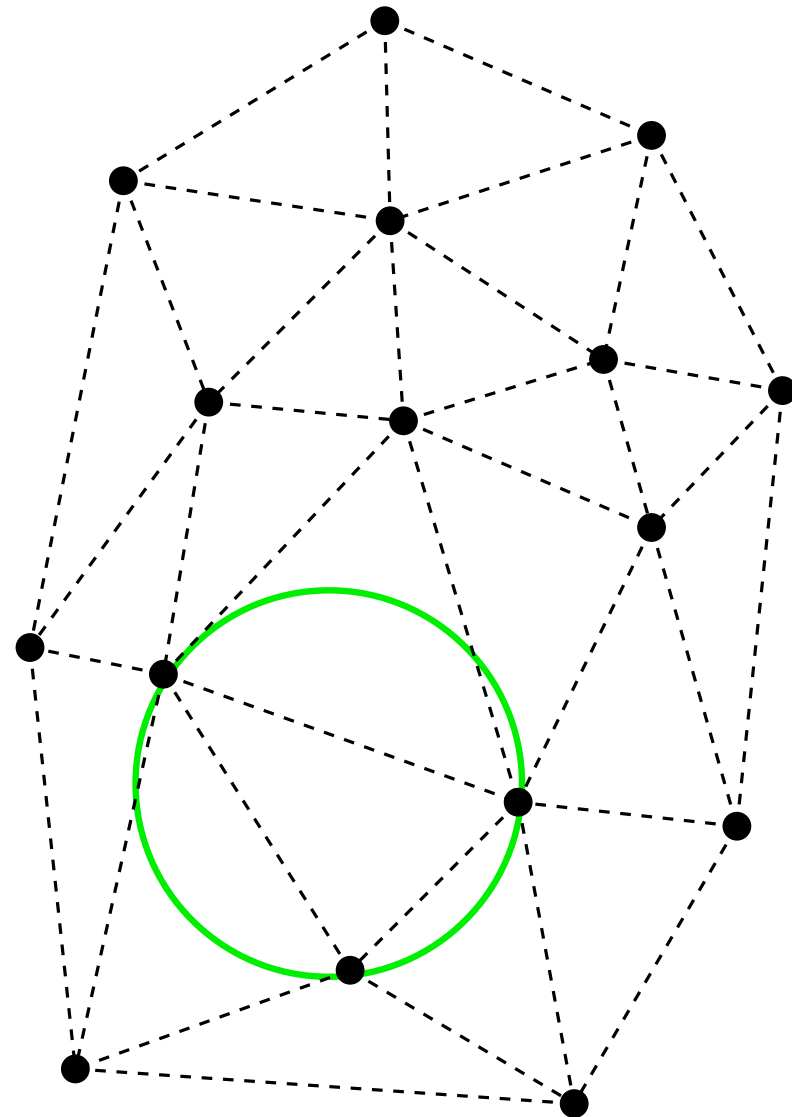
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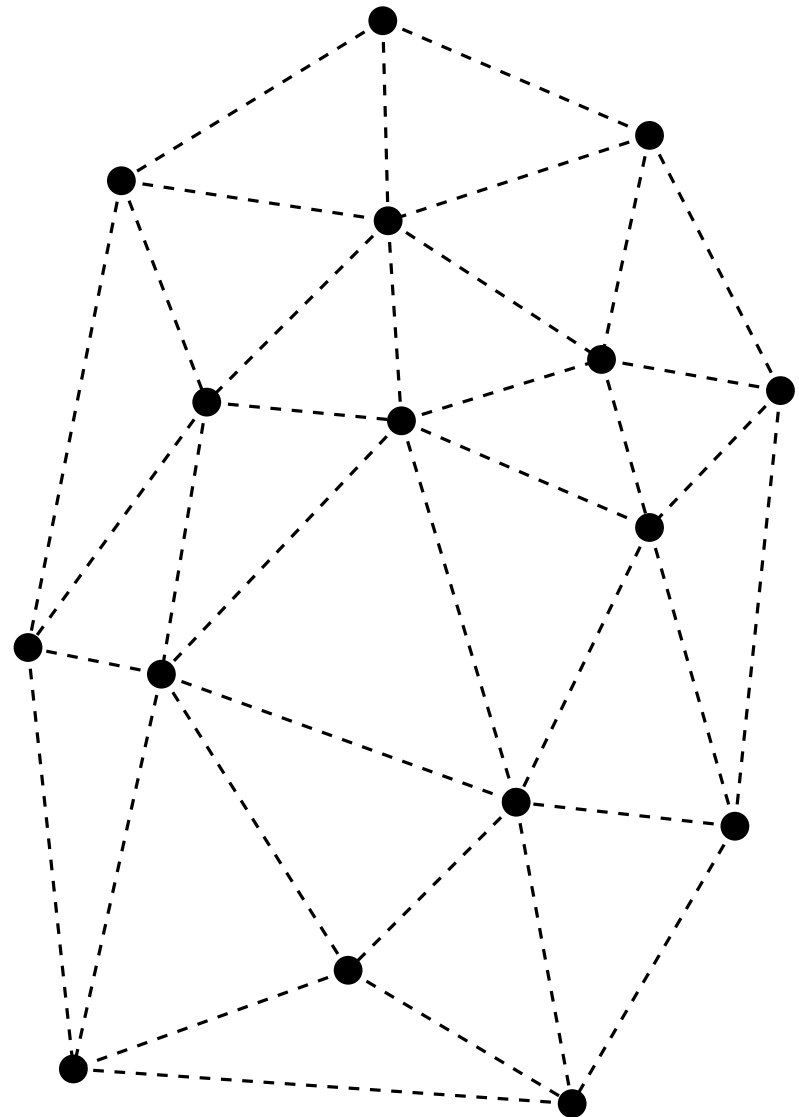
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Equivalent edge-based characterization

- Two points $p_i, p_j \in P$ form a *Delaunay edge* if and only if there exists a circle through p_i and p_j that does not contain any point of P in its interior.



DELAUNAY TRIANGULATION

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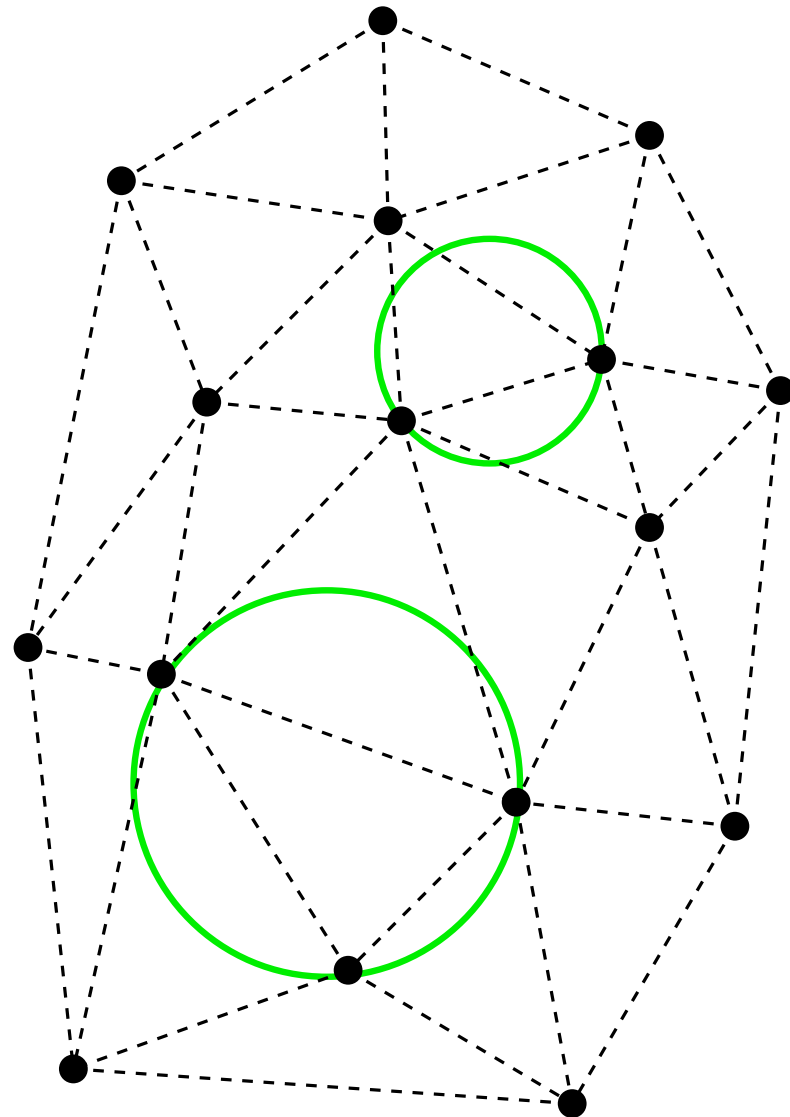
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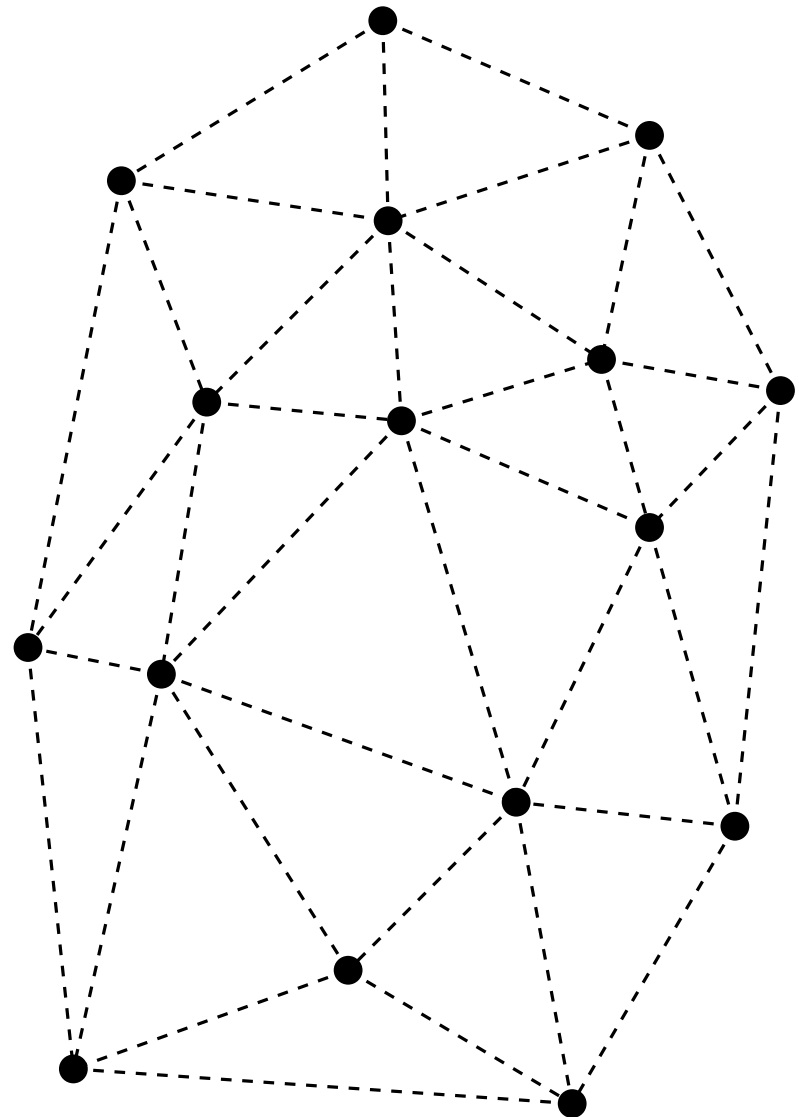
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Property: edges cannot cross

$Del(P)$ is a plane graph



DELAUNAY TRIANGULATION

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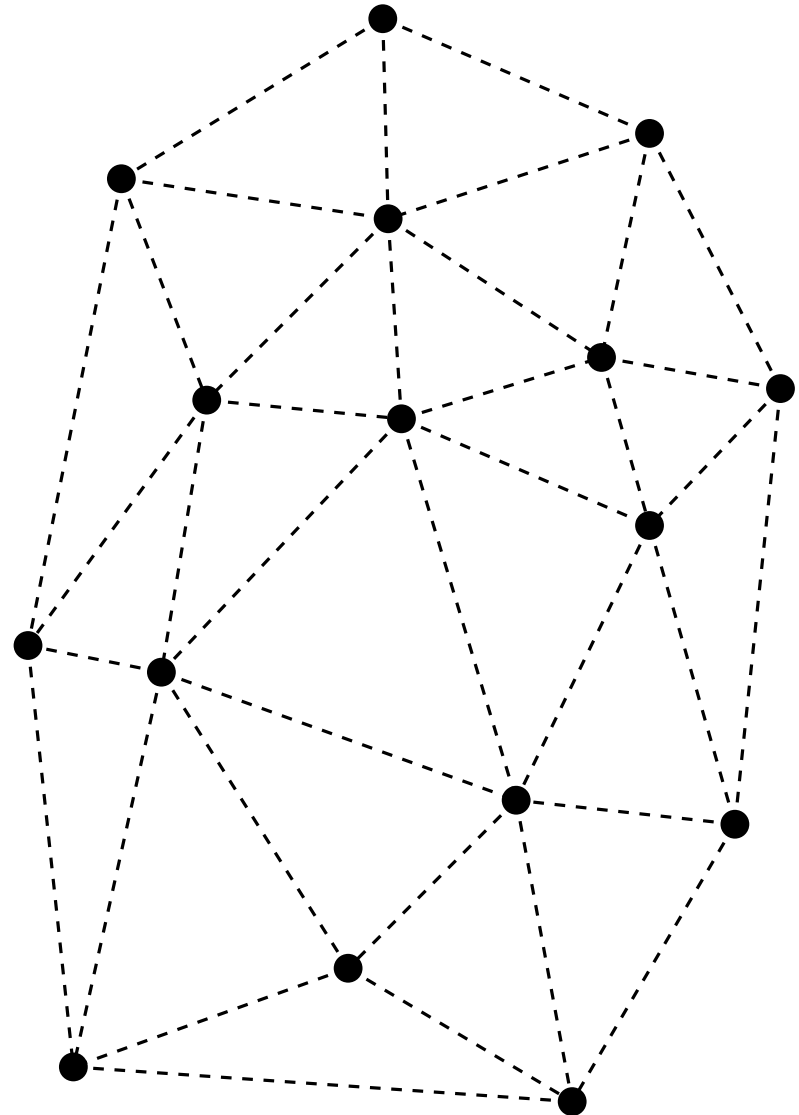
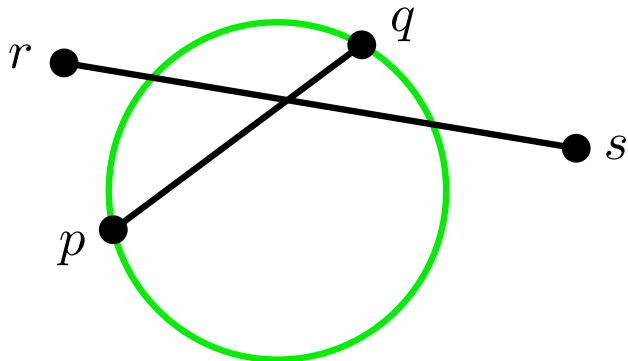
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If $\overline{pq}$ is a Delaunay edge, there exists an empty circle through p and q . If a segment $\overline{rs}$ intersects $\overline{pq}$, then every circle through r and s contains at least one of p or q .



DELAUNAY TRIANGULATION

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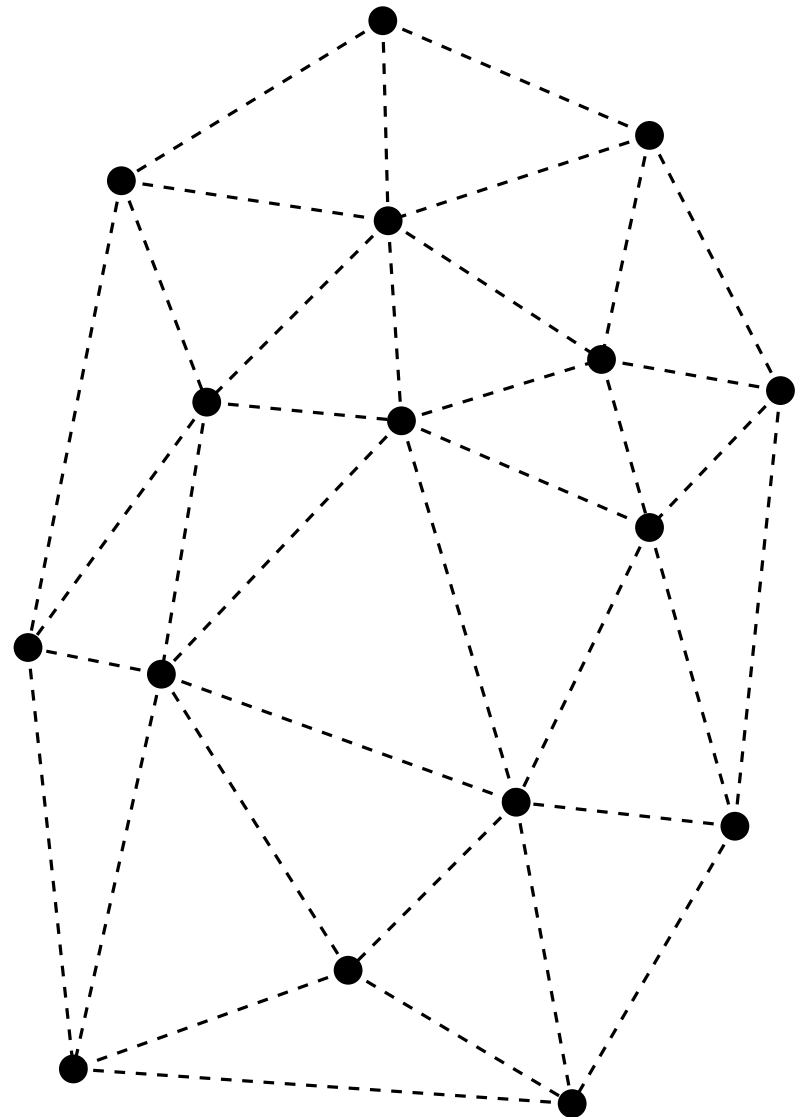
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Property: graph is a (pre-)triangulation

$Del(P)$ is a triangulation of P , except when P has three or more cocircular points. In this case, it is a *pre-triangulation* which can be trivially completed (although this can be done in several different ways).



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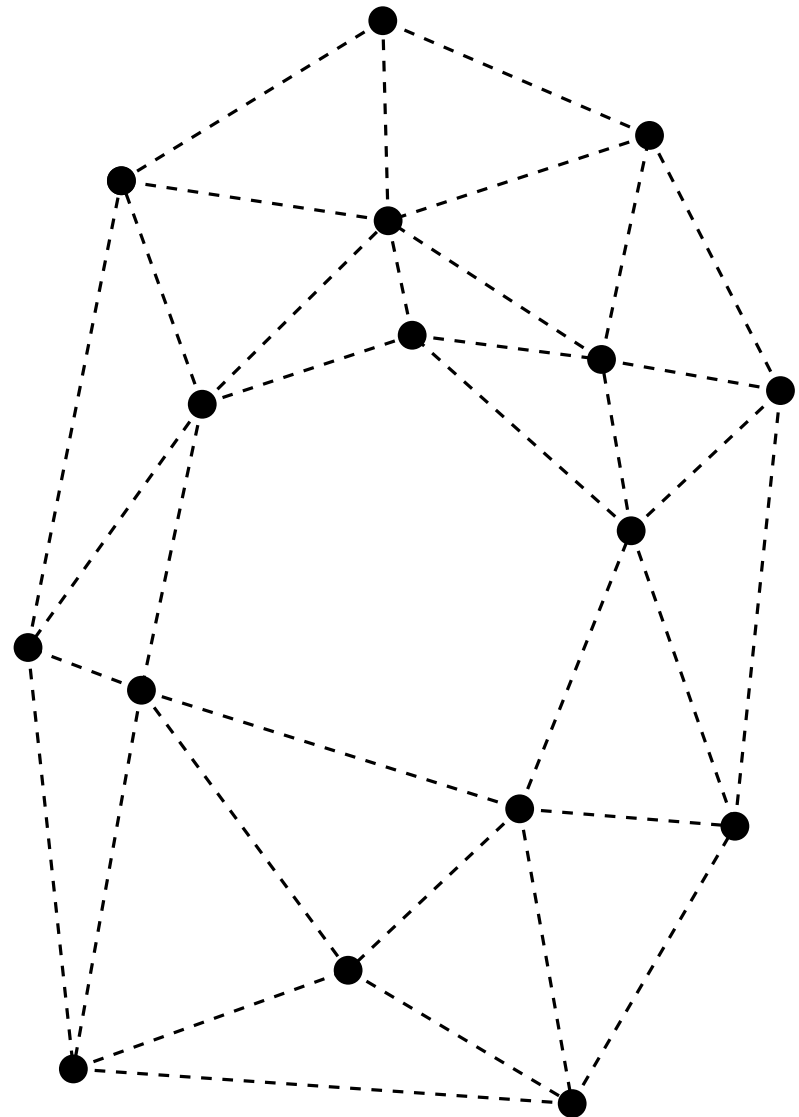
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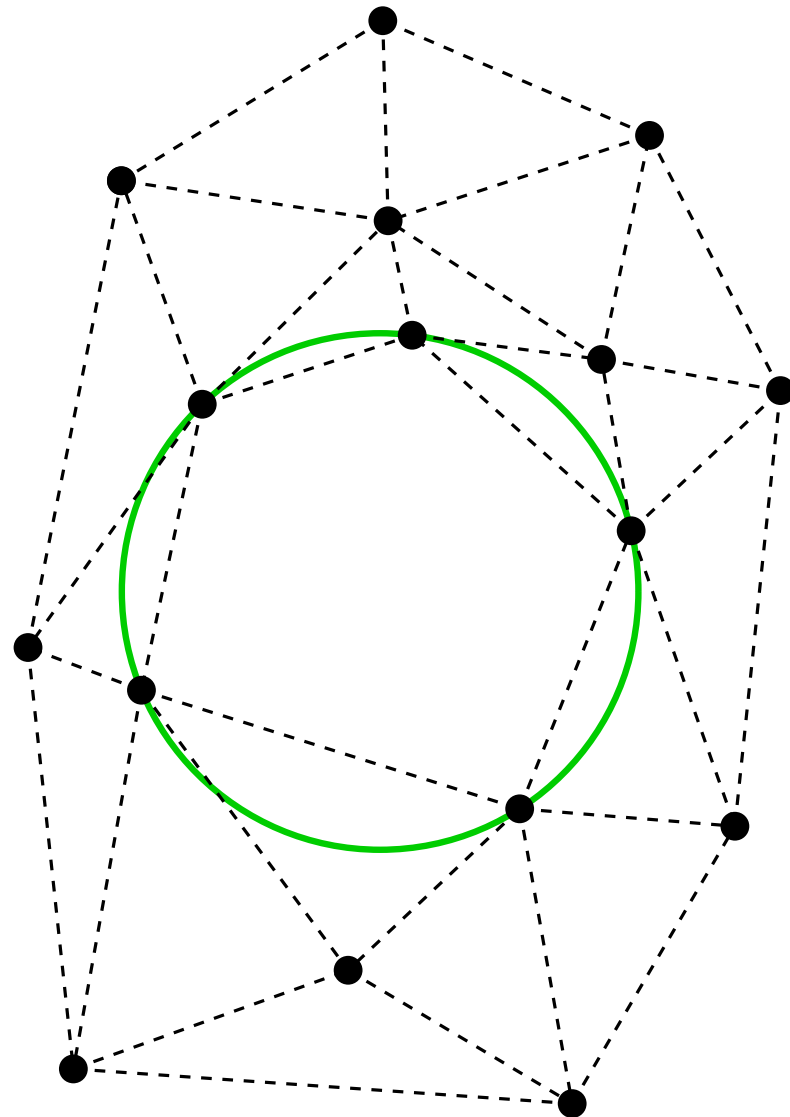
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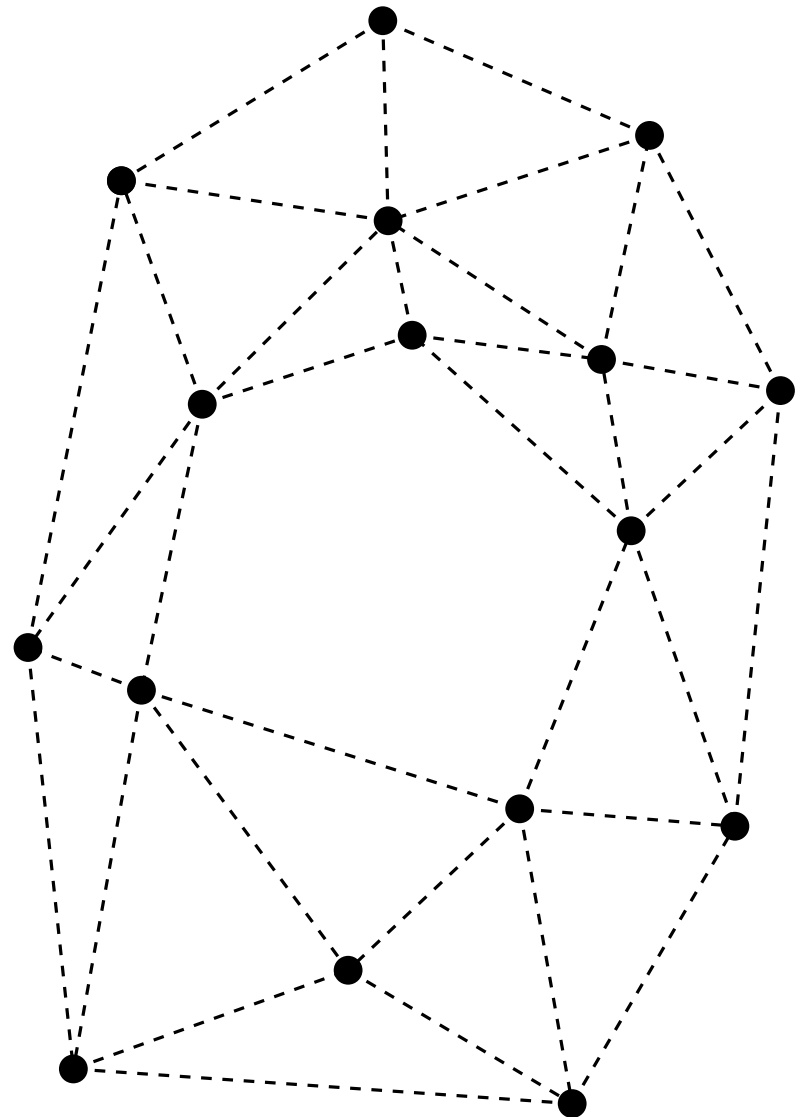
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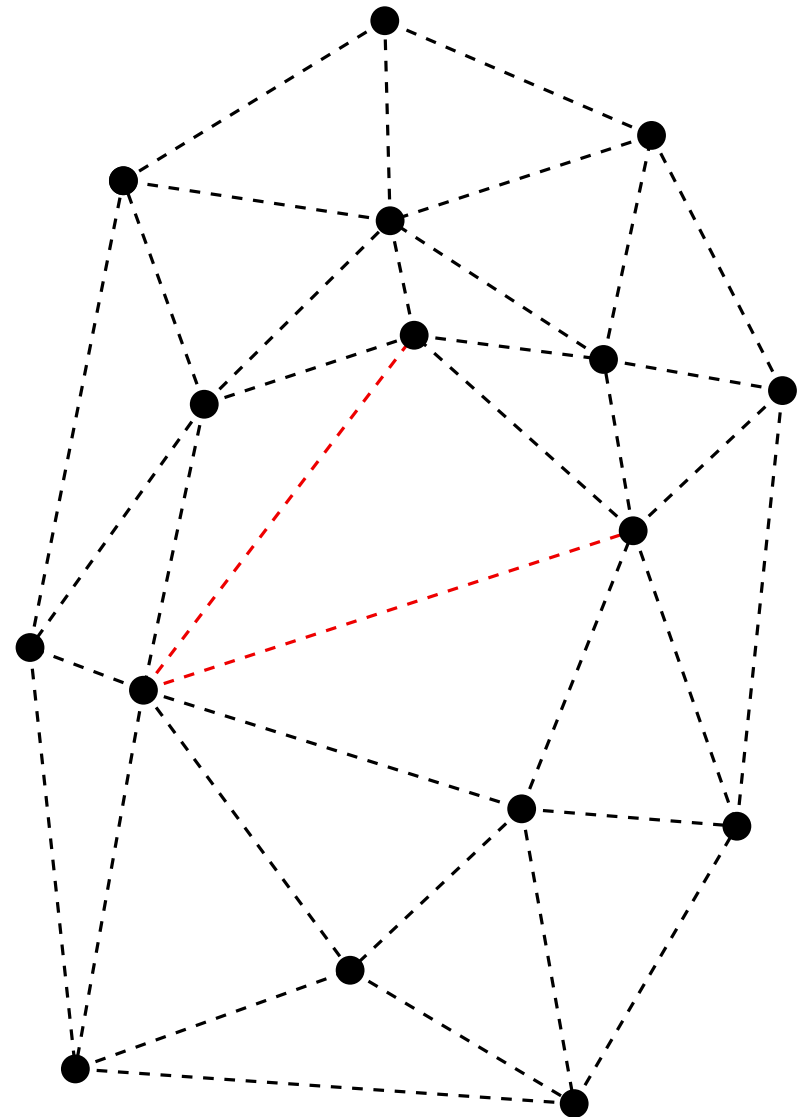
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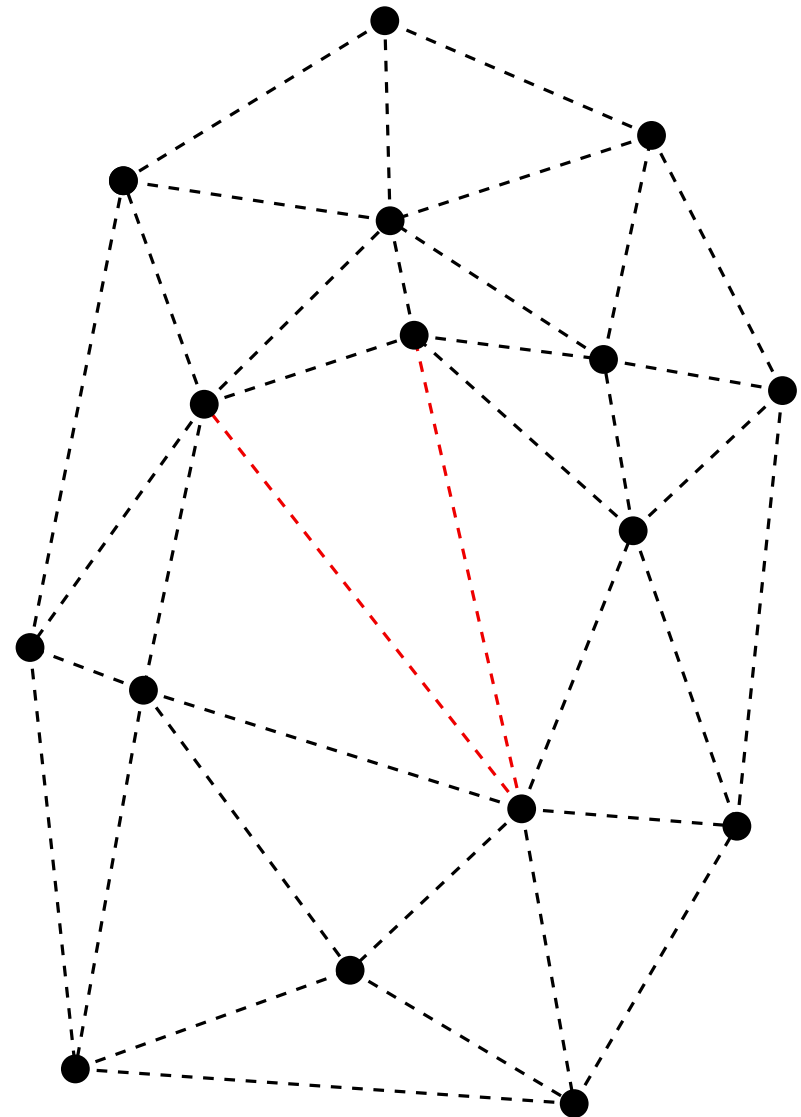
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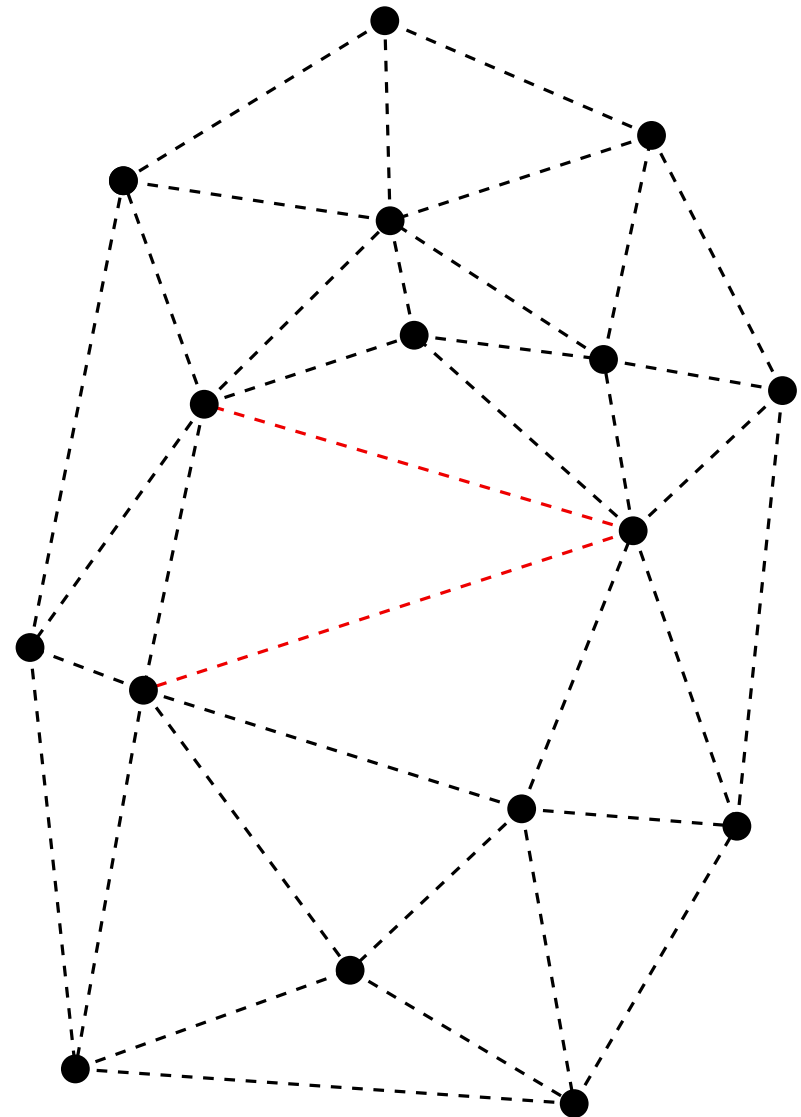
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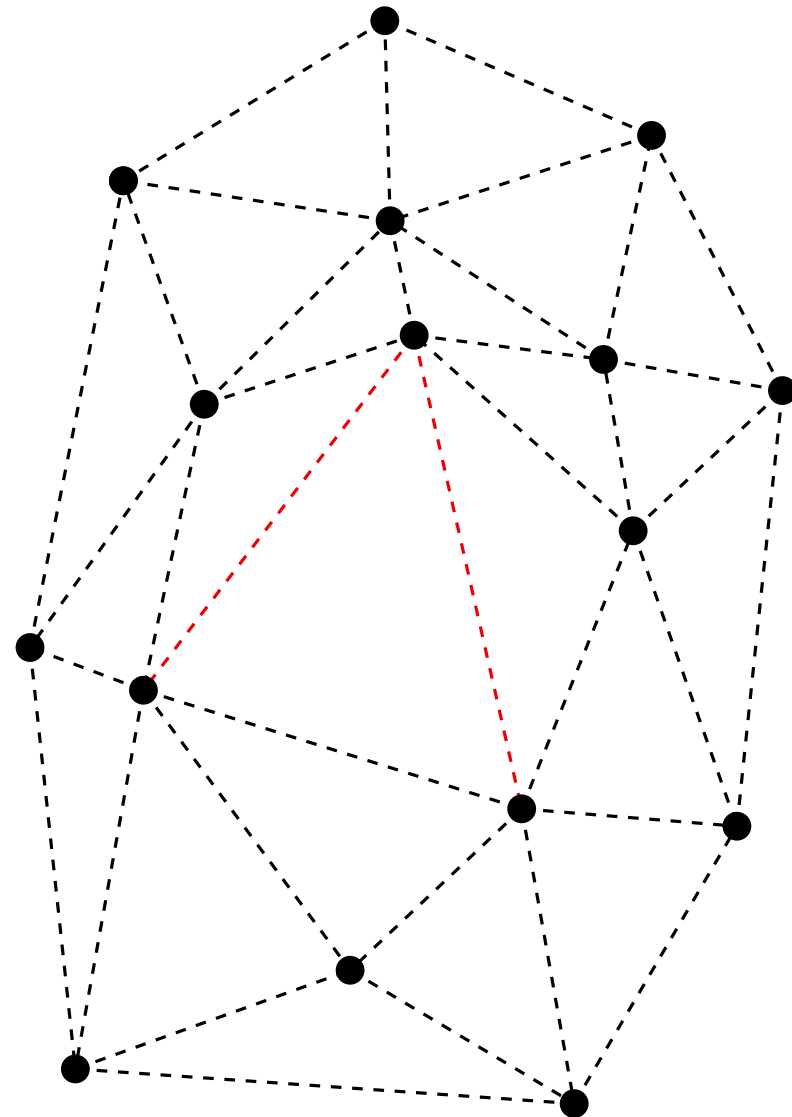
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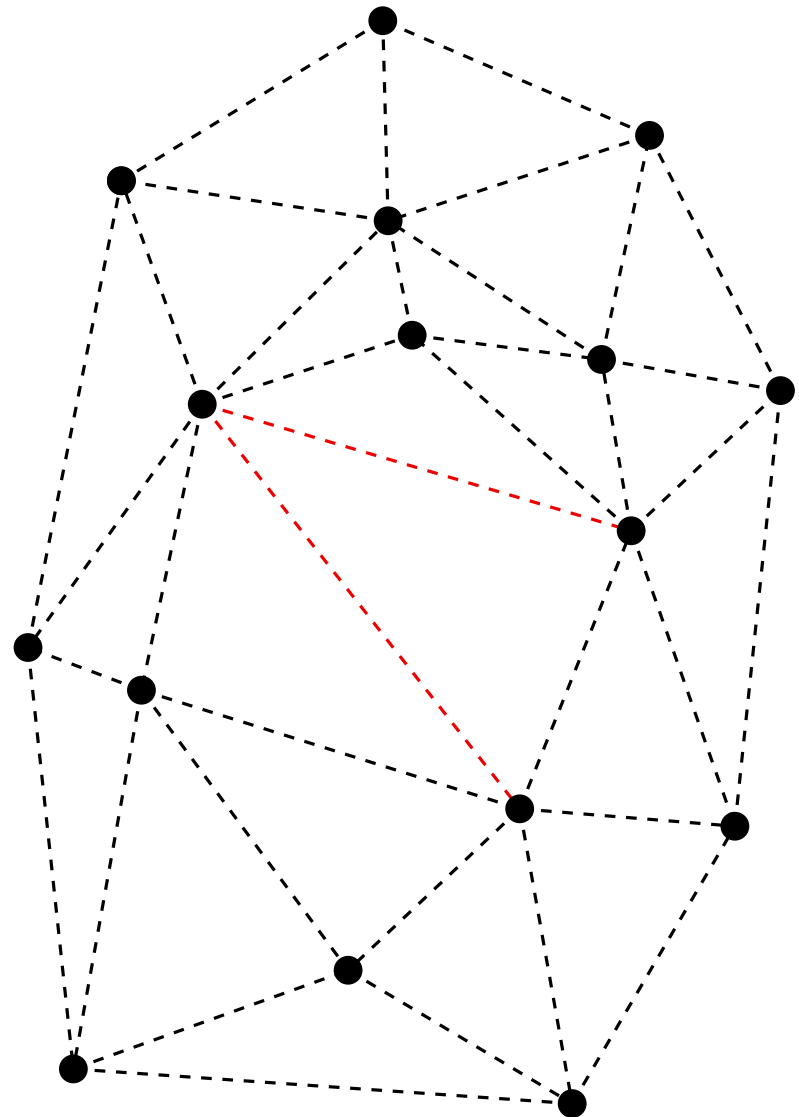
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DELAUNAY TRIANGULATION

DELAUNAY TRIANGULATION AND 3D CONVEX HULL

DELAUNAY TRIANGULATION

DELAUNAY TRIANGULATION AND 3D CONVEX HULL

Theorem

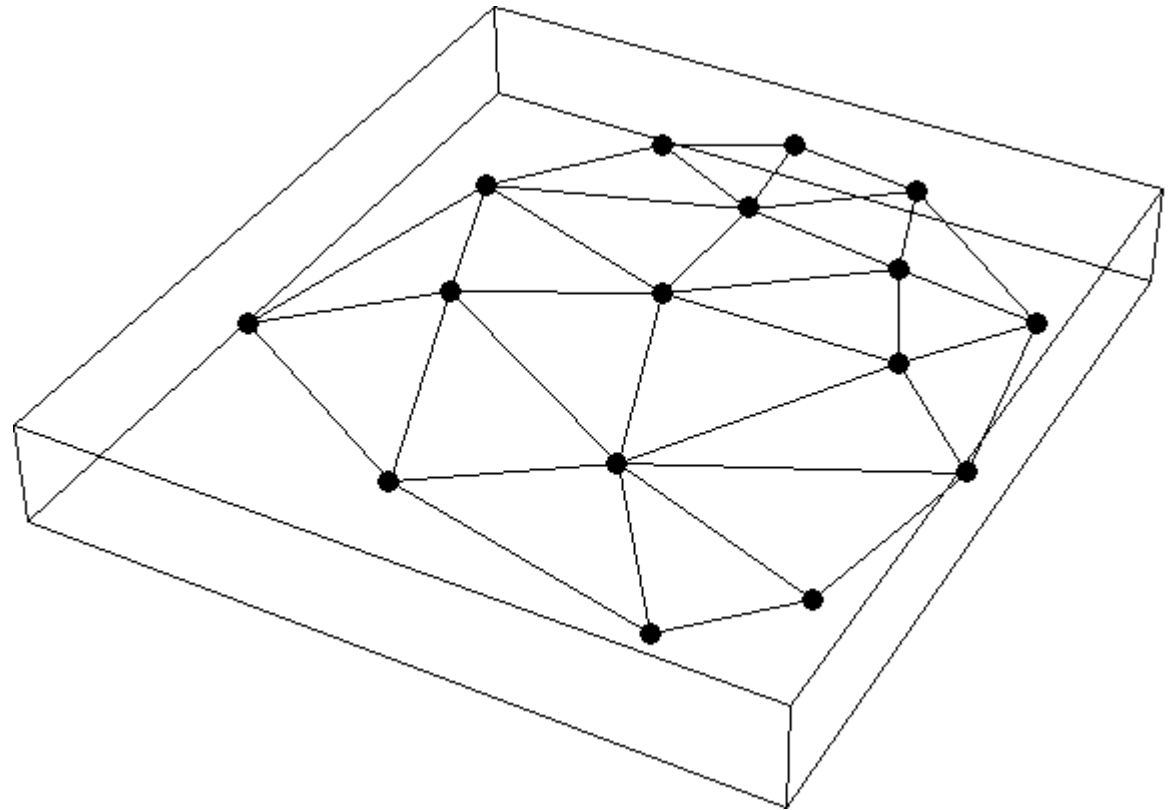
Let $P = \{p_1, \dots, p_n\}$ with $p_i = (a_i, b_i, 0)$. Let $p_i^* = (a_i, b_i, a_i^2 + b_i^2)$ be the vertical projection of each point p_i onto the paraboloid $z = x^2 + y^2$. Then $Del(P)$ is the orthogonal projection onto the plane $z = 0$ of the lower convex hull of P^* .

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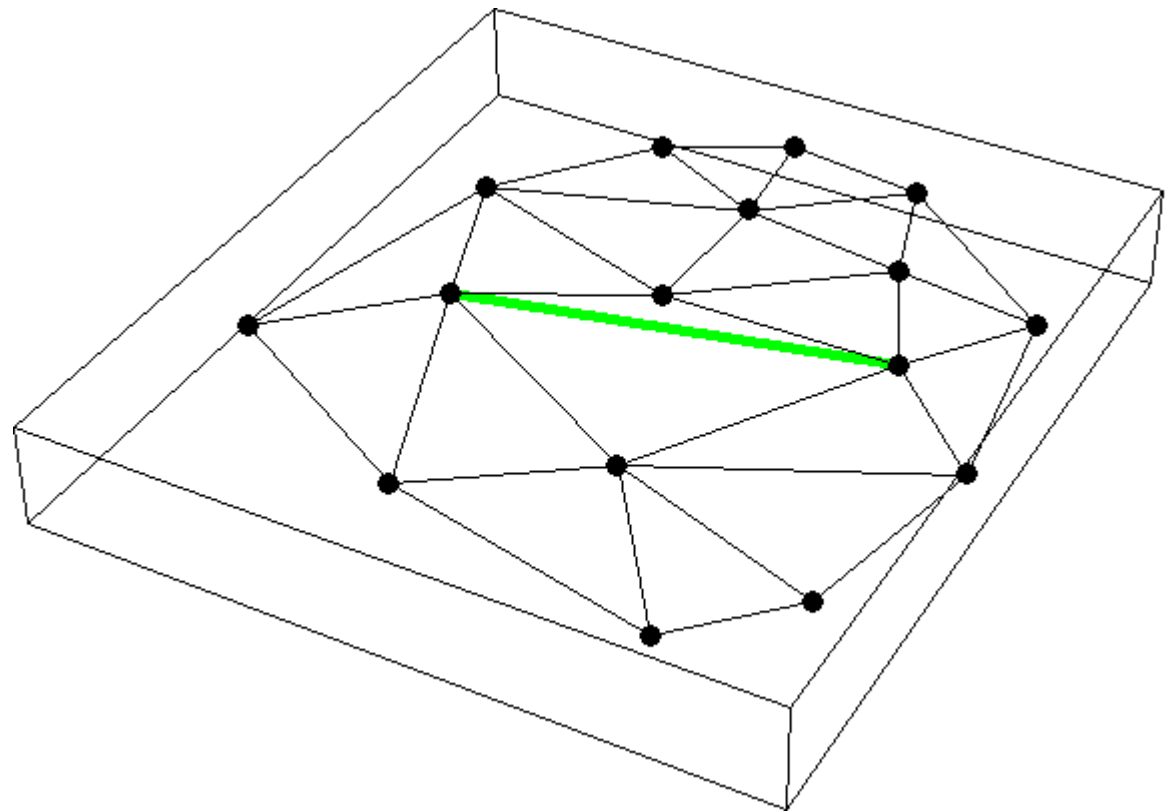


DELAUNAY TRIANGULATION

DELAUNAY TRIANGULATION AND 3D CONVEX HULL

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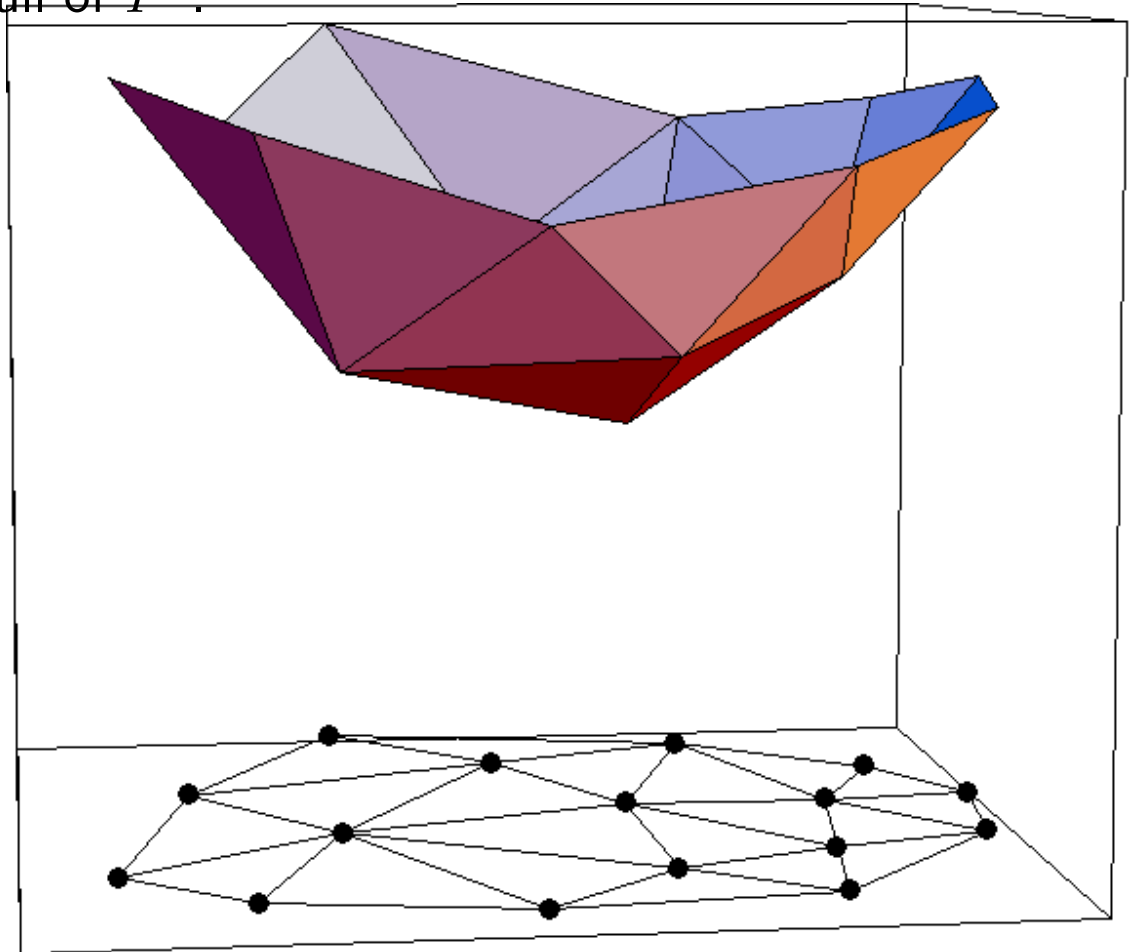


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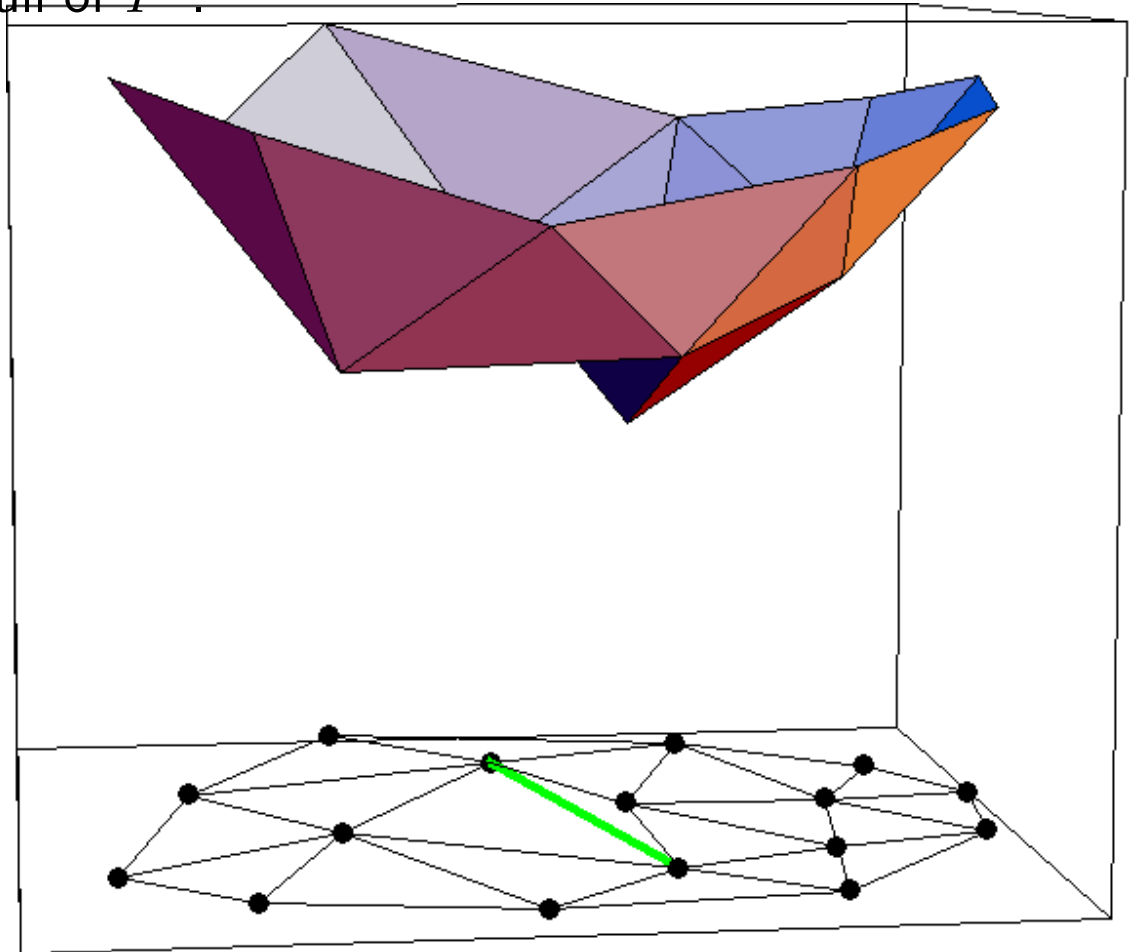


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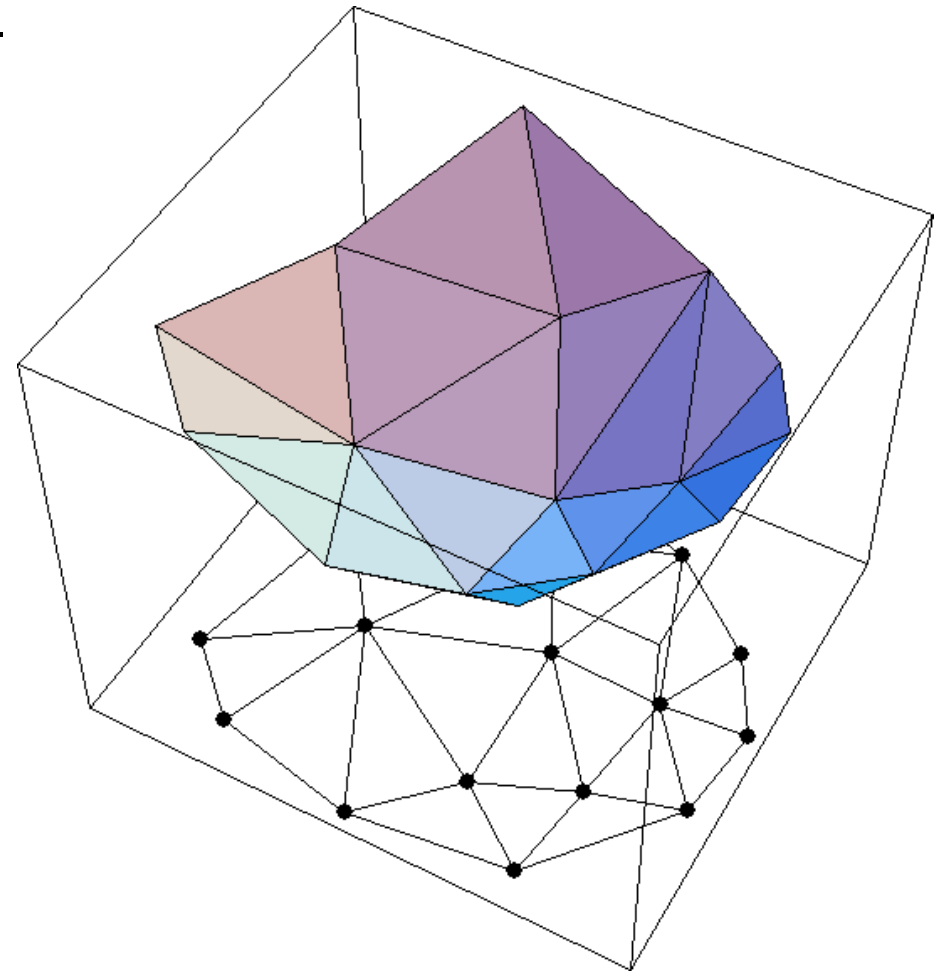


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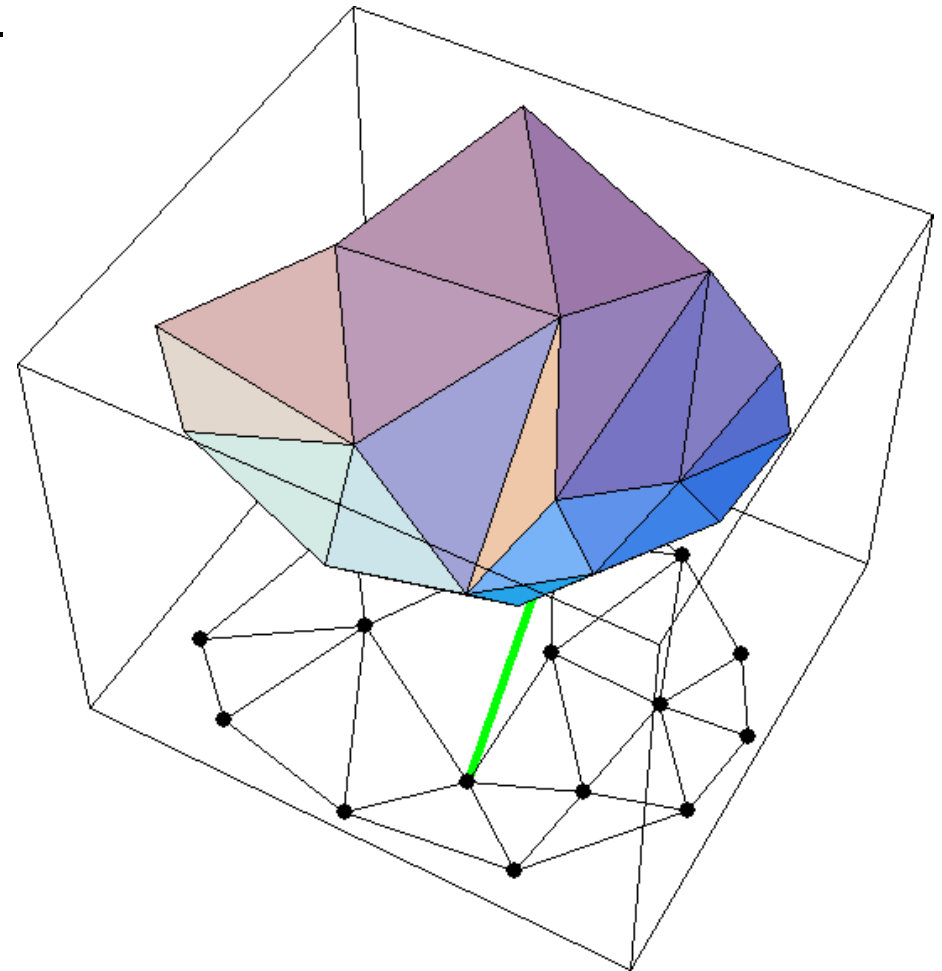


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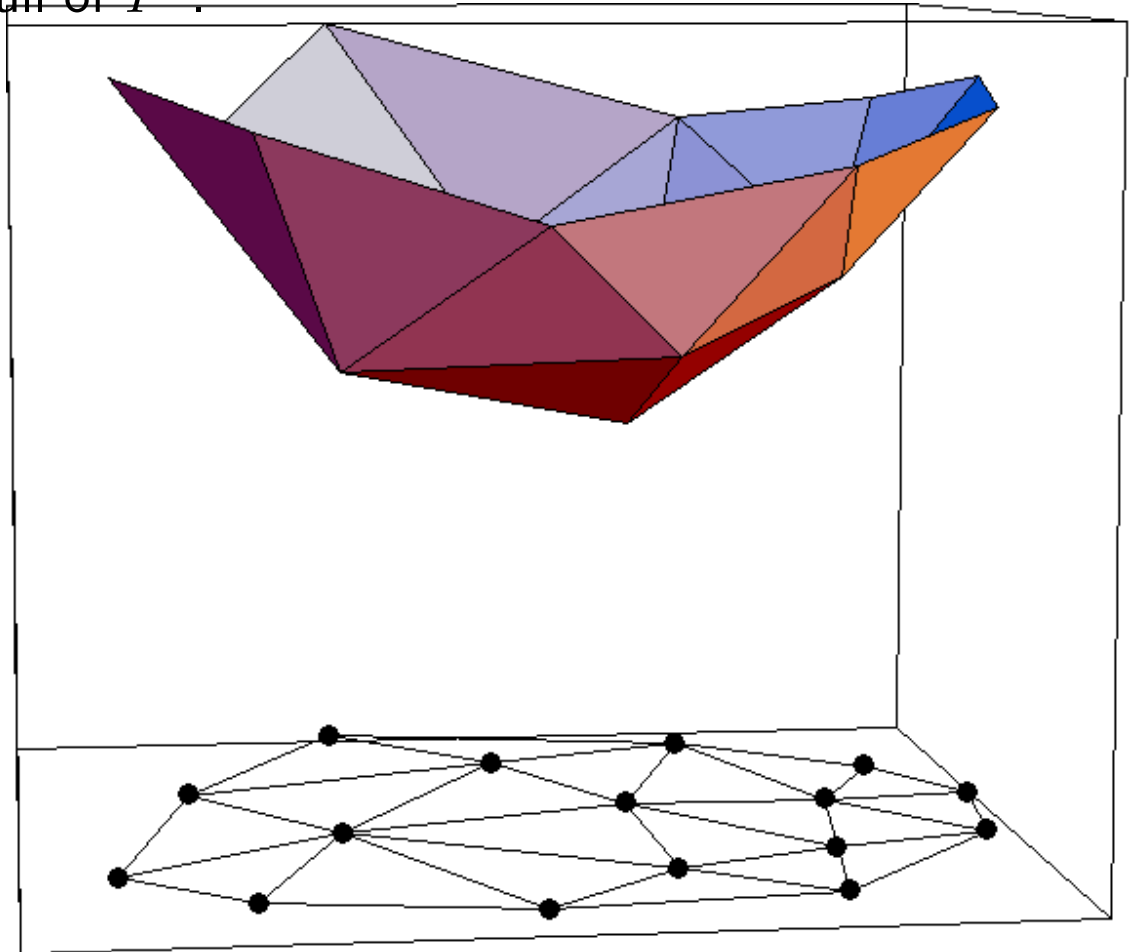


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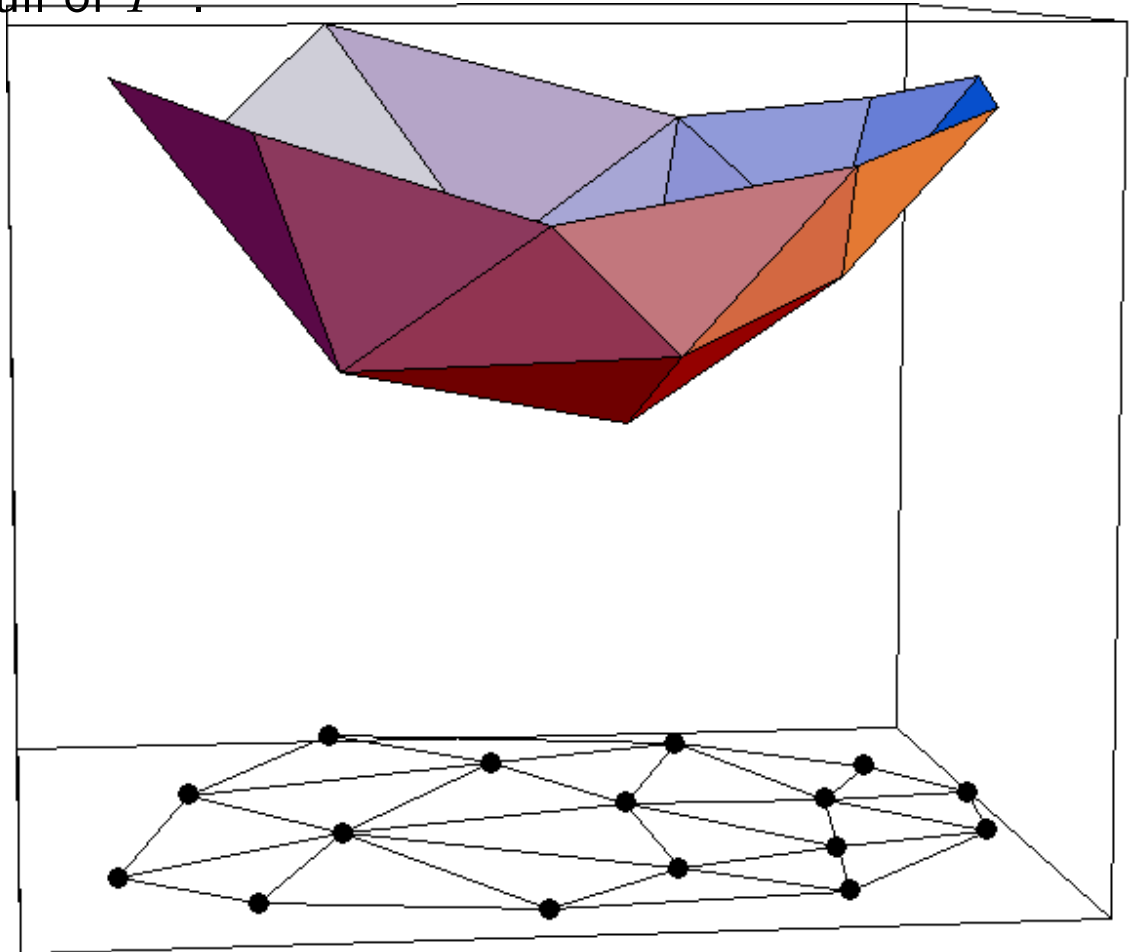
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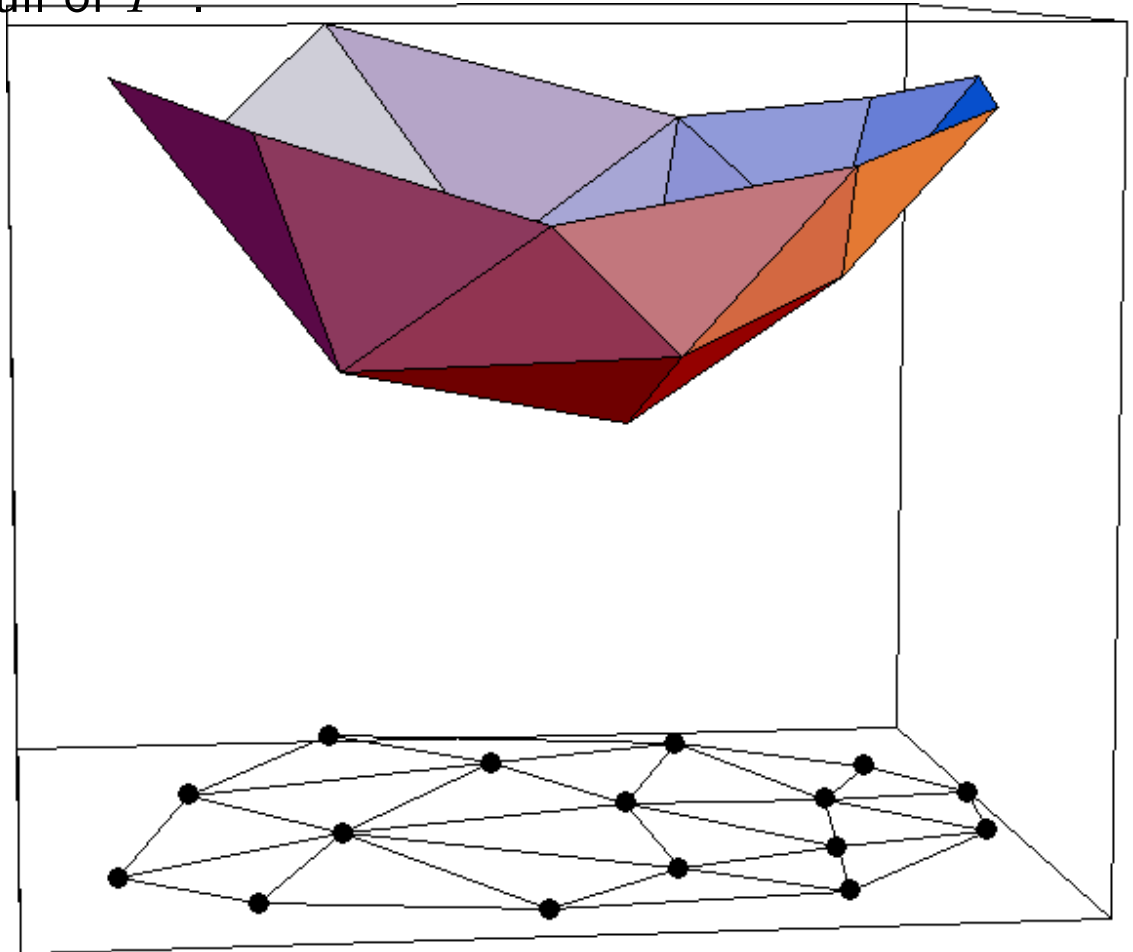
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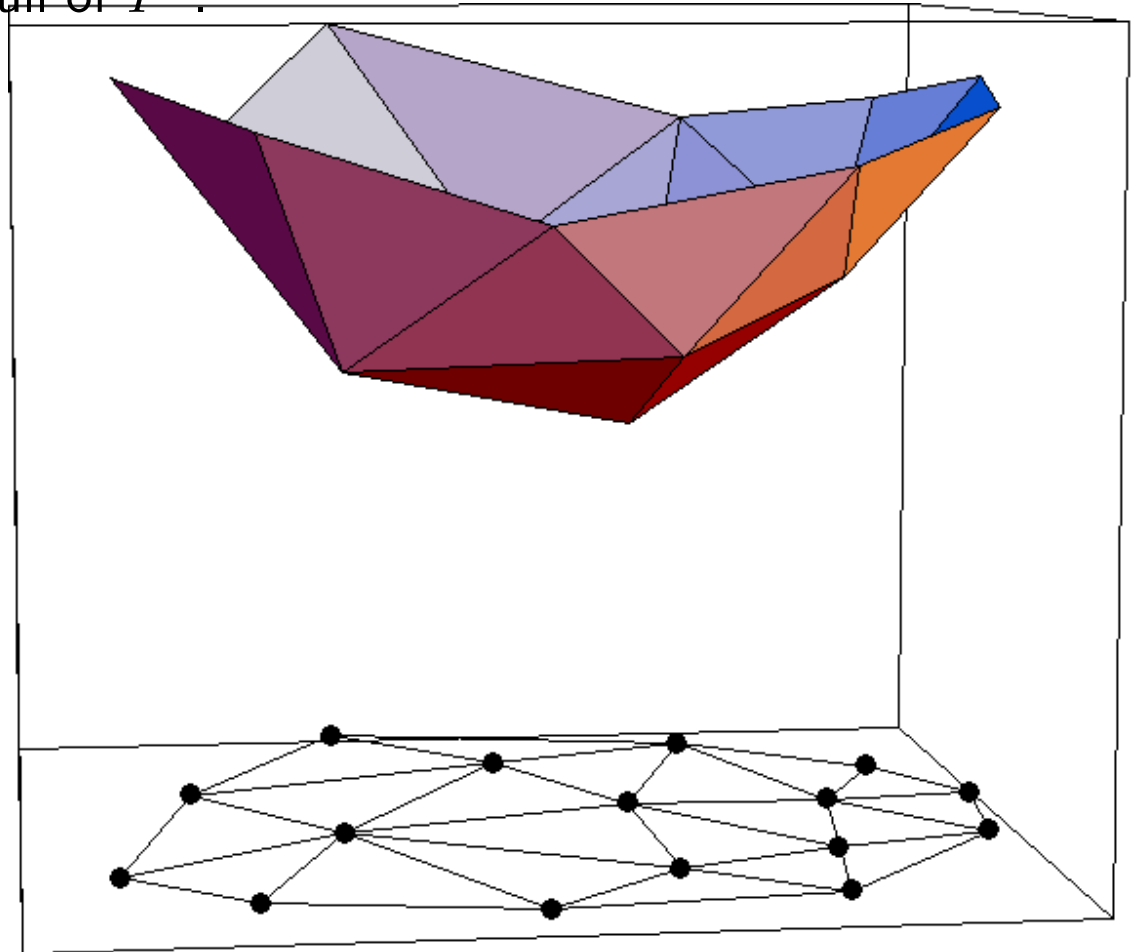
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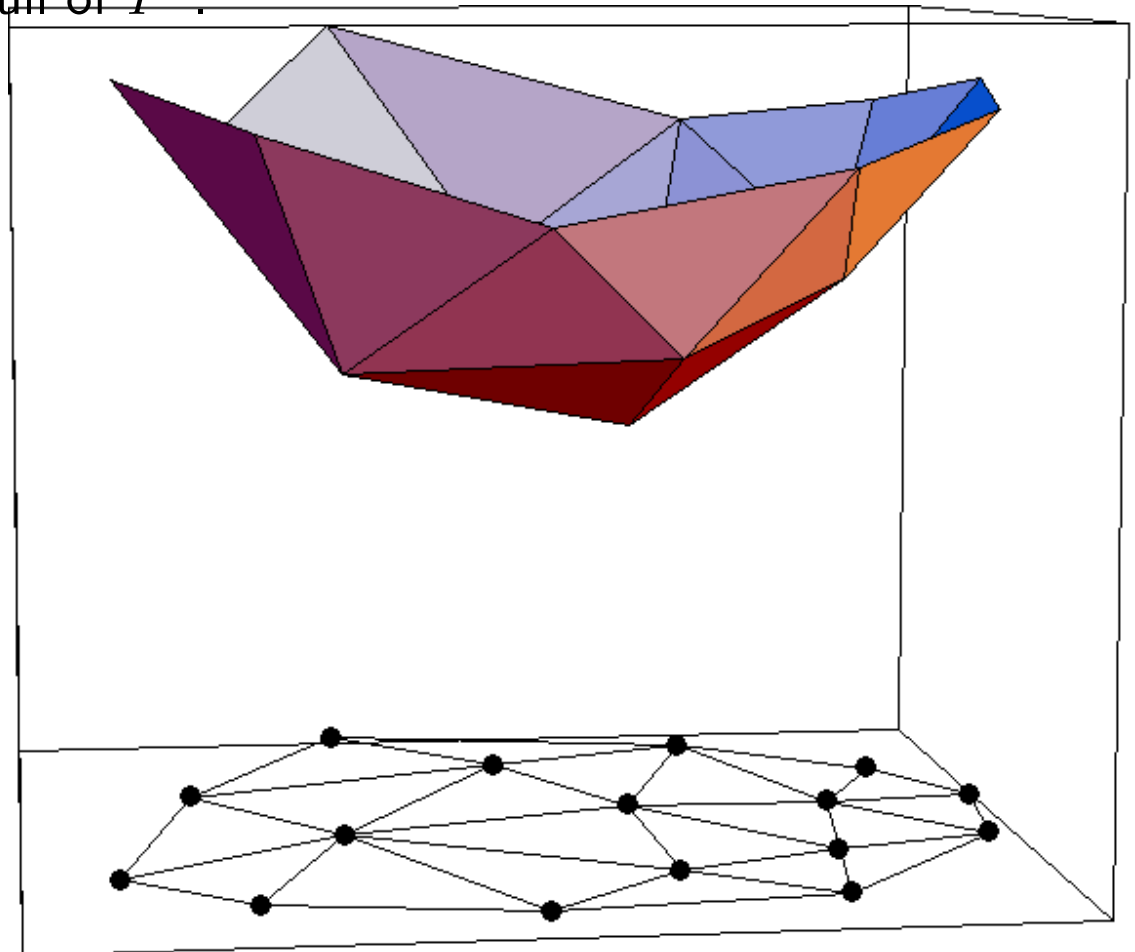
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p_i, p_j, p_k form a triangle of $Del(P)$



DELAUNAY TRIANGULATION

LOCAL CHARACTERIZATION

DELAUNAY TRIANGULATION

LOCAL CHARACTERIZATION

Definition

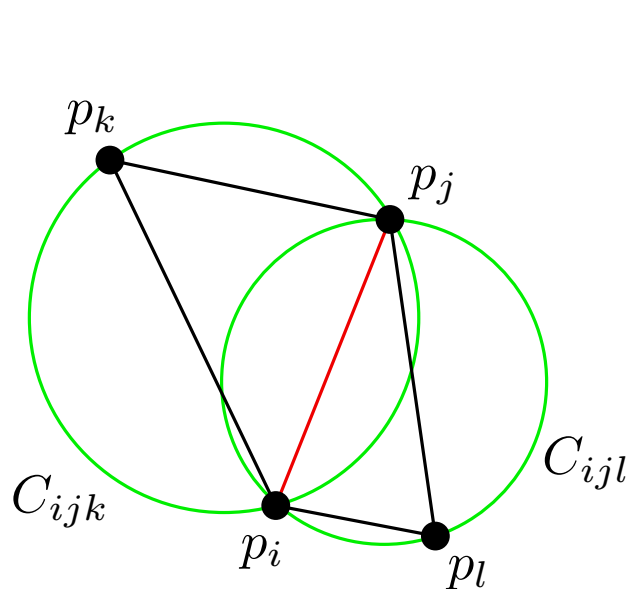
A triangulation $T(P)$ is **locally Delaunay** if each pair of triangles $p_i p_j p_k$ and $p_i p_j p_l$ sharing an edge $p_i p_j$ satisfies $p_l \notin C_{ijk}$ and $p_k \notin C_{ijl}$ (C_{ijk} denotes the circle through p_i , p_j and p_k)

DELAUNAY TRIANGULATION

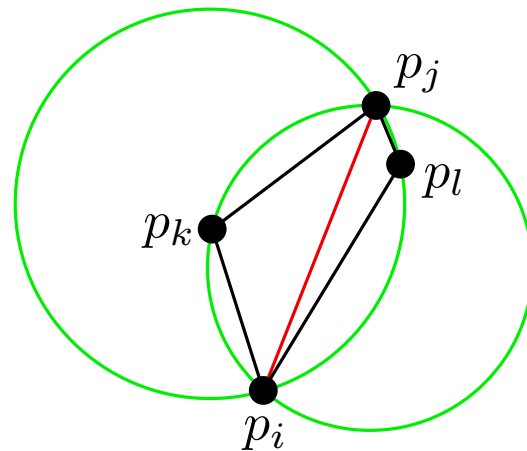
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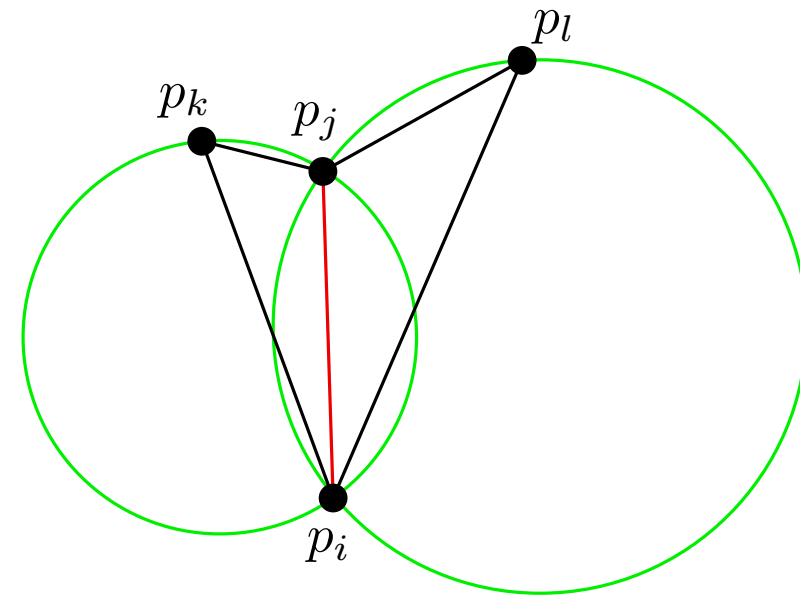
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The edge $p_i p_j$ **is**
locally Delaunay



The edge $p_i p_j$ **is not**
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The edge $p_i p_j$ **is**
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Note that quadrilateral
 $p_i p_l p_j p_k$ is not convex

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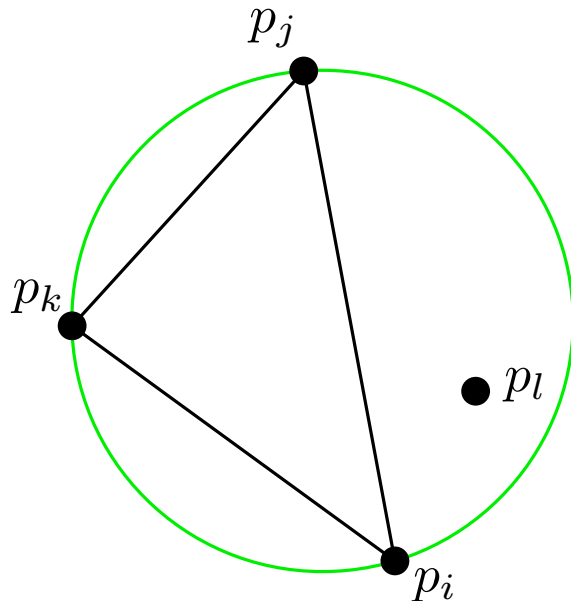
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DELAUNAY TRIANGULATION

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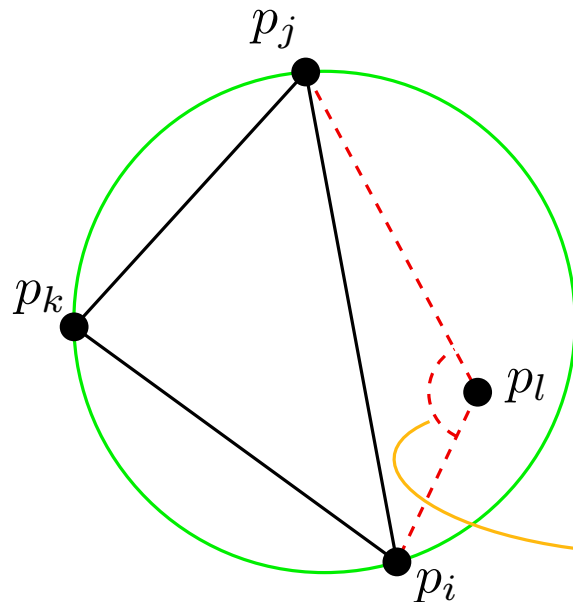
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Let $\overline{p_i p_j}$ be the edge of T_{ijk} separating p_l from T_{ijk} . Among all 4-tuples in this situation, let $ijkl$ maximize the angle $p_i p_l p_j$.



Assumption: i, j, k, l achieve the maximum angle among all groups of four points in this situation

DELAUNAY TRIANGULATION

LOCAL CHARACTERIZATION

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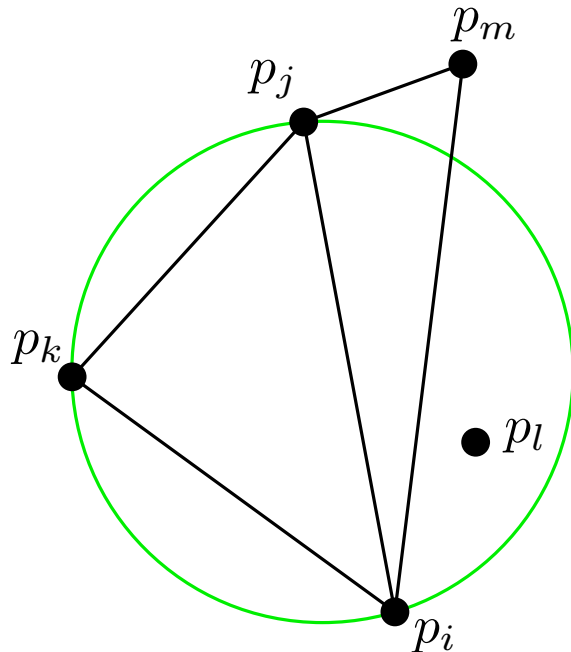
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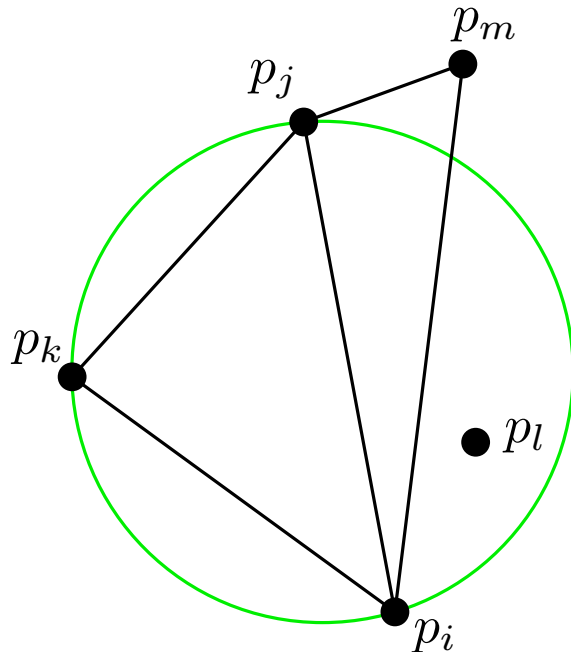
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Since $T(P)$ is locally Delaunay, $m \neq l$.

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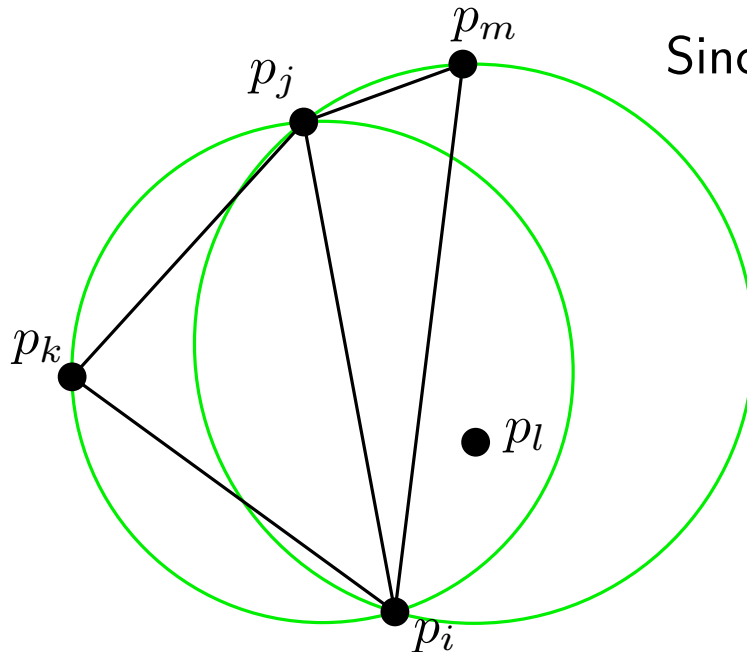
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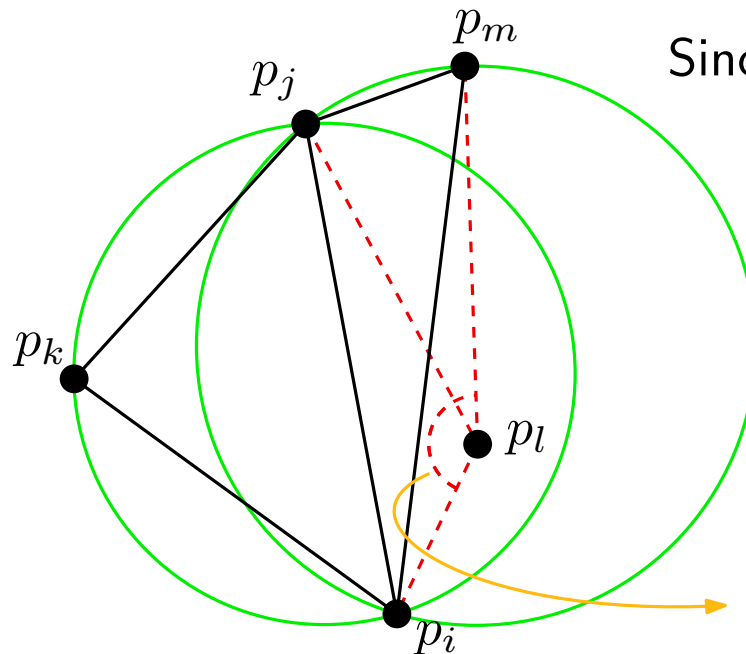
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Then $p_l \in C_{ijm}$.

Hence, one of the angles $p_i p_l p_m$ or $p_j p_l p_m$ would be greater than $p_i p_l p_j$.
Contradiction with assumption!

Assumption: i, j, k, l achieve the maximum angle among *all* groups of four points in this situation

DELAUNAY TRIANGULATION

DELAUNAY FLIPS

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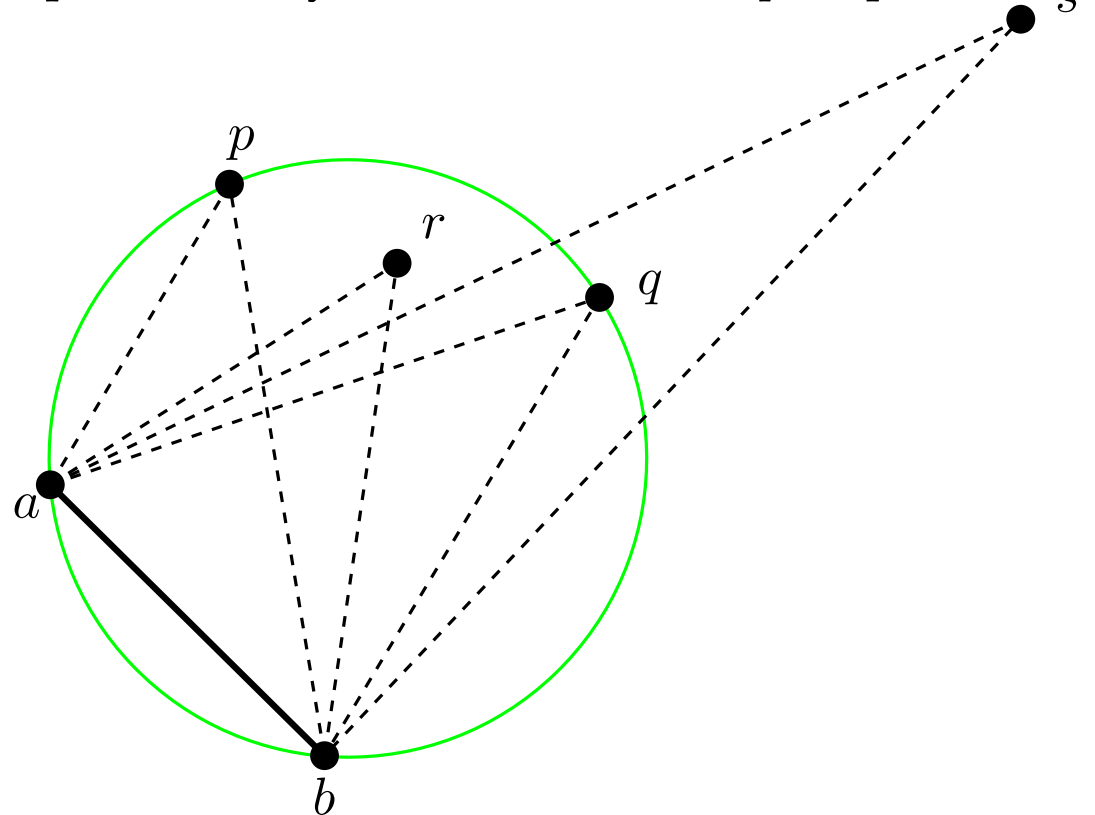
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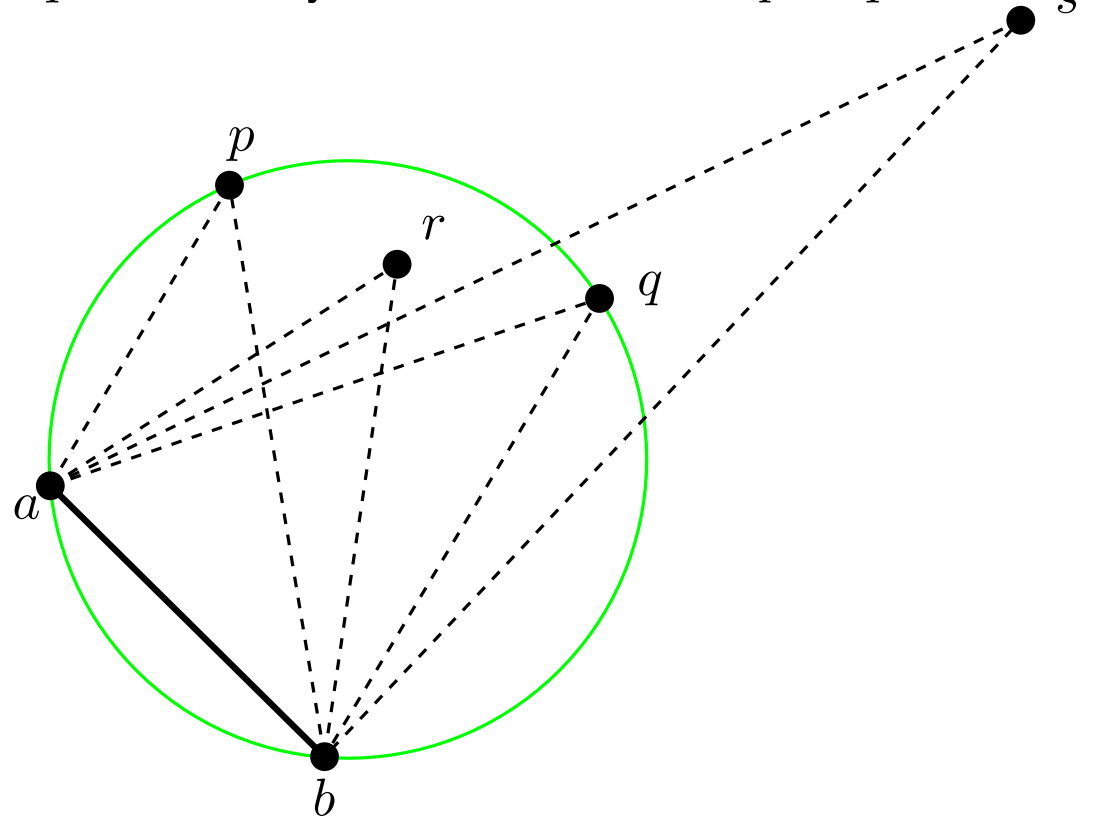
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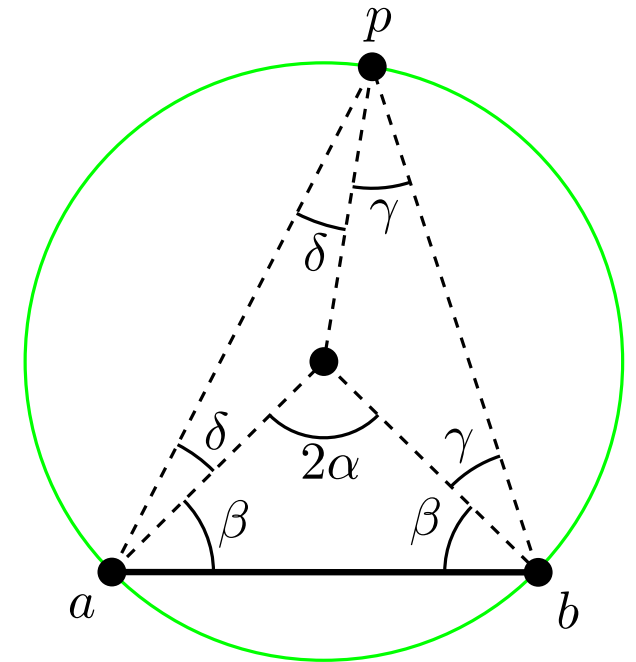
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First case (center of C inside $\triangle abp$):

$$\left. \begin{array}{l} 2\delta + 2\gamma + 2\beta = \pi \\ 2\alpha + 2\beta = \pi \end{array} \right\} \Rightarrow 2\alpha = 2\gamma + 2\delta \Rightarrow \alpha = \gamma + \delta \Rightarrow \hat{p} = \hat{q} = \alpha$$



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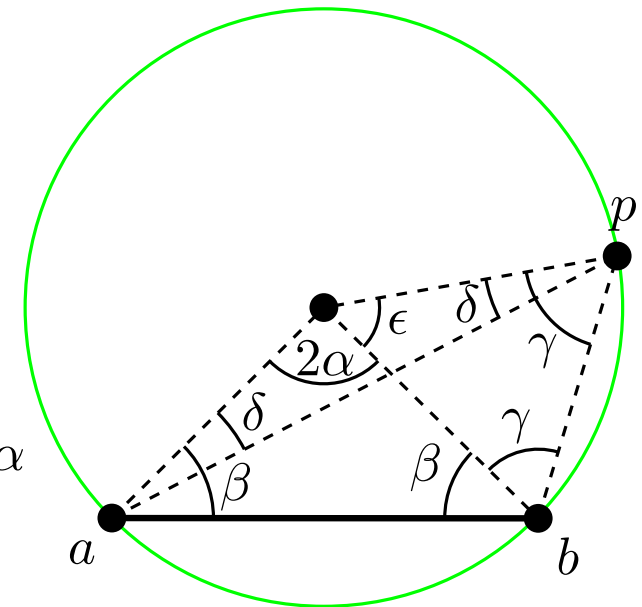
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Second case (center of C outside $\triangle abp$):

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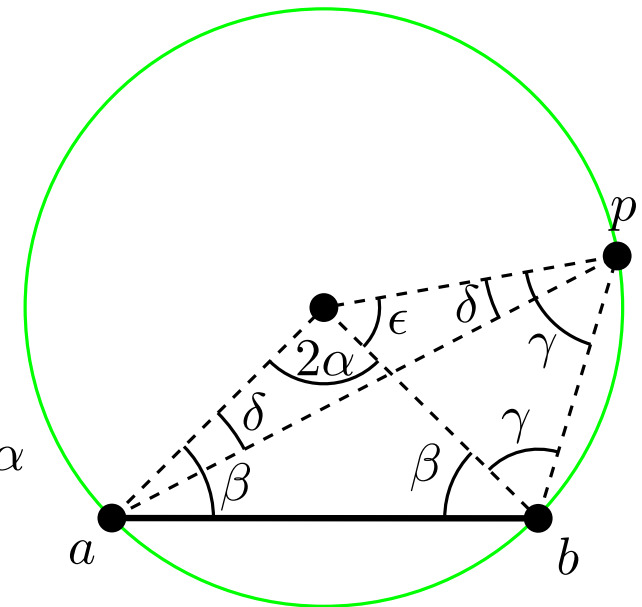
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FYI: the fact that $\hat{p} = \alpha$, is a classic result known as the *Inscribed Angle Theorem* (dating back to Euclid!)

DELAUNAY TRIANGULATION

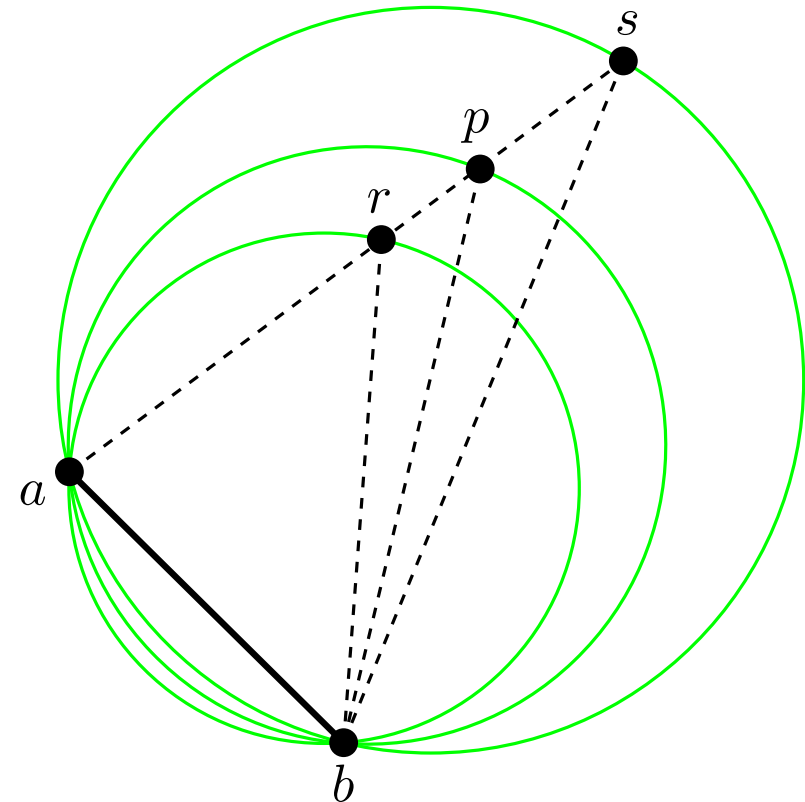
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The remaining relations follow immediately



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Lemma 2 [Thale's theorem]. If $\overline{ab}$ is a diameter of C , the angle $\hat{p}$ for any $p \in C$ is $\pi/2$.

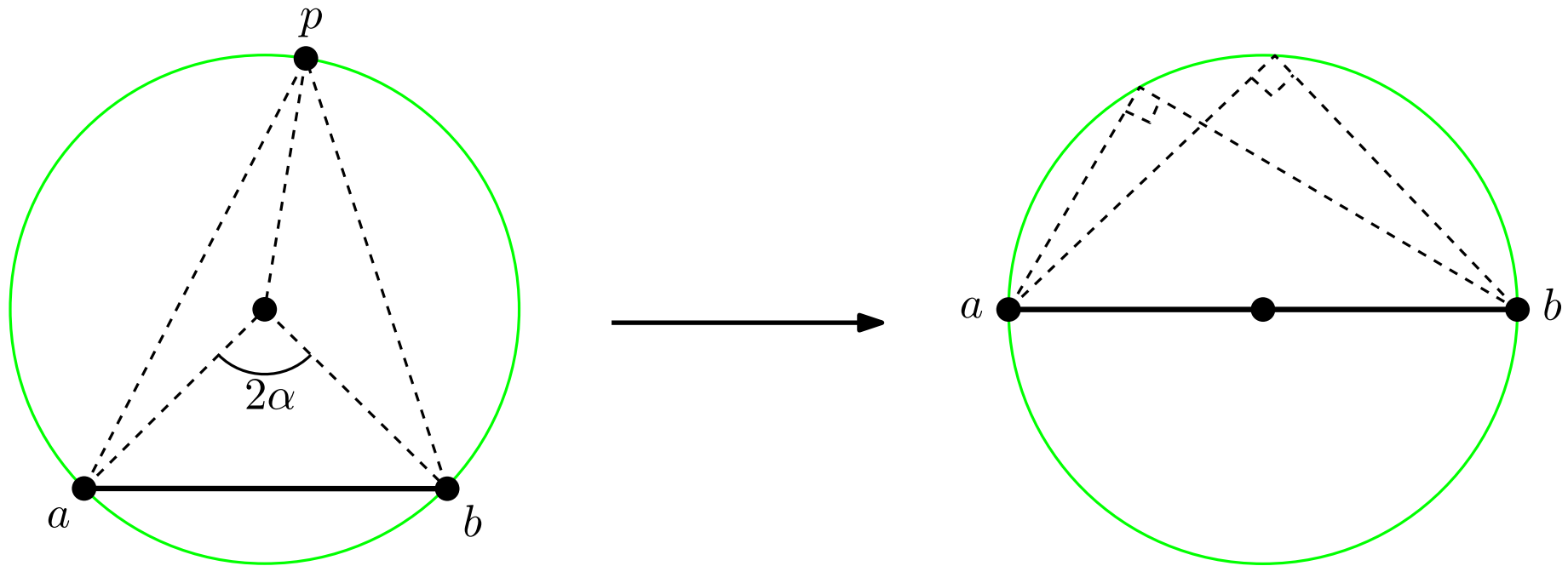
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We will see that $Del(P)$ can be obtained from *any* triangulation of P by **Delaunay flips**, which consist in deleting the diagonal of a convex quadrilateral if it is not locally Delaunay, and replacing it by the other diagonal of the quadrilateral.

Lemma 2 [Thale's theorem]. If $\overline{ab}$ is a diameter of C , the angle $\hat{p}$ for any $p \in C$ is $\pi/2$.

Since in this case $2\alpha = \pi$.



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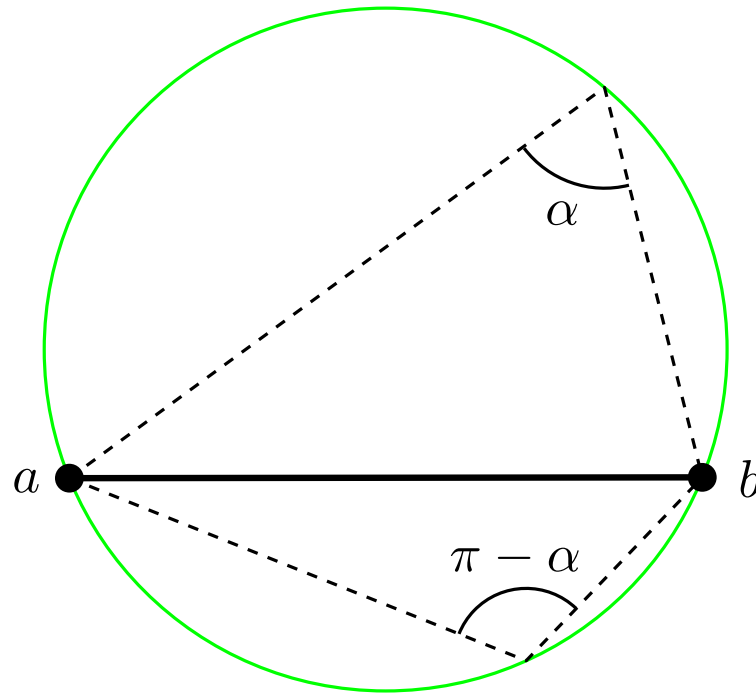
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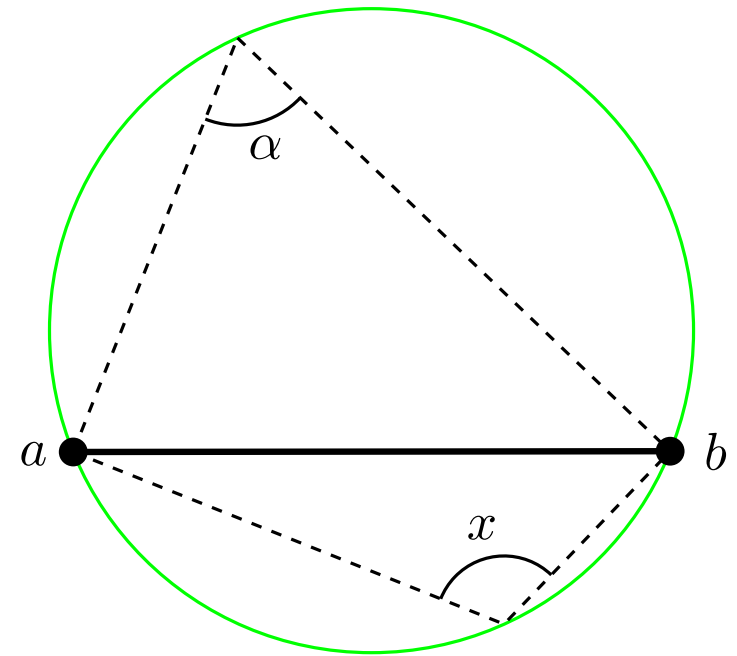
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The *Inscribed Quadrilateral Theorem* (a consequence of the *Inscribed Angle Theorem*) says that a quadrilateral is inscribed in a circle if and only if both pairs of opposite angles add up to π .



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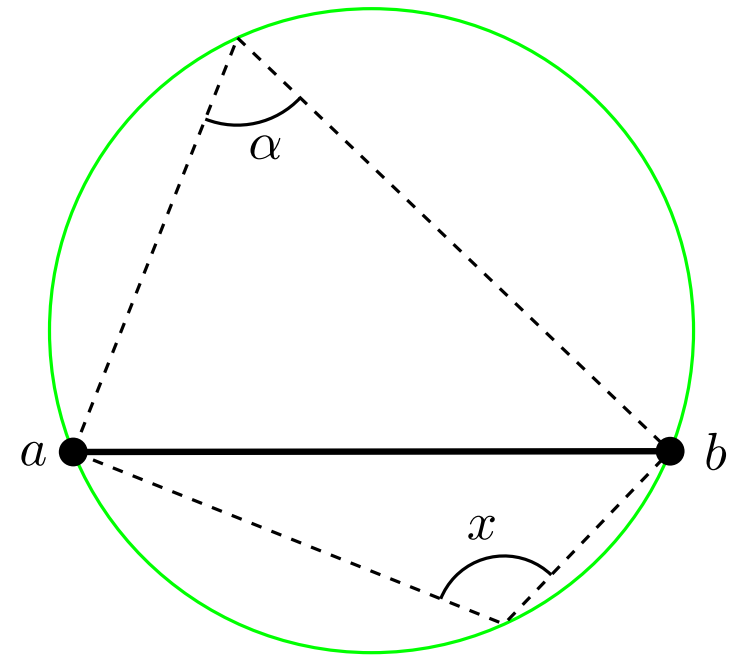
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Our quadrilateral is inscribed in a circle, thus opposite angles, like α and x are supplementary. Thus $x + \alpha = \pi$, and thus $x = \pi - \alpha$.



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Lemma 4. Let $\overline{pq}$ be the common edge of the triangles pqa and pqb , forming a convex quadrilateral. Then:

$$a \in ext(C_{pqb}) \iff b \in ext(C_{pqa})$$

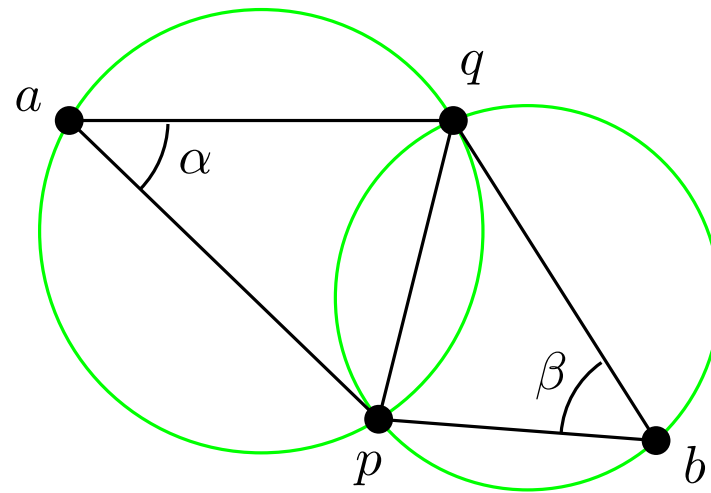
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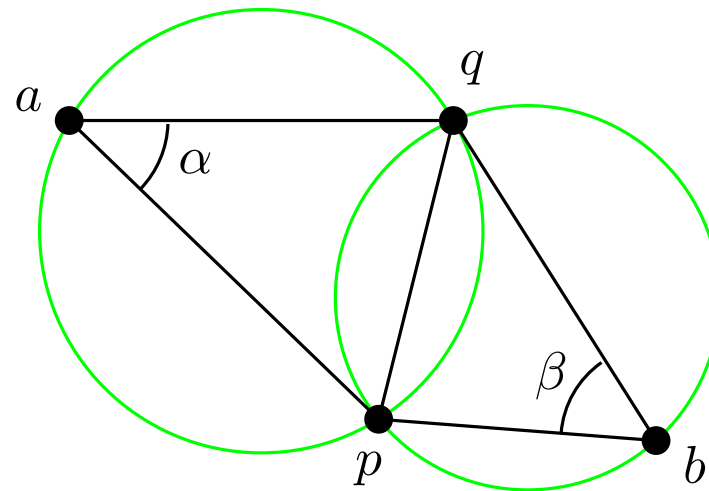
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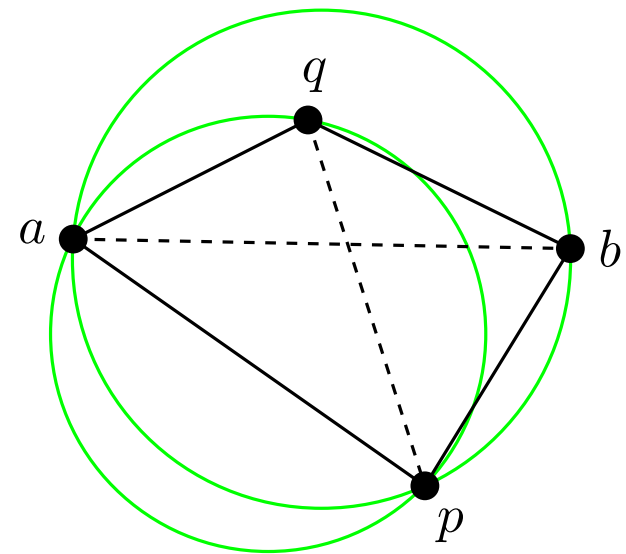
$\overline{ab}$ is not locally Delaunay

$$\iff q \in \text{int}(C_{abp})$$

$$\iff \widehat{aqp} > \widehat{abp}$$

$$\iff b \in \text{ext}(C_{apq})$$

$$\iff \overline{pq} \text{ is locally Delaunay}$$



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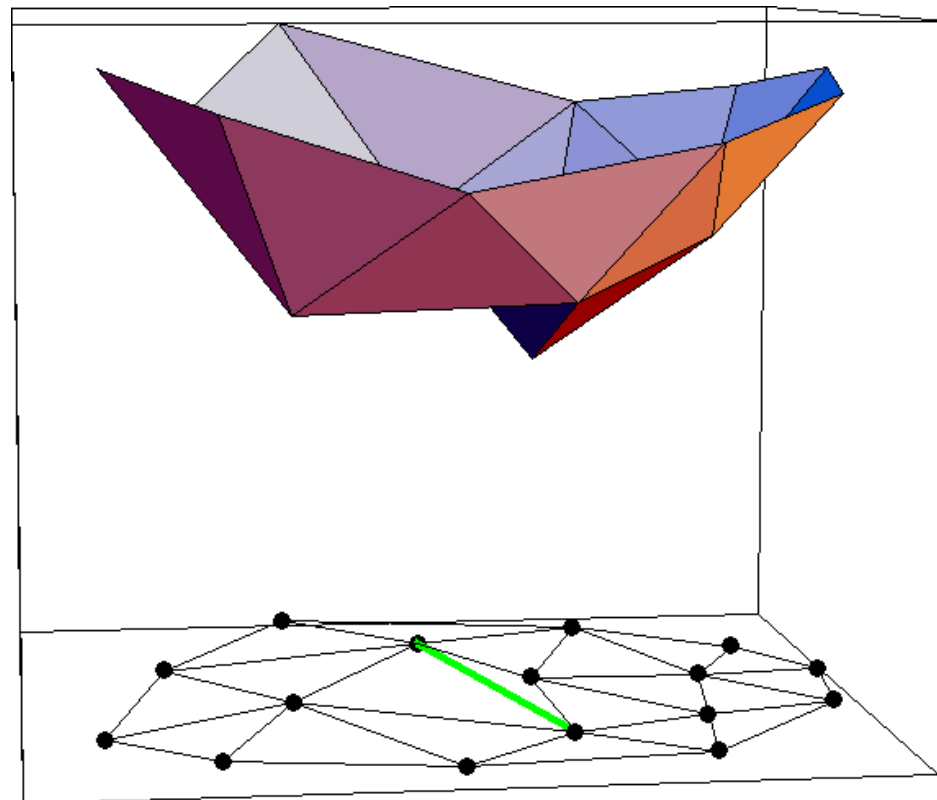
Observation. Let P be a set of points $p_i = (x_i, y_i, 0)$ in the plane, and let P^* be the set of their vertical projections $p^* = (x_i, y_i, x_i^2 + y_i^2)$ onto the unit paraboloid. Producing a Delaunay flip in a triangulation of P corresponds to “sticking” a tetrahedron from below to the corresponding polyhedrization of P^* .

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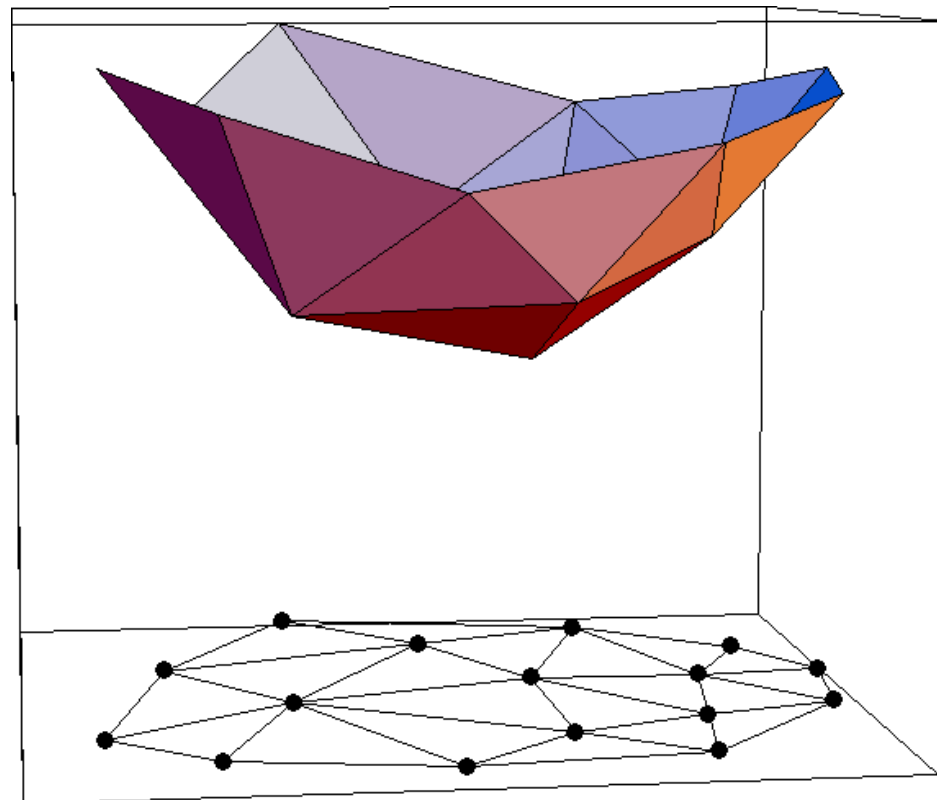


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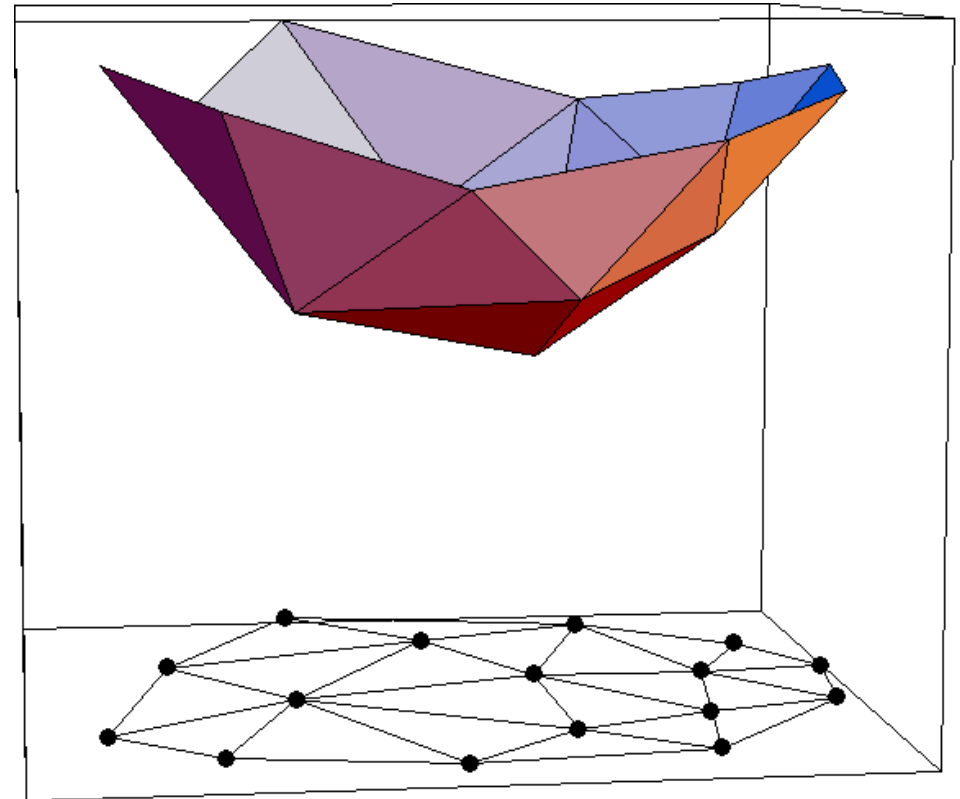
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Once flipped, the quadrilateral is locally Delaunay: the fourth point lies in the exterior of the circumcircle of the triangle.

In the paraboloid, this means that the fourth point lies above the triangular face of the polyhedrization.



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Important Corollary. Given any triangulation of P , performing locally Delaunay flips is a procedure converging to $Del(P)$.

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2. Compute a triangulation, by any of the known methods, and apply Delaunay flips.
3. Incremental algorithm
 Compute an enclosing triangle for $\{p_1, \dots, p_n\}$
 Compute $Del(p_1, \dots, p_{i+1})$ from $Del(p_1, \dots, p_i)$

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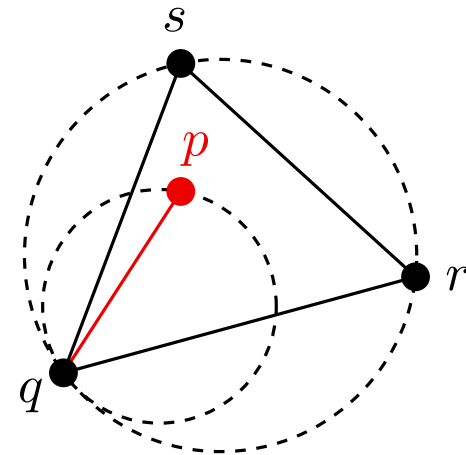
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As C_{qrs} is empty, there exist empty circles C_{pq} , such as the circle through p and q tangent to C_{qrs} in q . Similarly for r and s .



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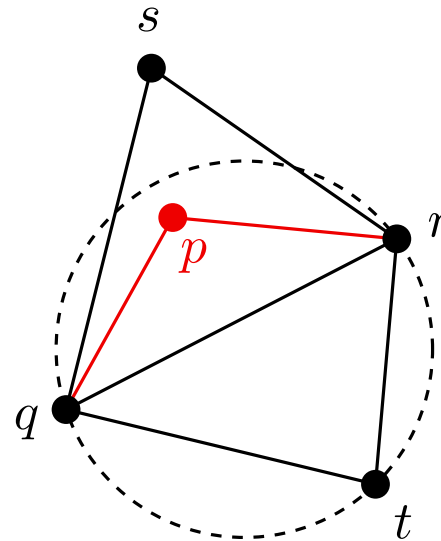
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Since p may lie in the interior of C_{qrt} .



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Obvious, because the property is local: it affects only quadrilaterals formed by two triangles sharing an edge.

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Algorithm

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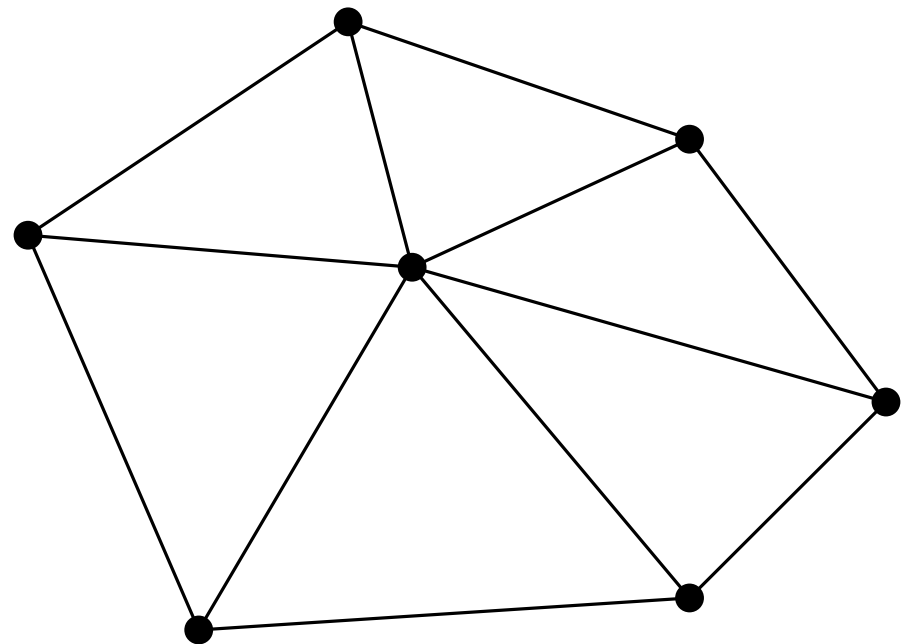
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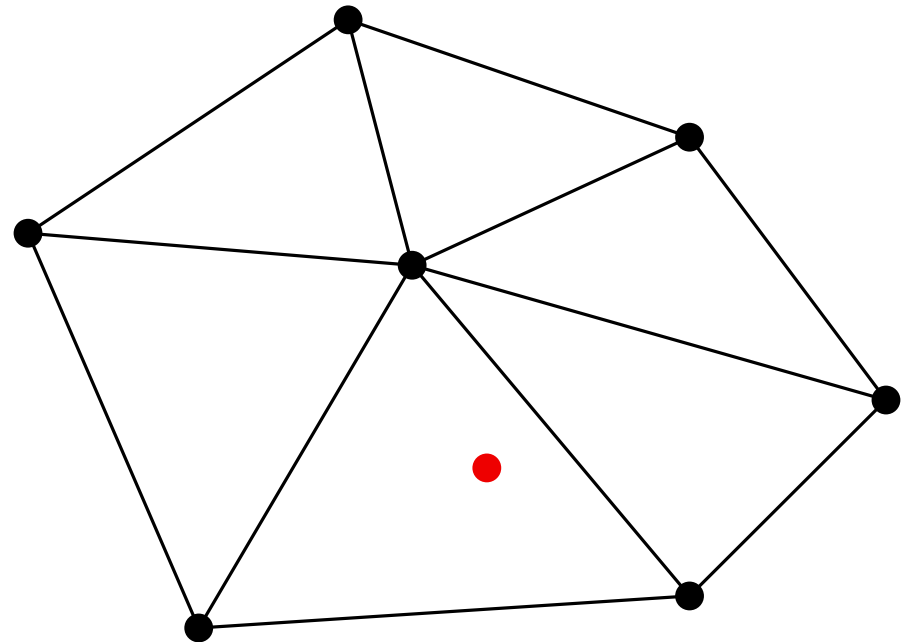
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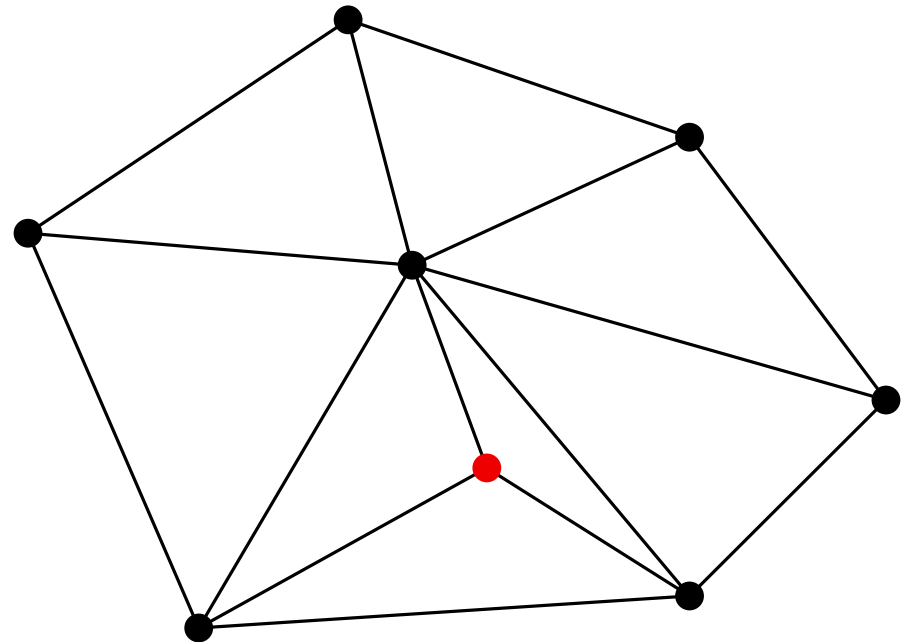
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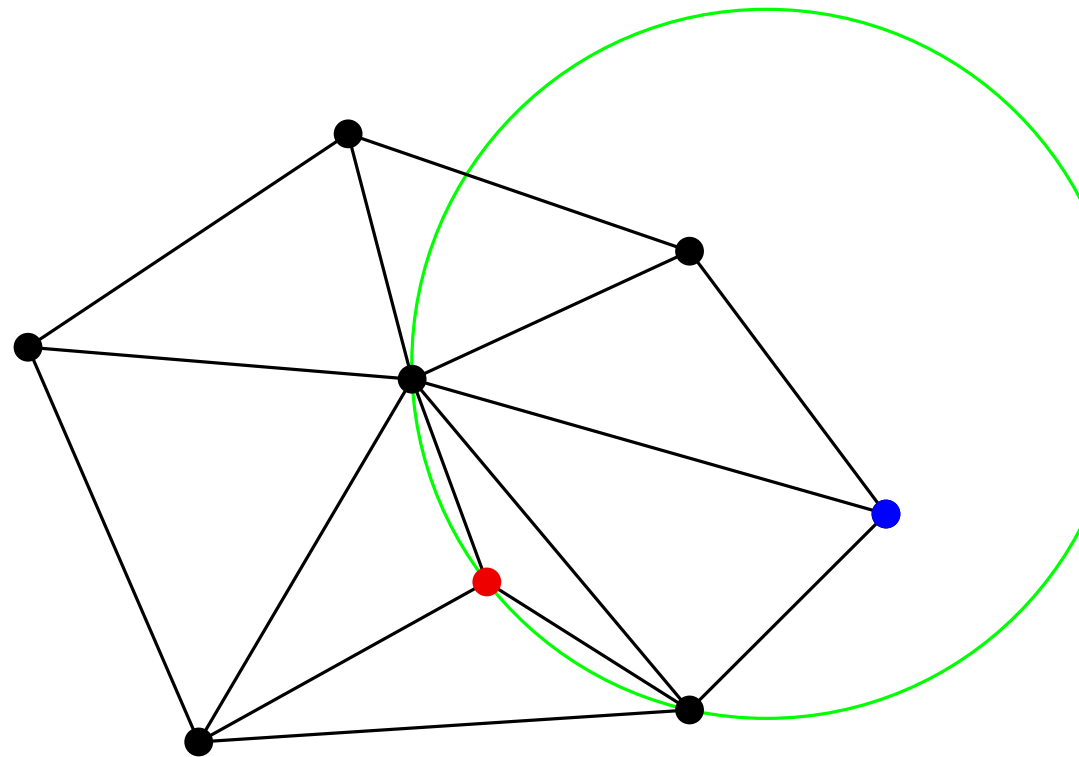
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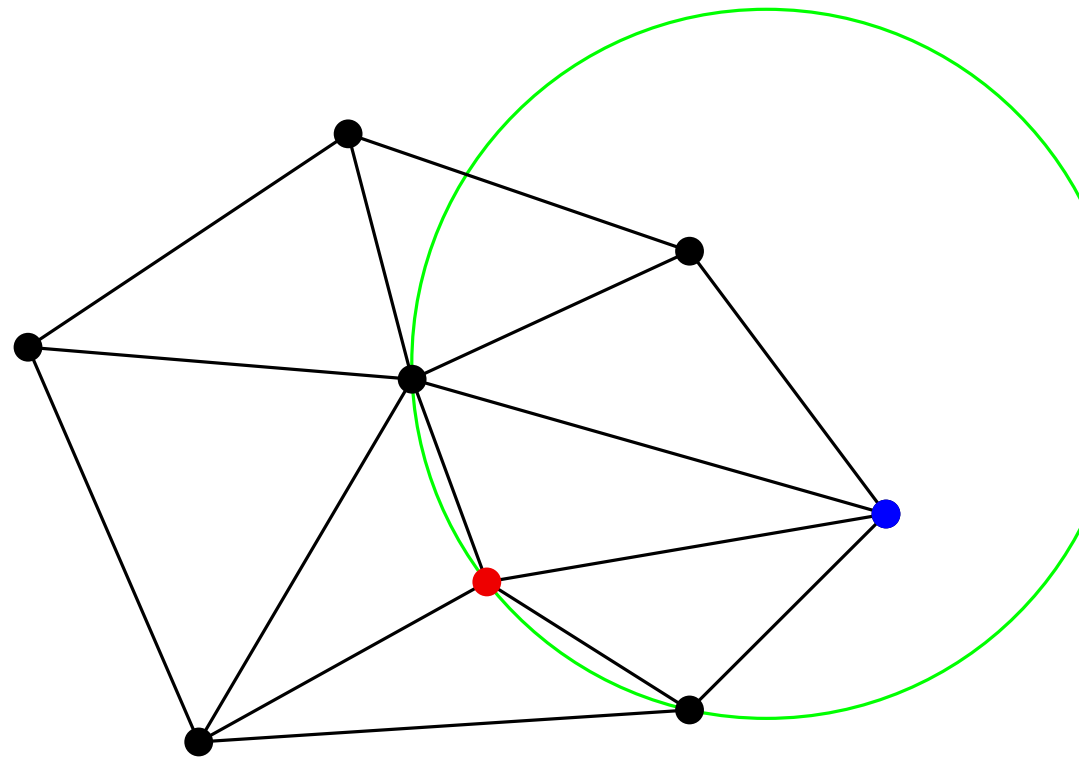
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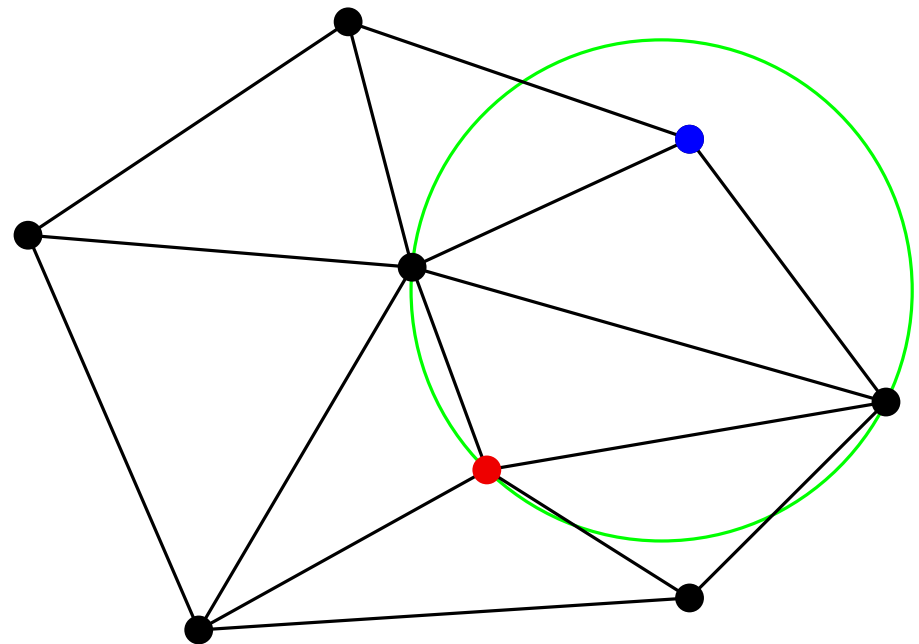
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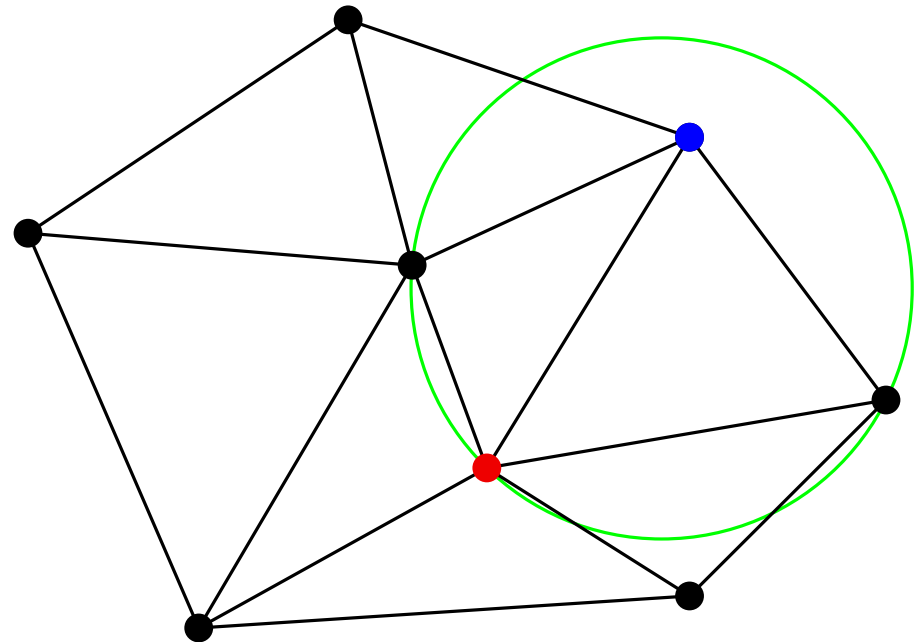
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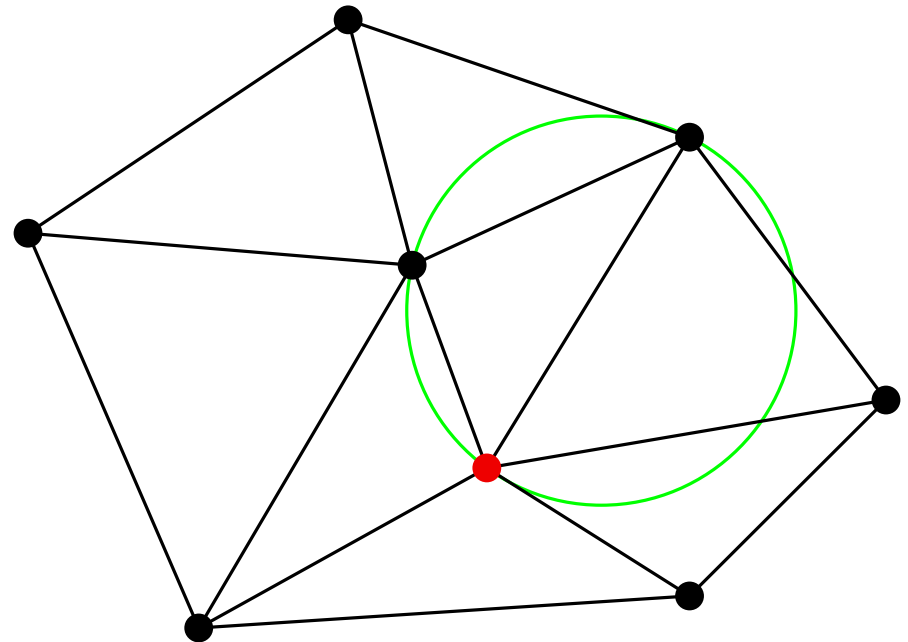
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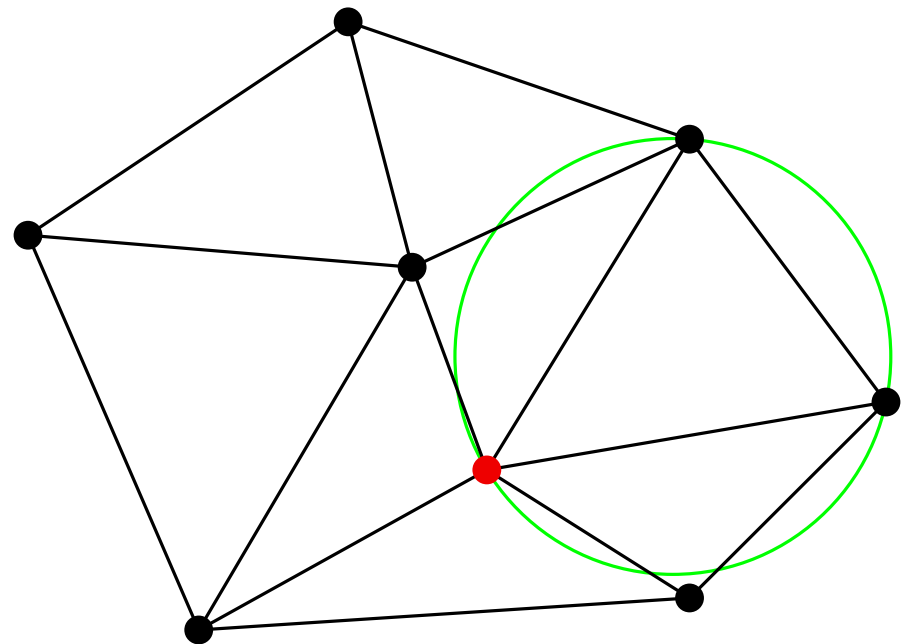
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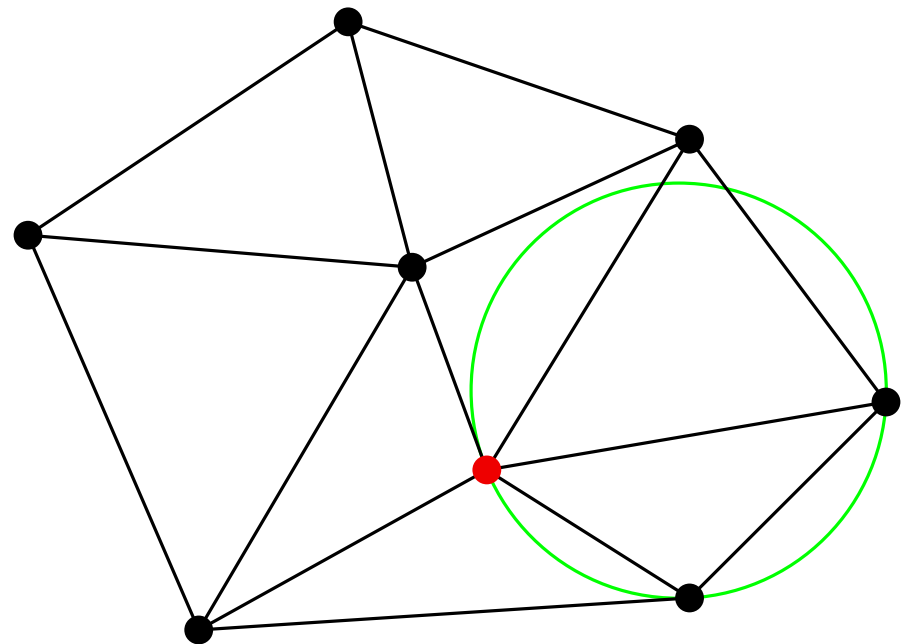
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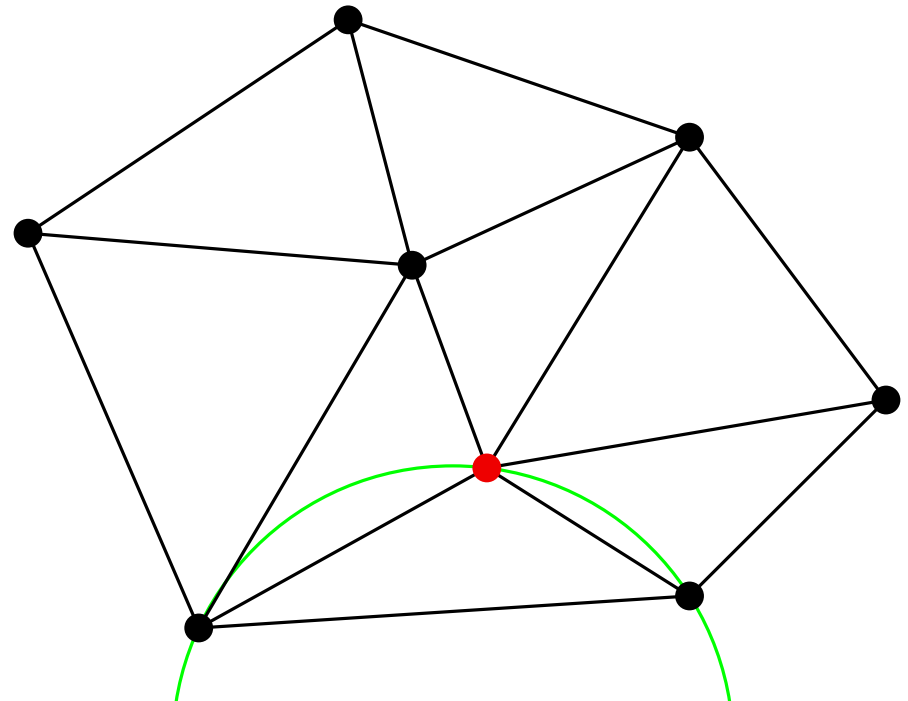
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Each time a new point is added to the triangulation, and before adding the next point, the following routine is executed:

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While there is triangle incident to p not locally Delaunay:
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DELAUNAY TRIANGULATION

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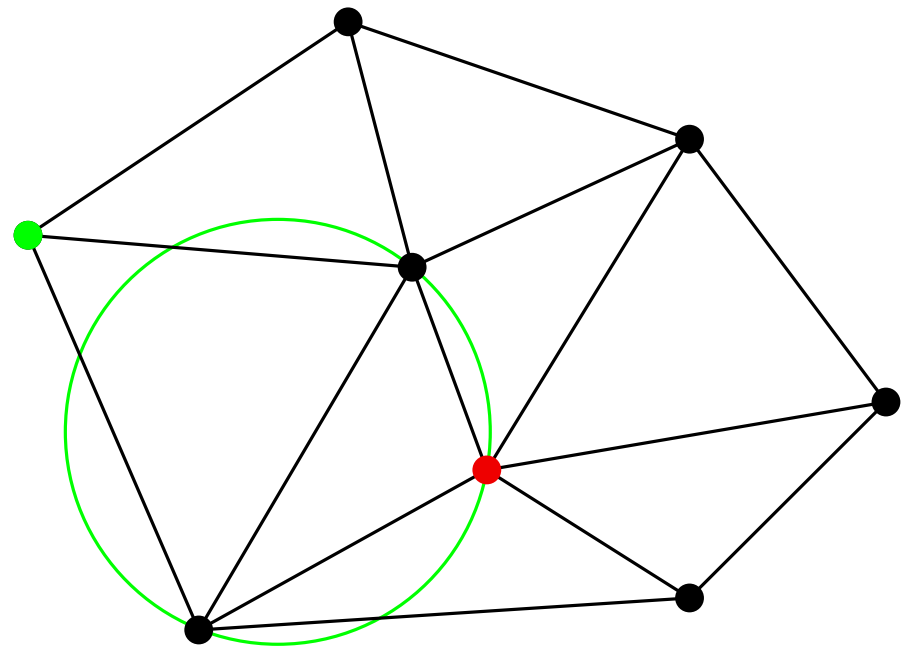
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As the average order is smaller than 6, the expected added running time is not $O(n^2)$ but simply $O(n)$.

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Let us be more precise:

If $\mathcal{T} = \{T_1, \dots, T_t\}$ is a triangulation of P , the **fineness** (a.k.a., *angle vector*) of $\mathcal{T}$ is the increasingly sorted list of the angles of all the triangles T_i of $\mathcal{T}$: $F(\mathcal{T}) = (\alpha_1, \dots, \alpha_{3t})$.

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The proof of this statement requires a last lemma.

DELAUNAY TRIANGULATION

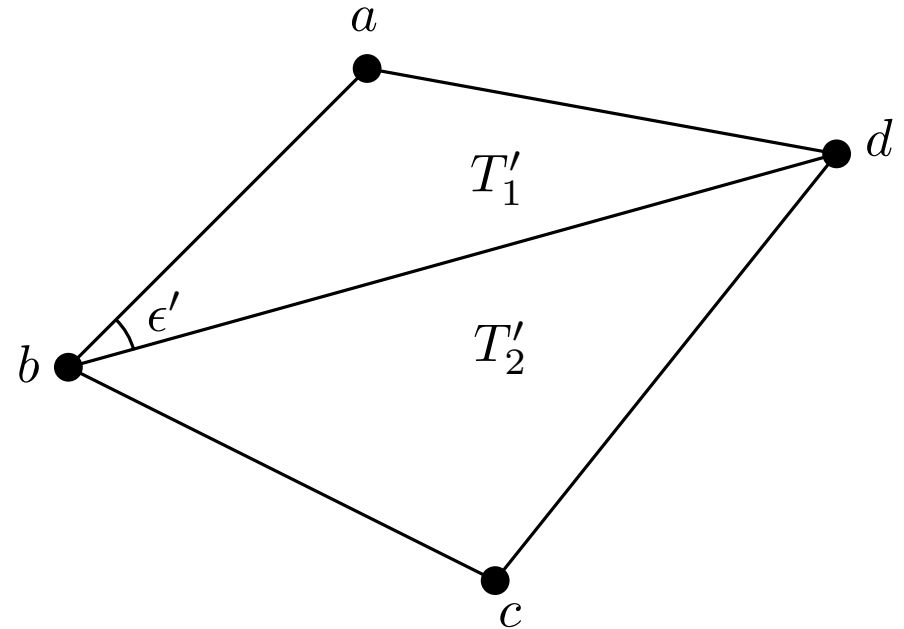
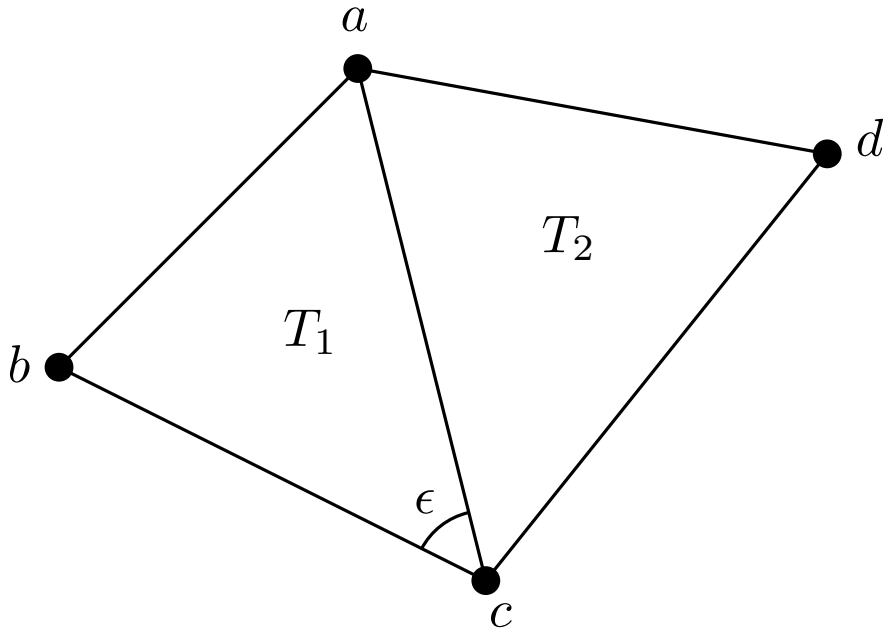
DELAUNAY TRIANGULATION AND EQUIANGULARITY

Lemma 7. Let a, b, c and d be four points forming a convex quadrilateral, in counterclockwise order. Let $\mathcal{T}$ and $\mathcal{T}'$ be the two possible triangulations of the quadrilateral: $\mathcal{T}$ uses the diagonal $\overline{ac}$ and $\mathcal{T}'$ uses $\overline{bd}$. Let ϵ and ϵ' respectively be the minimum angles of $\mathcal{T}$ and $\mathcal{T}'$. Then:

$$\epsilon > \epsilon' \iff d \in \text{ext}(C_{abc})$$

$$\epsilon = \epsilon' \iff d \in \partial(C_{abc})$$

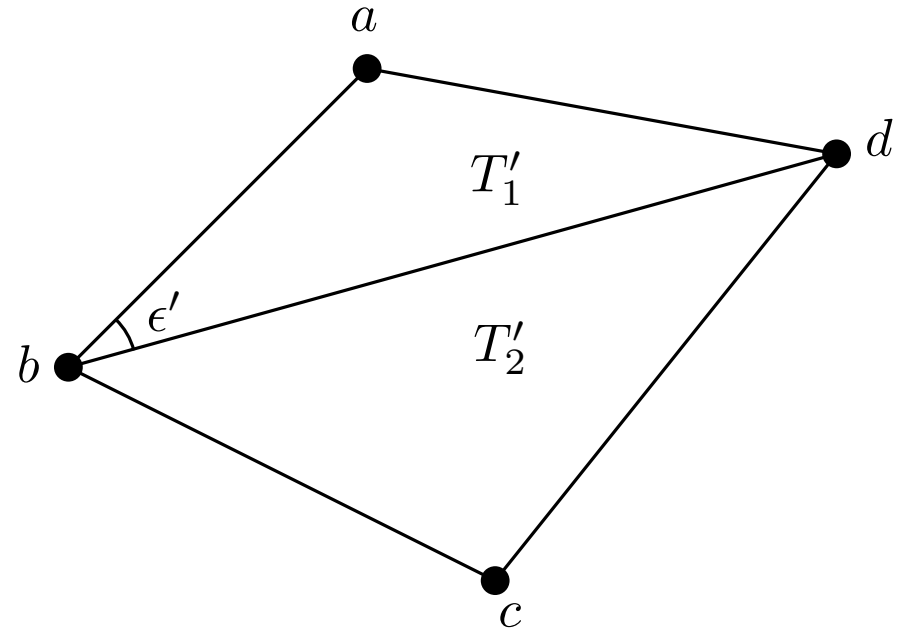
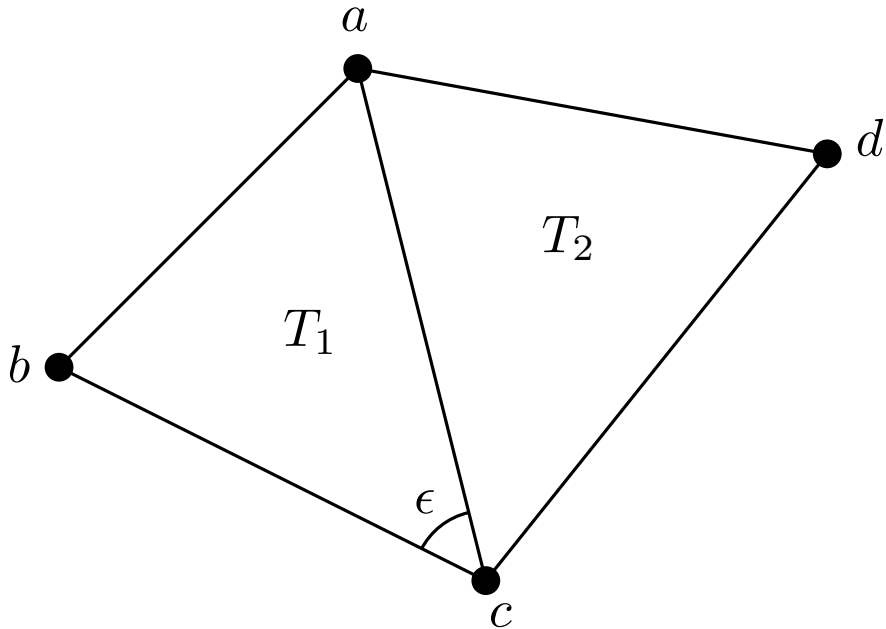
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DELAUNAY TRIANGULATION

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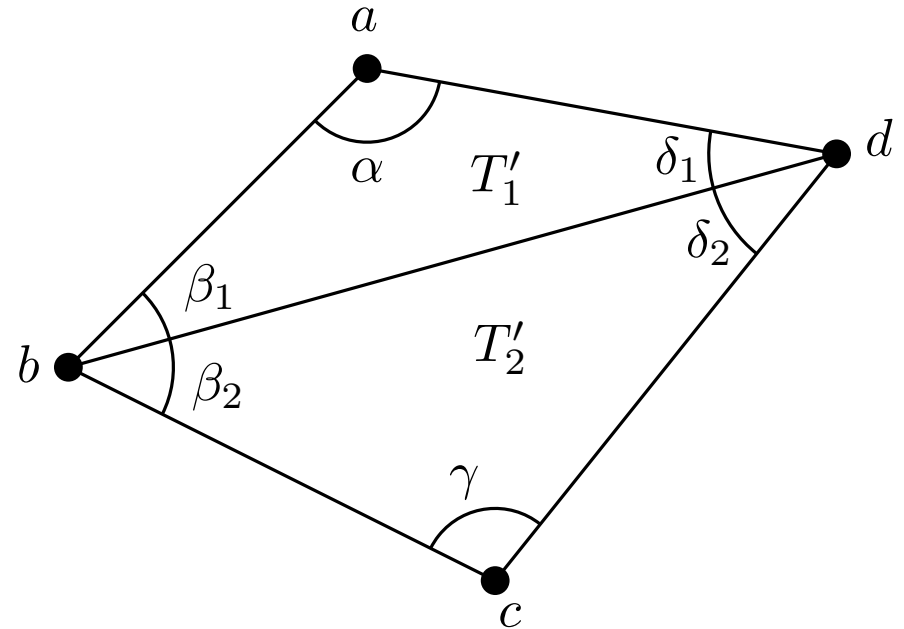
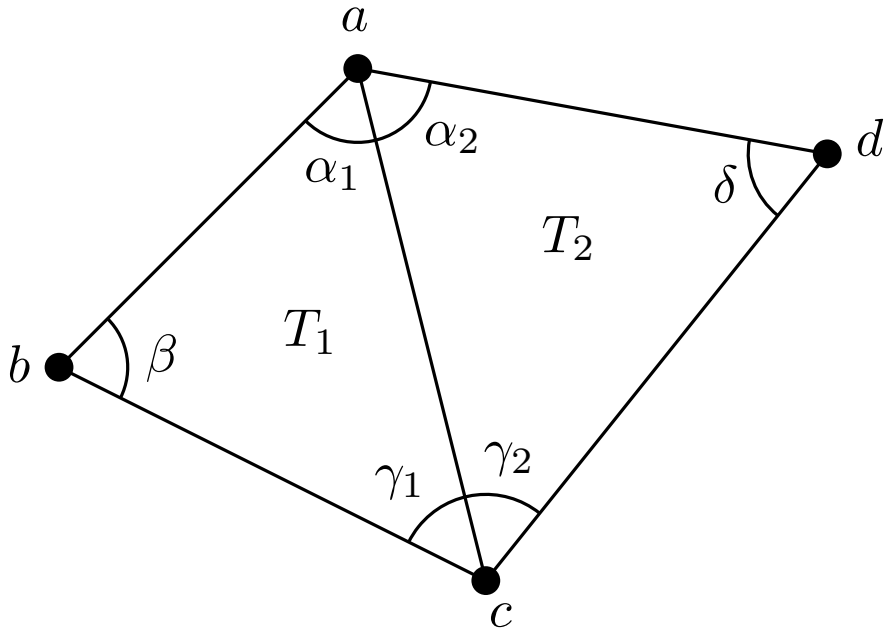


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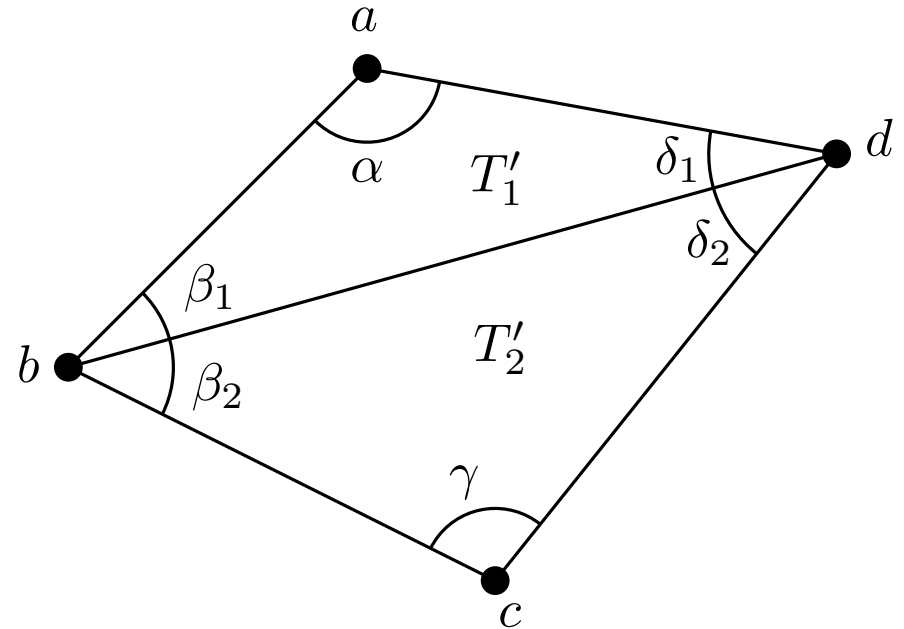
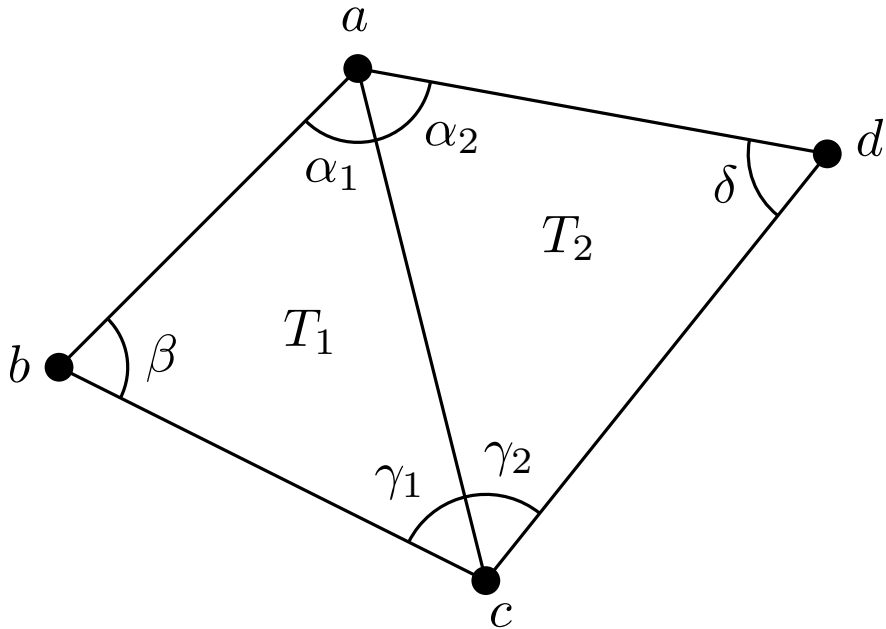
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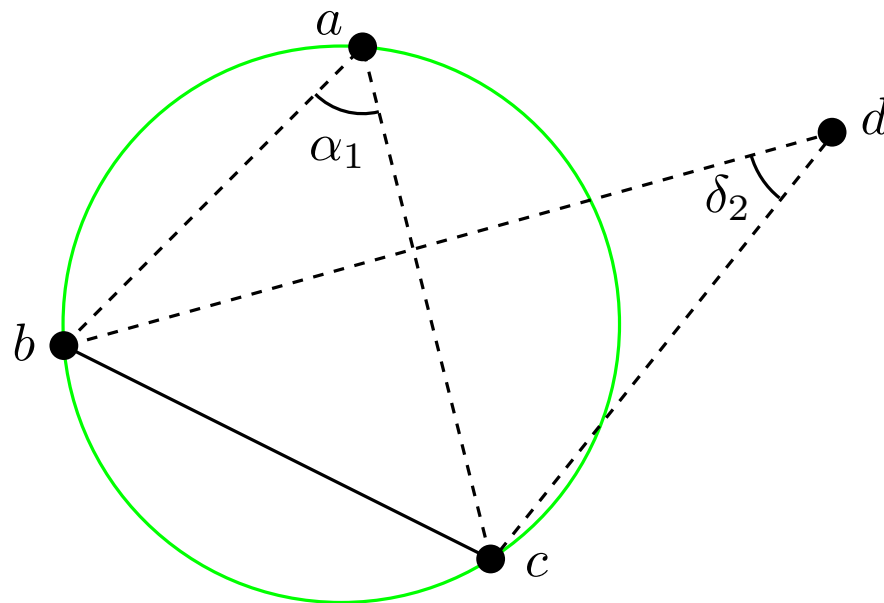
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If $\epsilon' = \delta_2$, then $\delta_2 = \epsilon' < \epsilon \leq \alpha_1$ and, therefore, $d \in \text{ext}(C_{abc})$.



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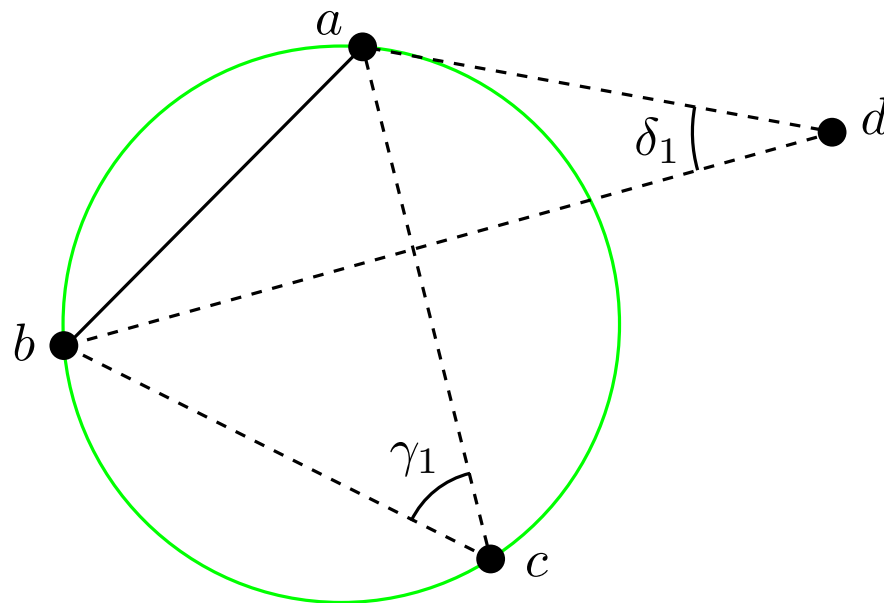
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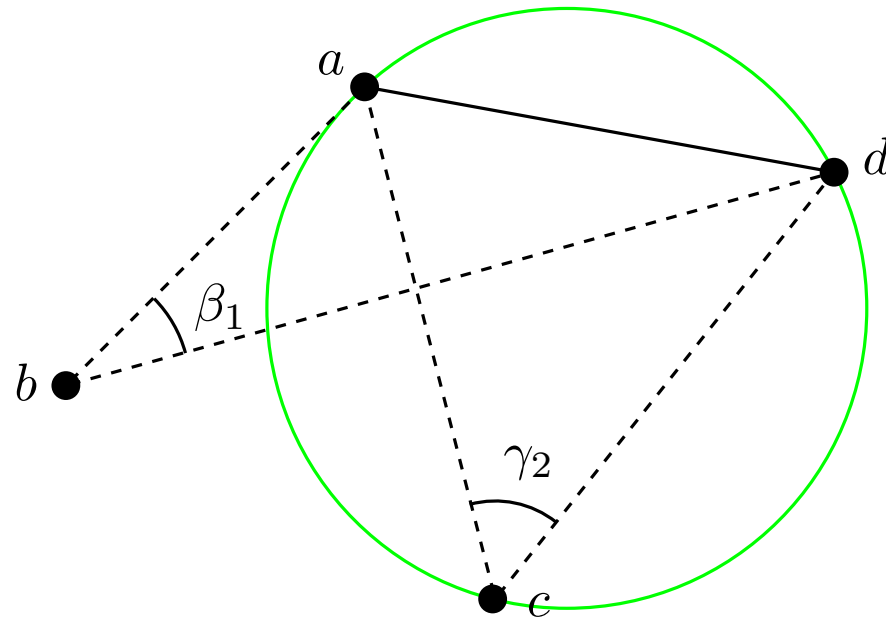
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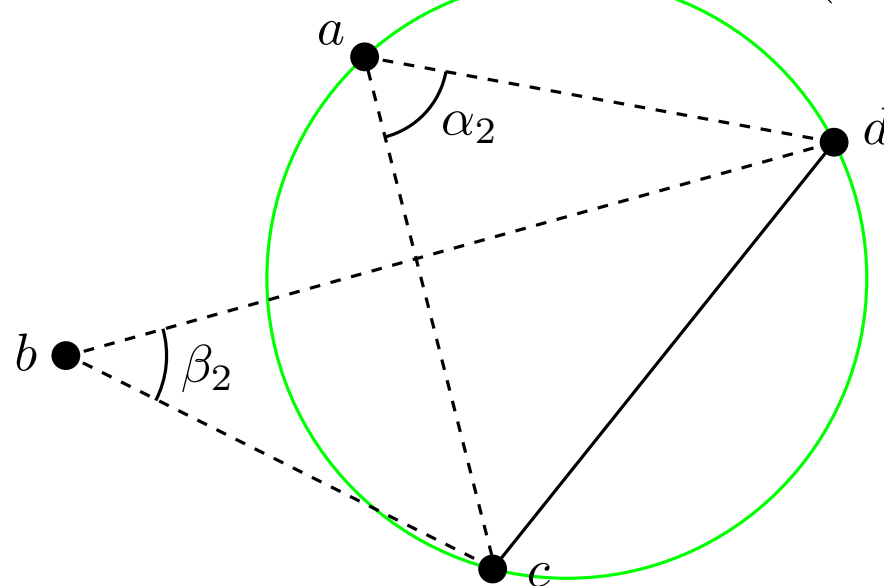
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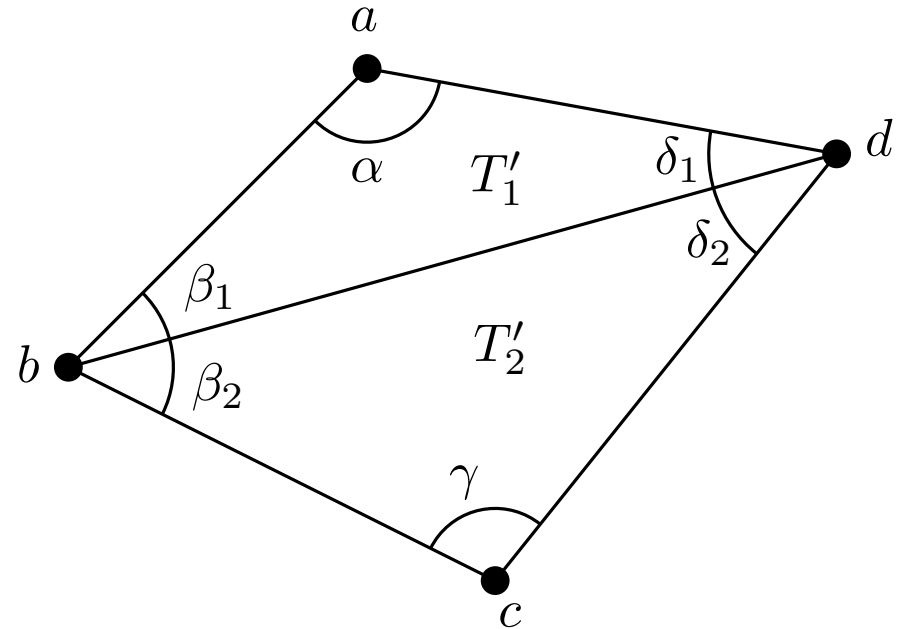
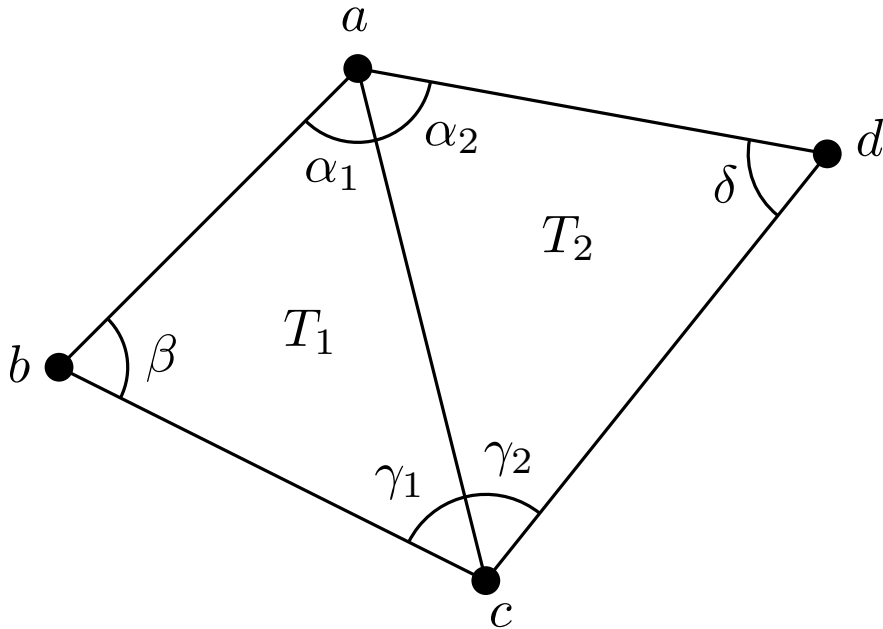
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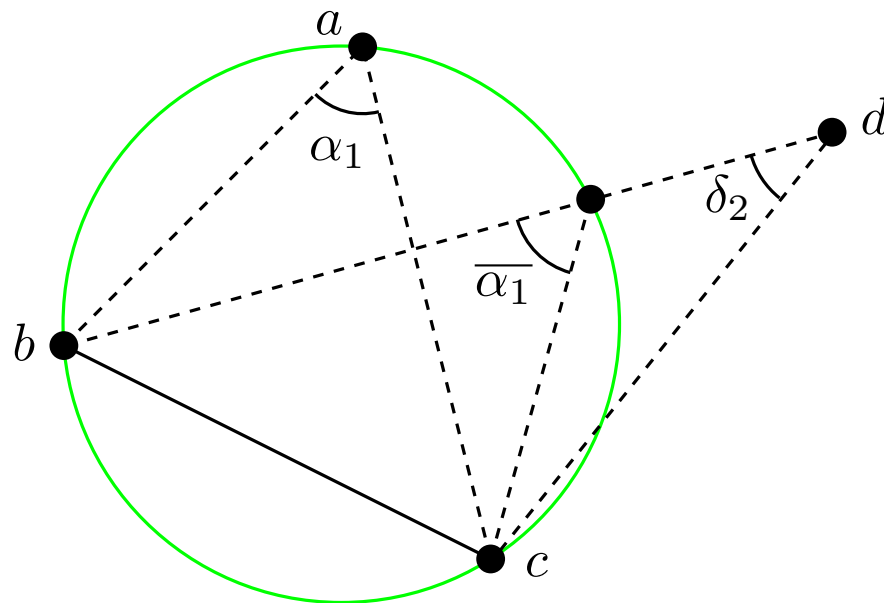
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Corollary. The Delaunay triangulation is the most equiangular among all triangulations of a given set of points.

If P does not contain four or more concyclic points, it follows from the previous lemma.

If P contains four or more concyclic points, $Del(P)$ contains a polygon inscribed in a circle which can be triangulated in several ways. Nevertheless, Lemma 1 (on the geometrical locus of all the points from which a segment is seen under a given angle) guarantees that every triangulation of a polygon inscribed in a circle has the same fineness, since each edge of the polygon belongs to a triangle, and every possible triangle gives rise to the same angle.

DELAUNAY TRIANGULATION

DELAUNAY TRIANGULATION AND OPTIMAL TRIANGULATIONS

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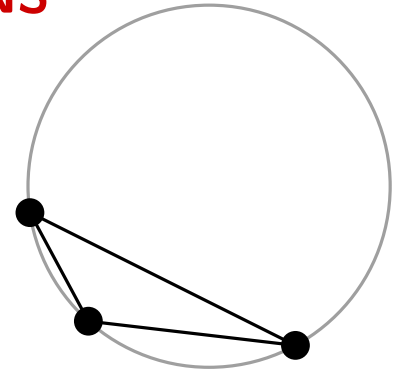
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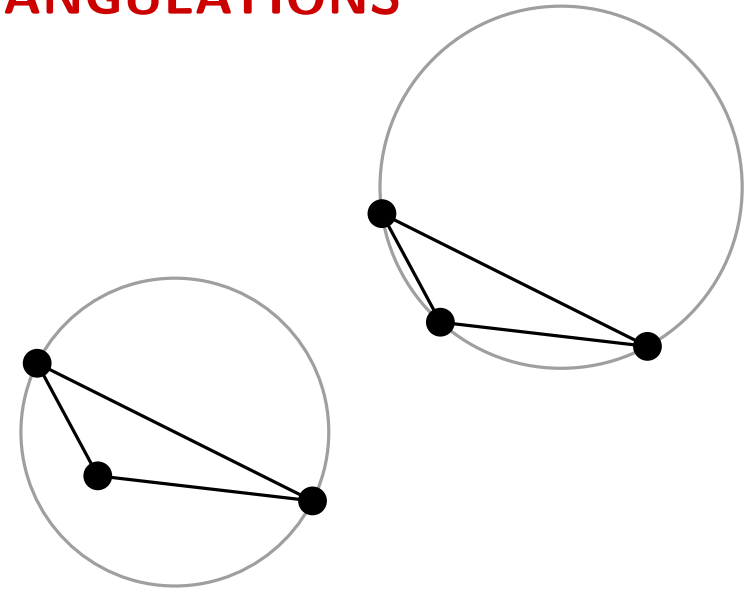


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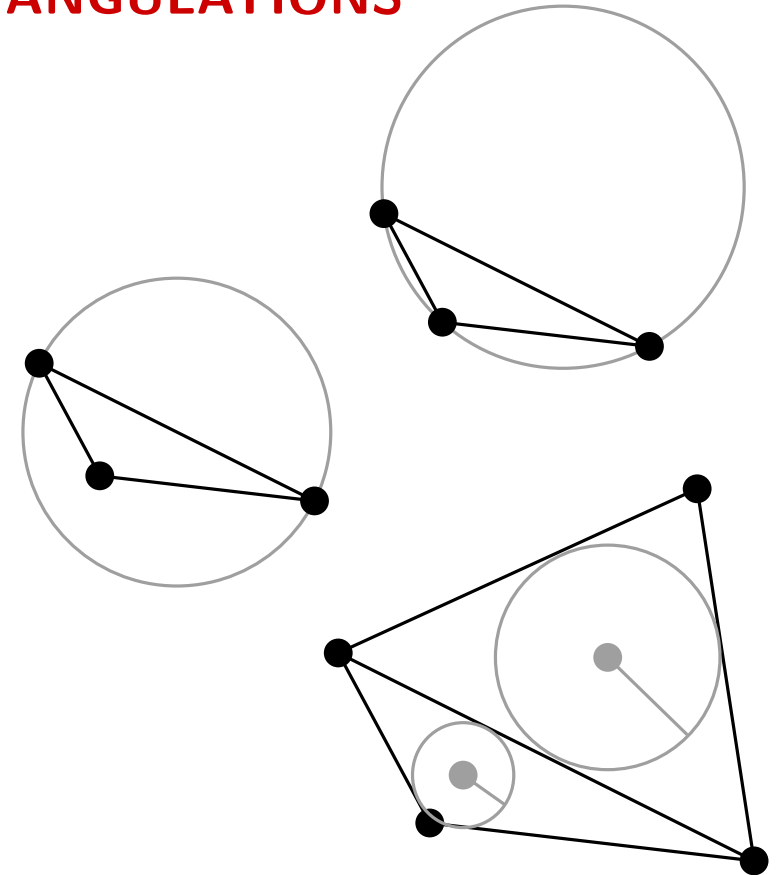


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DELAUNAY TRIANGULATION

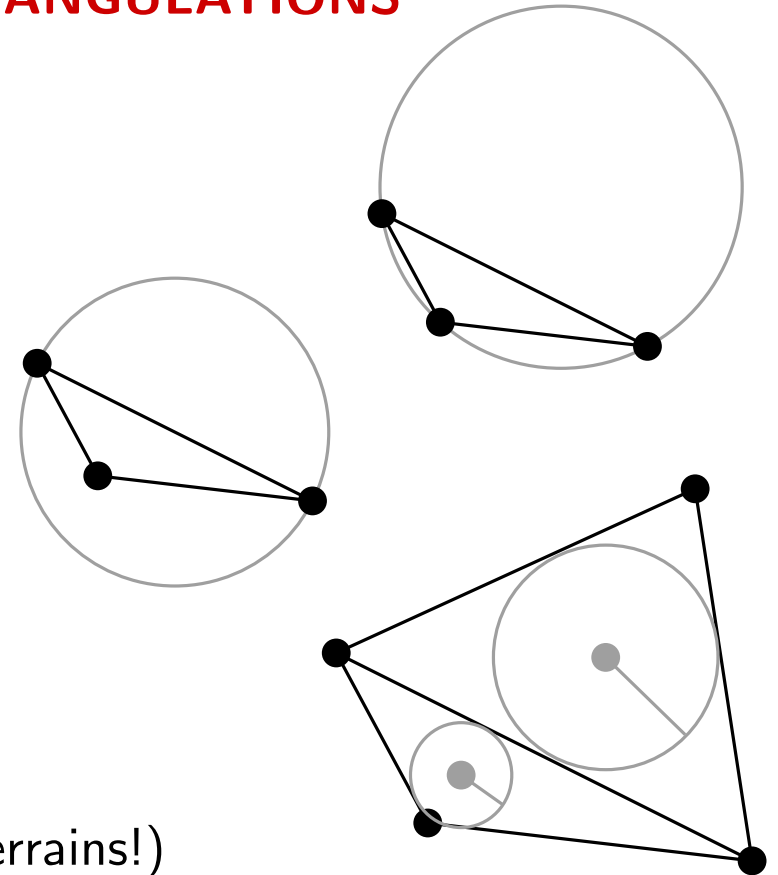
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- Maximizes the minimum angle
- Maximizes the whole angle vector
- Minimizes the maximum circumcircle
- Minimizes the maximum circumscribed circle
- Maximizes the sum of the inradii
- Minimizes the integral of the gradient squared (for terrains!)

This is known as the *roughness* of the terrain

Note that this property is independent from the data, in other words, it is independent from the values of the z -coordinates of the input points!



DELAUNAY TRIANGULATION

TWO BOOKS WITH MUCH MORE INFORMATION

A. Okabe, B. Boots, K. Sugihara, S. N. Chiu

Spatial Tessellations

2nd ed., J. Wiley & Sons, 2000.

F. Aurenhammer, R. Klein, D.-T. Lee

Voronoi Diagrams and Delaunay Triangulations

World Scientific, 2013.